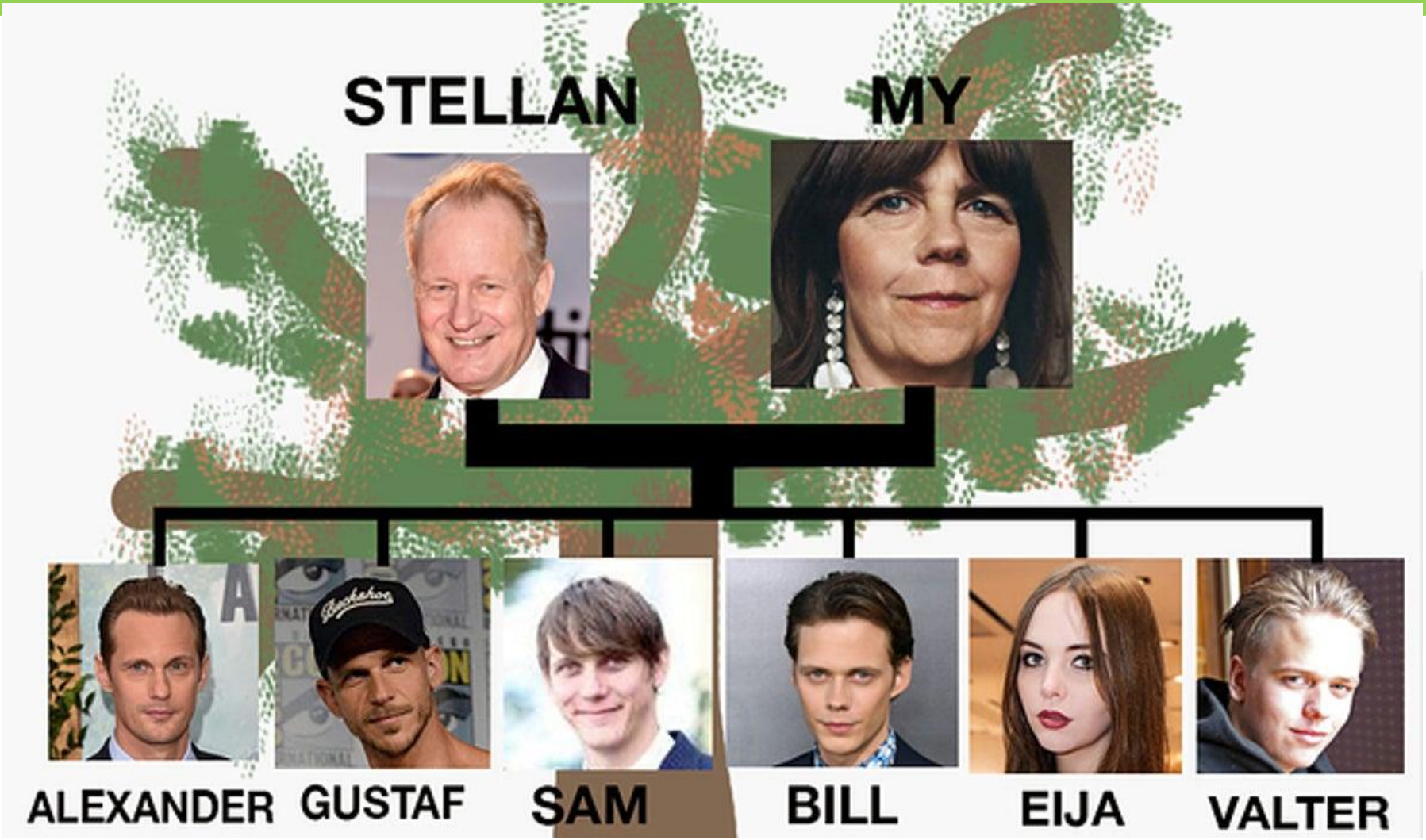
MBG 1201 Introduction to Modern Biology

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Outline

Comparison of Asexual and Sexual Reproduction
Auotosomal and sex chromosomes
Sexual Life Cycles
Meiosis
Origins of Genetic Variation
Crossing Over
Independent Assortment
Random Fertilization
The Evolutionary Significance of Genetic Variation Within Populations
Mendelian Genetics

Offspring acquire genes from parents by inheriting chromosomes.

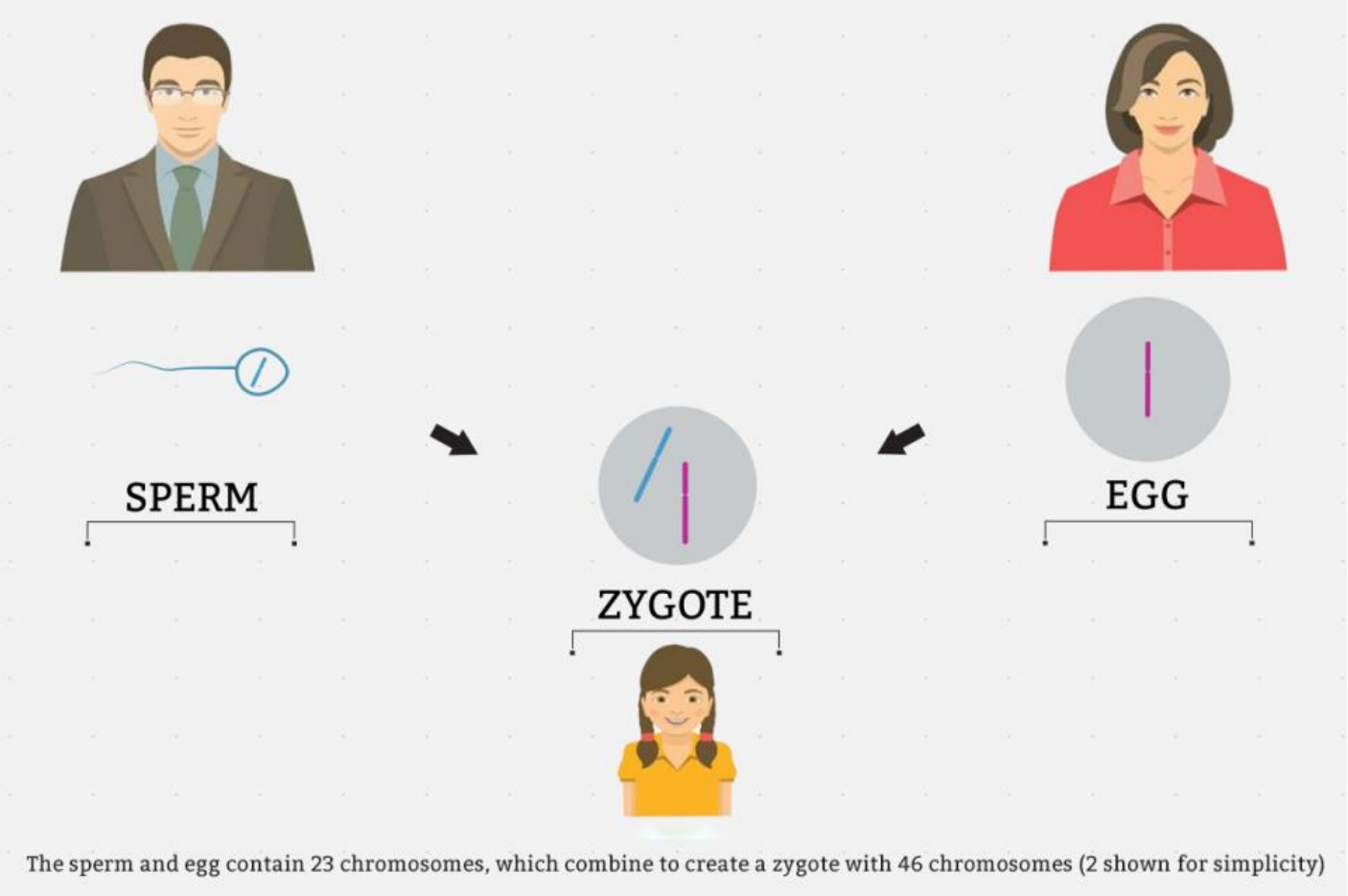


The genes we inherit from our mothers and fathers are our genetic link to our parents, and they account for family resemblances such as shared eye color or freckles.

Offspring acquire genes from parents by inheriting chromosomes.

• Genes are the units of heredity, and are made up of segments of DNA

• In animals and plants, genes are passed to the next generation through reproductive cells called **gametes** (sperm and eggs)
During fertilization, male and female gametes (sperm and eggs) unite, passing on
genes of both parents to their offspring.



The sperm and egg contain 23 chromosomes, which combine to create a zygote with 46 chromosomes (2 shown for simplicity)

Offspring acquire genes from parents by inheriting chromosomes.

- Every species has a characteristic number of chromosomes.

Humans have 46 chromosomes in their **somatic cells**, all cells of the body except gametes and their precursors.

Each chromosome consists of a single long DNA molecule, elaborately coiled in association with various proteins. One chromosome includes several hundred to a few thousand genes, each gene being a precise sequence of nucleotides along the DNA molecule.

Organism	Chromosomes in Gamete	Chromosomes in Somatic Cell
Human	23	46
Mouse	20	40
Flatworm	8	16
Corn	10	20

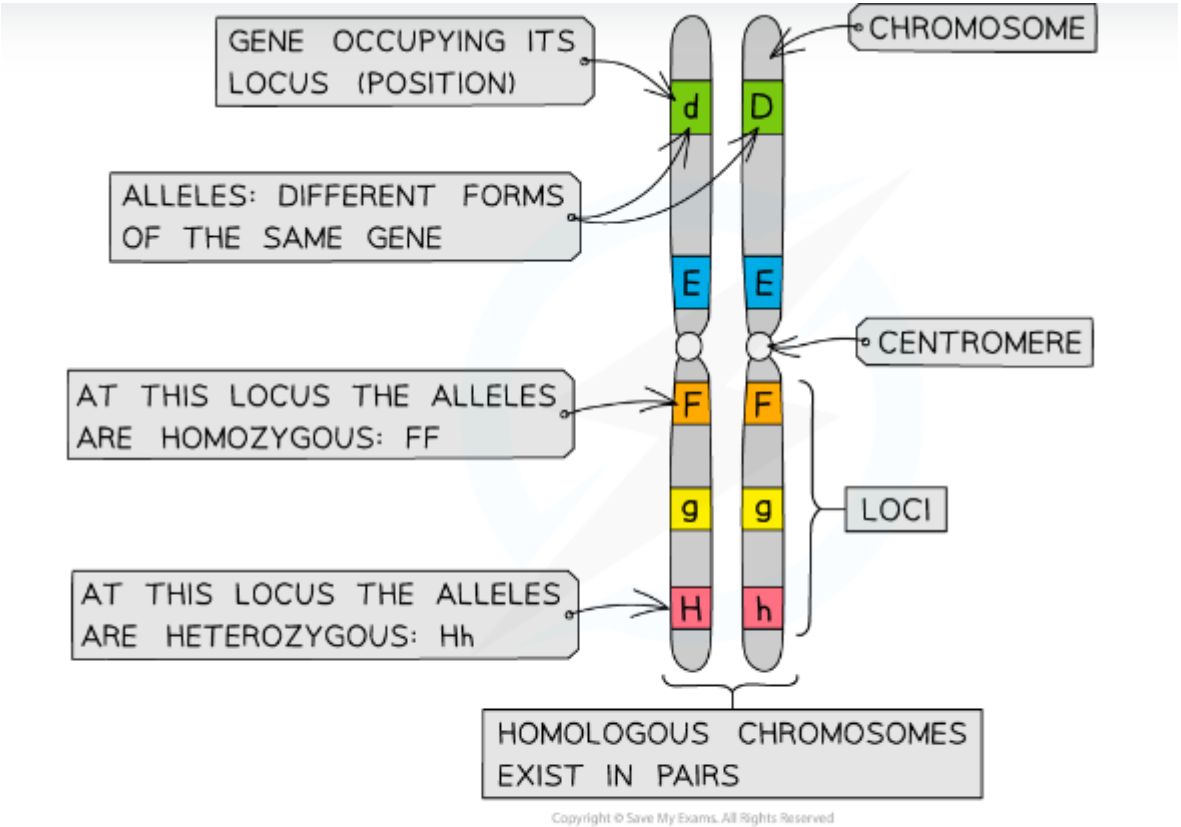
Offspring acquire genes from parents by inheriting chromosomes.

- **locus:** A gene's specific location along the length of a chromosome.

- Except for small amounts of DNA in mitochondria and chloroplasts, most DNA is packaged into chromosomes.

- One set of chromosomes is inherited from each parent.

Our genome consists of the genes and other DNA that make up the chromosomes we inherited from our parents.



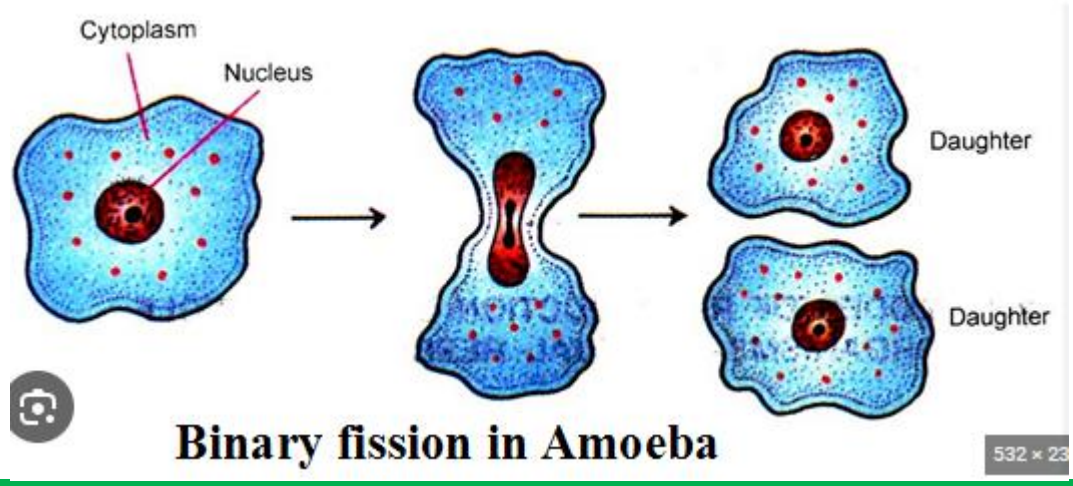
Five different genes found at five different loci

Comparison of Asexual and Sexual Reproduction

• In **asexual reproduction**, one parent produces genetically identical offspring by mitosis e.g. A yeast cell or an amoeba is the sole parent and passes copies of all its genes to its offspring without the fusion of gametes.

• Asexual reproduction gives rise to **clones**, a group of genetically identical individuals

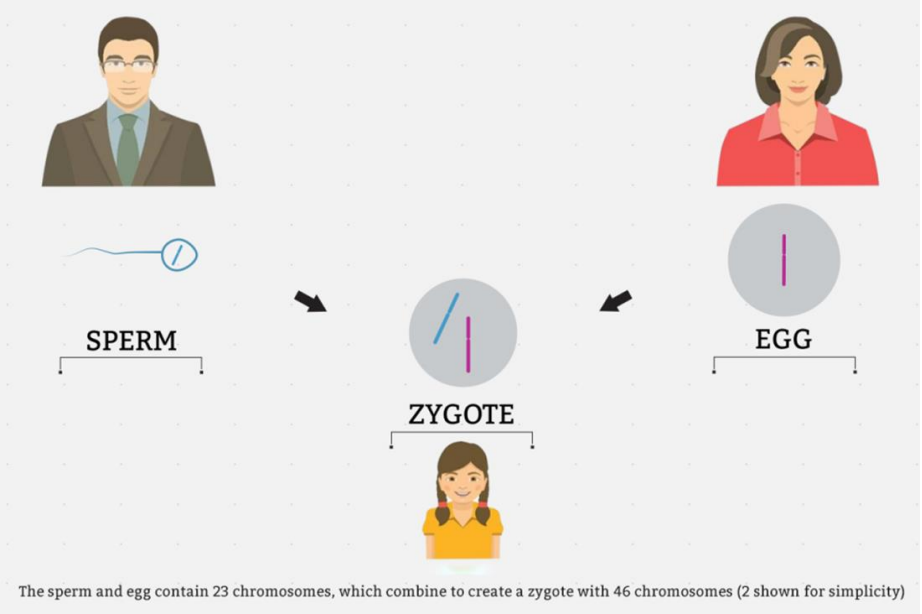
• Genetic differences occasionally arise in asexually reproducing organisms as a result of changes in the DNA called **mutations**.



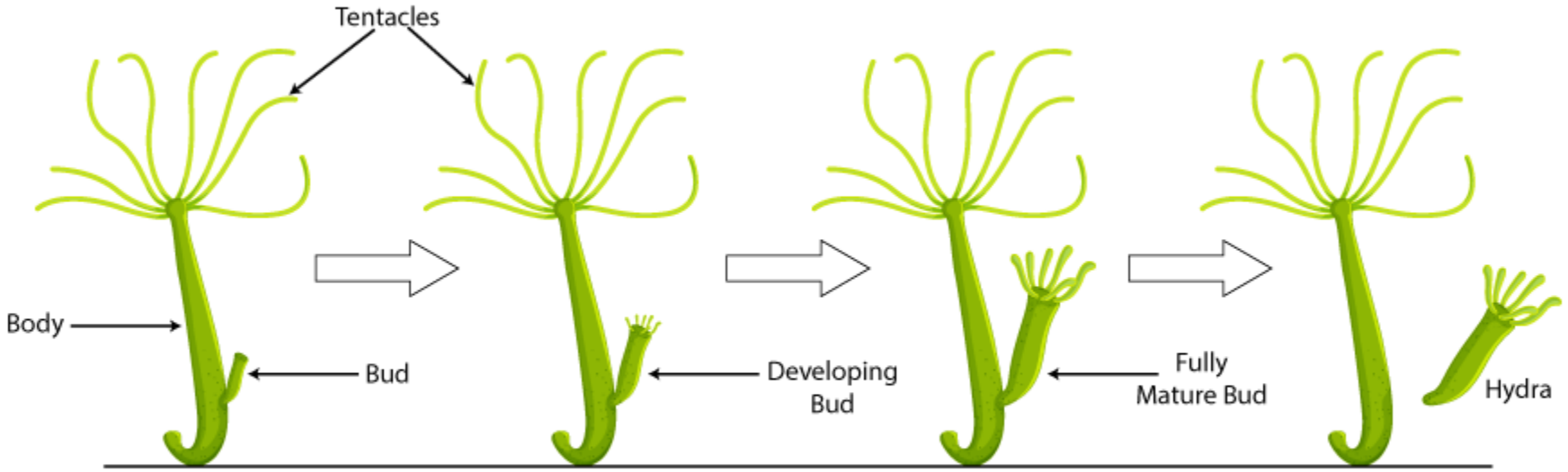
Comparison of Asexual and Sexual Reproduction

- In **sexual reproduction**, two parents give rise to offspring that have unique combinations of genes inherited from the two parents

- Offspring of sexual reproduction **vary genetically** from their parents and siblings



Asexual Reproduction in Multicellular Organisms



Budding In Hydra

Some multicellular organisms are also capable of reproducing asexually.
This relatively simple animal, a hydra, reproduces by budding.
The bud, a localized mass of mitotically dividing cells, develops into a small hydra, which detaches from the parent.

A bud is generally formed due to cell division at one particular site.
Redwoods can reproduce asexually by sprouting from the stump.

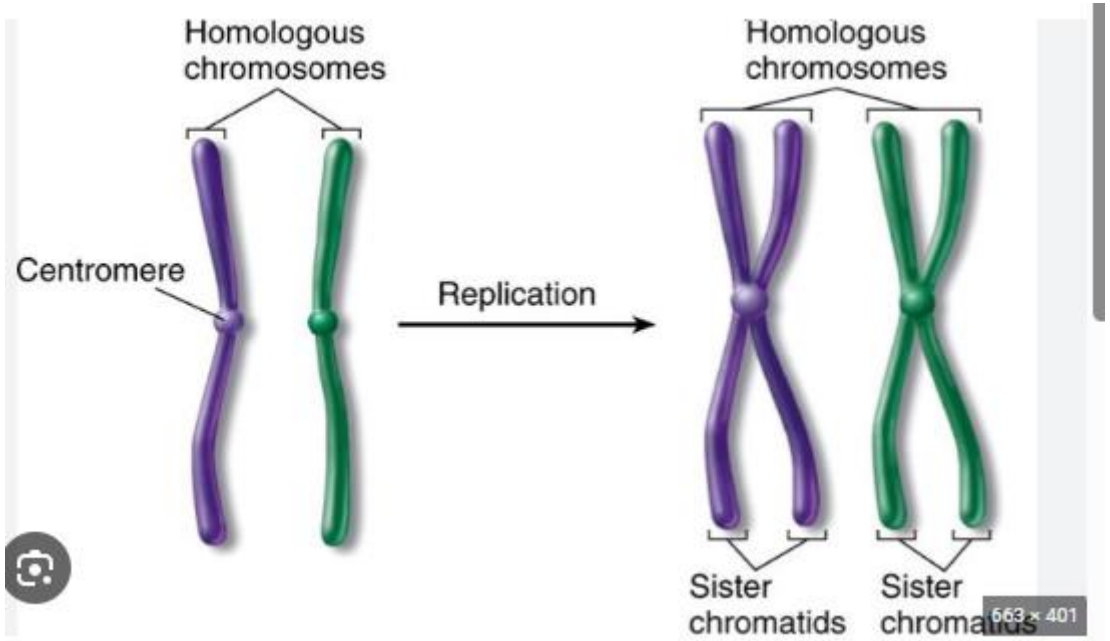


• Human somatic cells (any cell other than a gamete) have **23 pairs** of chromosomes.

• The two chromosomes in each pair are called homologous chromosomes, or homologs.

• Chromosomes in a homologous pair are the same length, centromere position, and staining pattern and carry genes controlling the same inherited characters.

• Specific Chromosomes are identified based on : size, the position of their centromeres, and the pattern of colored bands produced by certain chromatin-binding stains.





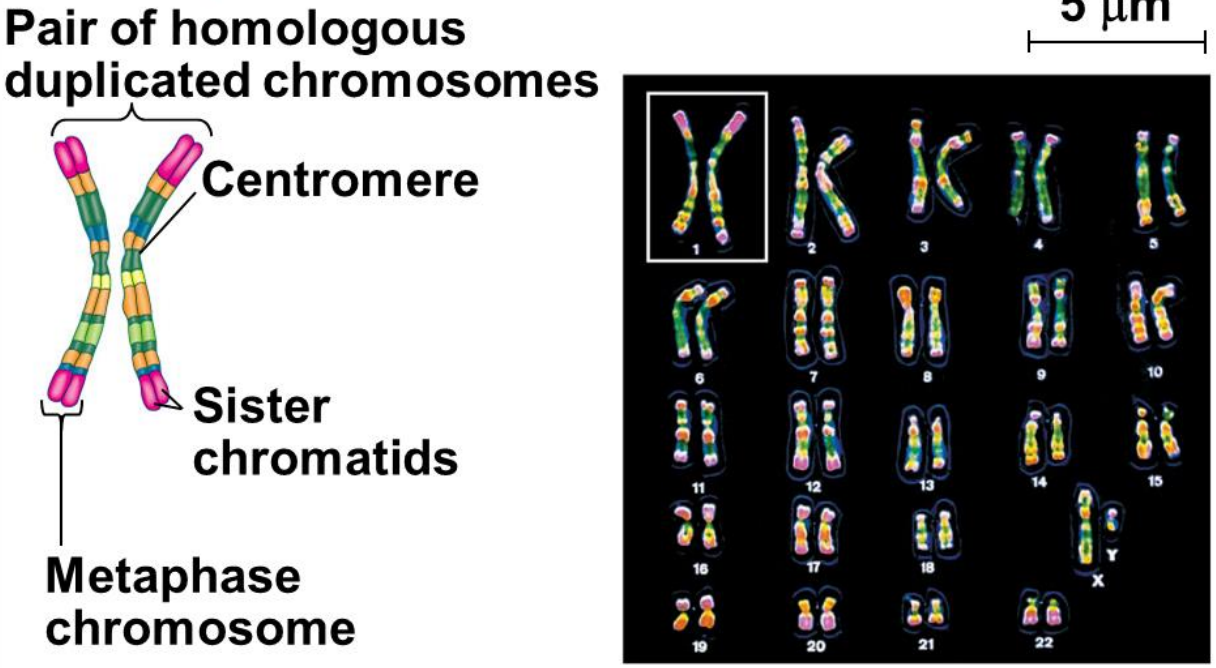
APPLICATION



- **karyotype:** a display of condensed chromosomes arranged in pairs. Karyotyping can be used to screen for defective chromosomes or abnormal numbers of chromosomes associated with certain congenital disorders, such as Down syndrome

Karyotypes are prepared from isolated somatic cells, which are treated with a drug to stimulate mitosis and then grown in culture for several days. Cells arrested when the chromosomes are most highly condensed—at metaphase—are stained and then viewed with a microscope equipped with a digital camera. An image of the chromosomes is displayed on a computer monitor, and digital software is used to arrange them in pairs according to their appearance.

TECHNIQUE



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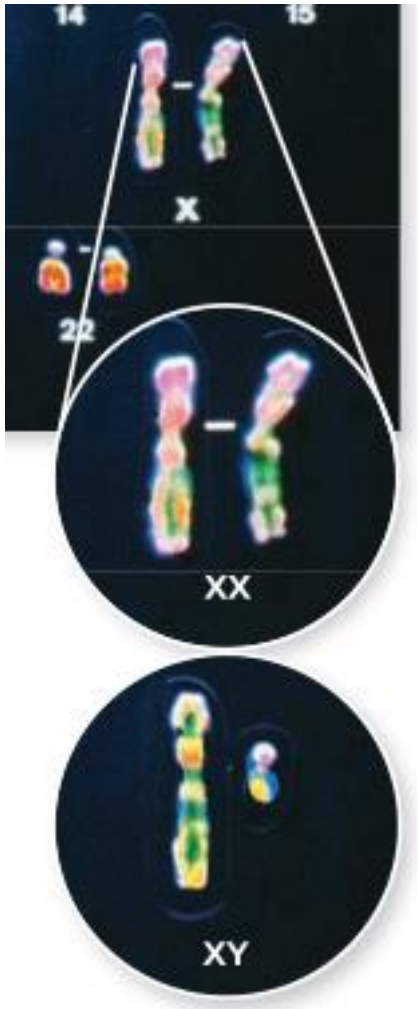
The **sex chromosomes** are called X and Y

- Human females have a homologous pair of X chromosomes (XX)

- Human males have one X and one Y chromosome (XY)

The X and Y chromosomes are not homologous except in small regions.

- The 22 pairs of chromosomes that do not determine sex are called **autosomes**.

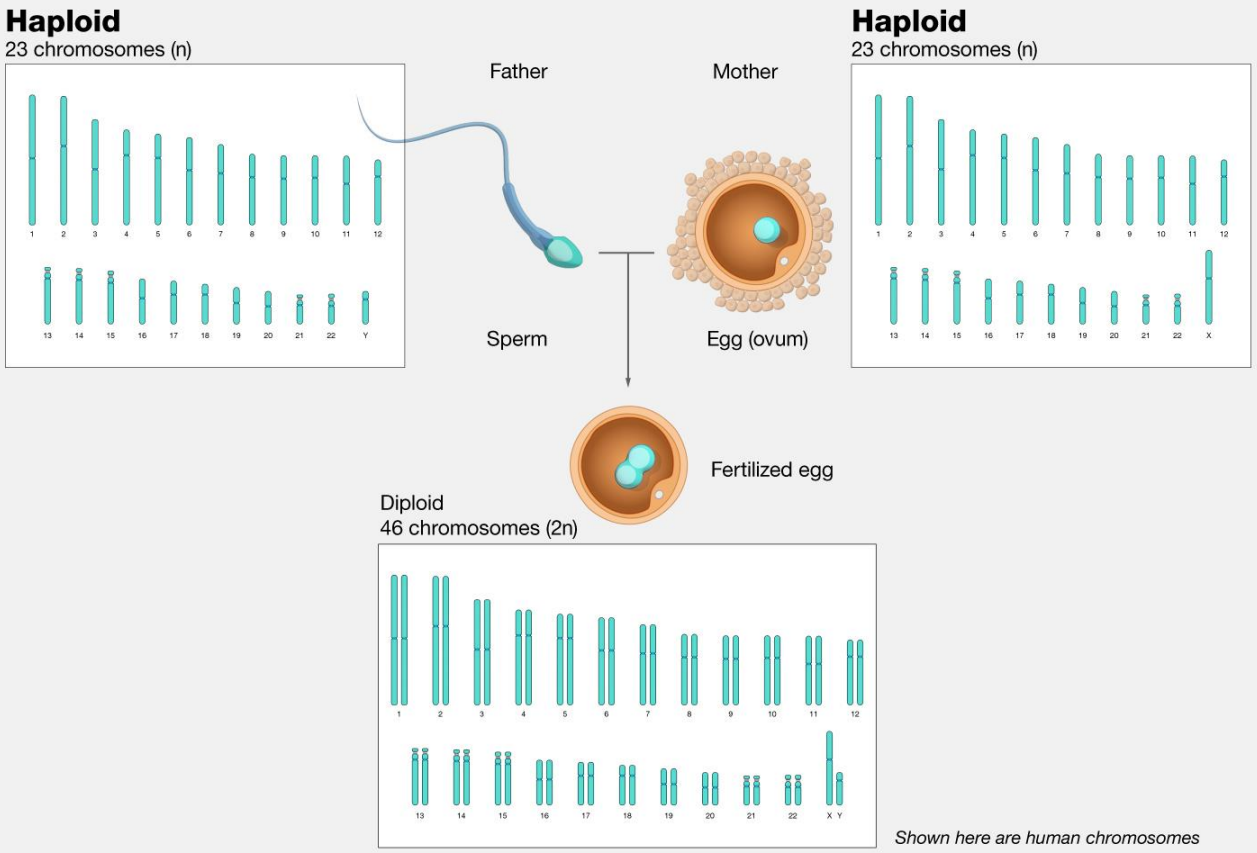


• Each pair of homologous chromosomes arises from sexual reproduction and includes one chromosome from each parent

• The 46 chromosomes in a human somatic cell are actually two sets of 23:

one from the mother (maternal) and one from the father (paternal)

• The number of chromosomes in a single set is represented by n.

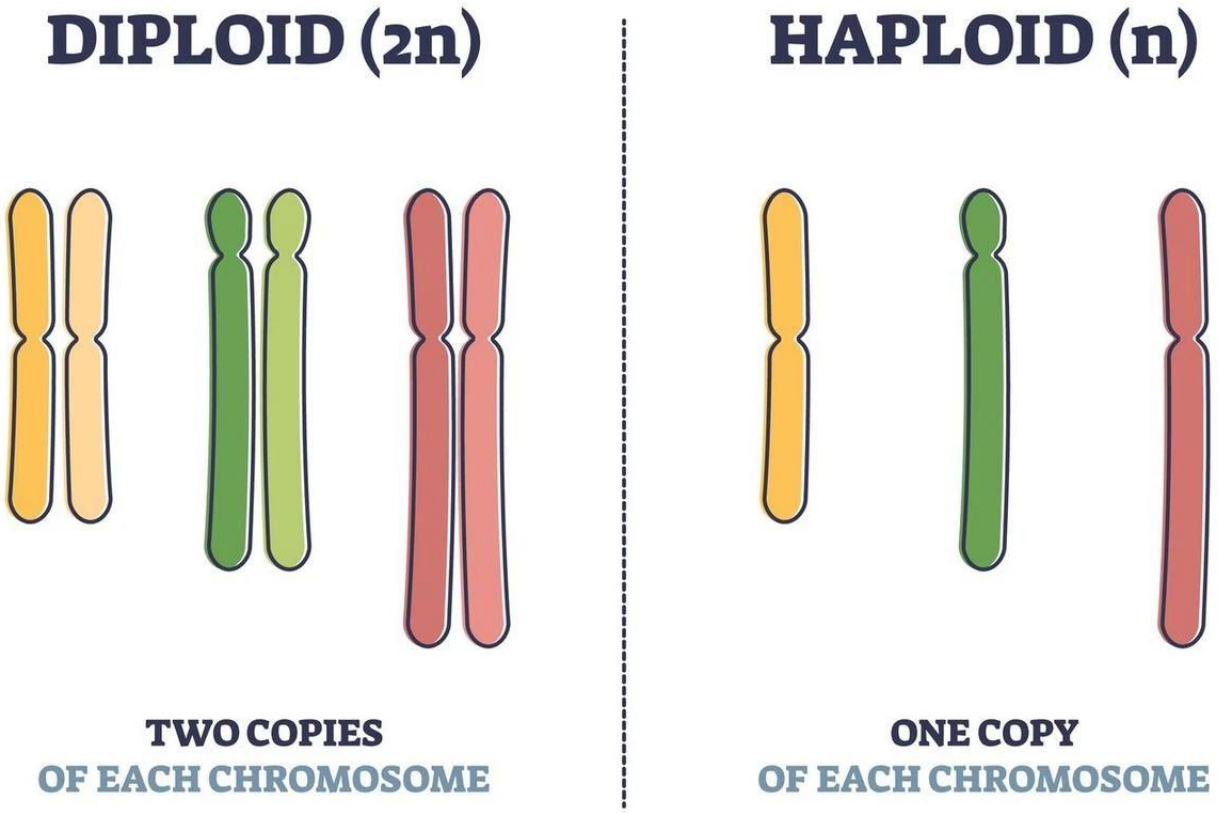


Somatic Cells

- **A diploid cell (2n)** has two sets of chromosomes

- For humans, the diploid number is 46 (2n = 46)

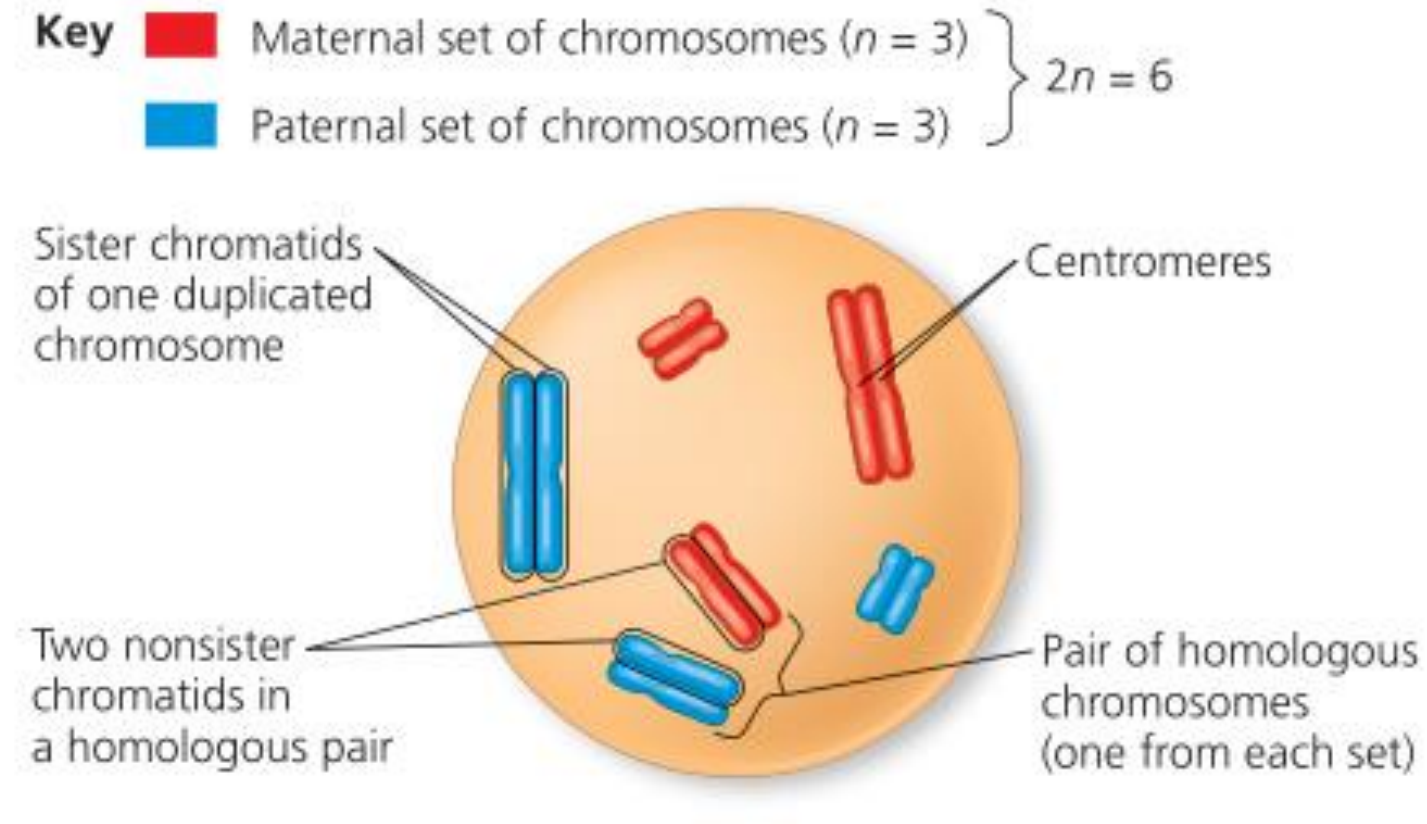
The chromosome number generally does not correlate with the size or complexity of a species' genome; it simply reflects how many linear pieces of DNA make up the genome, which is a function of the evolutionary history of that species



Describing Chromosomes

- In a cell in which DNA synthesis has occurred, each chromosome is replicated
- Each replicated chromosome consists of two identical sister chromatids

▼ Figure 13.4 Describing chromosomes. A cell from an organism with a diploid number of 6 ($2n = 6$) is depicted here following chromosome duplication and condensation. Each of the six duplicated chromosomes consists of two sister chromatids associated closely along their lengths. Each homologous pair is composed of one chromosome from the maternal set (red) and one from the paternal set (blue). Each set is made up of three chromosomes in this example (long, medium, and short). Together, one maternal and one paternal chromatid in a pair of homologous chromosomes are called nonsister chromatids.



Reproductive cell-Gametes (Sperm and Eggs)

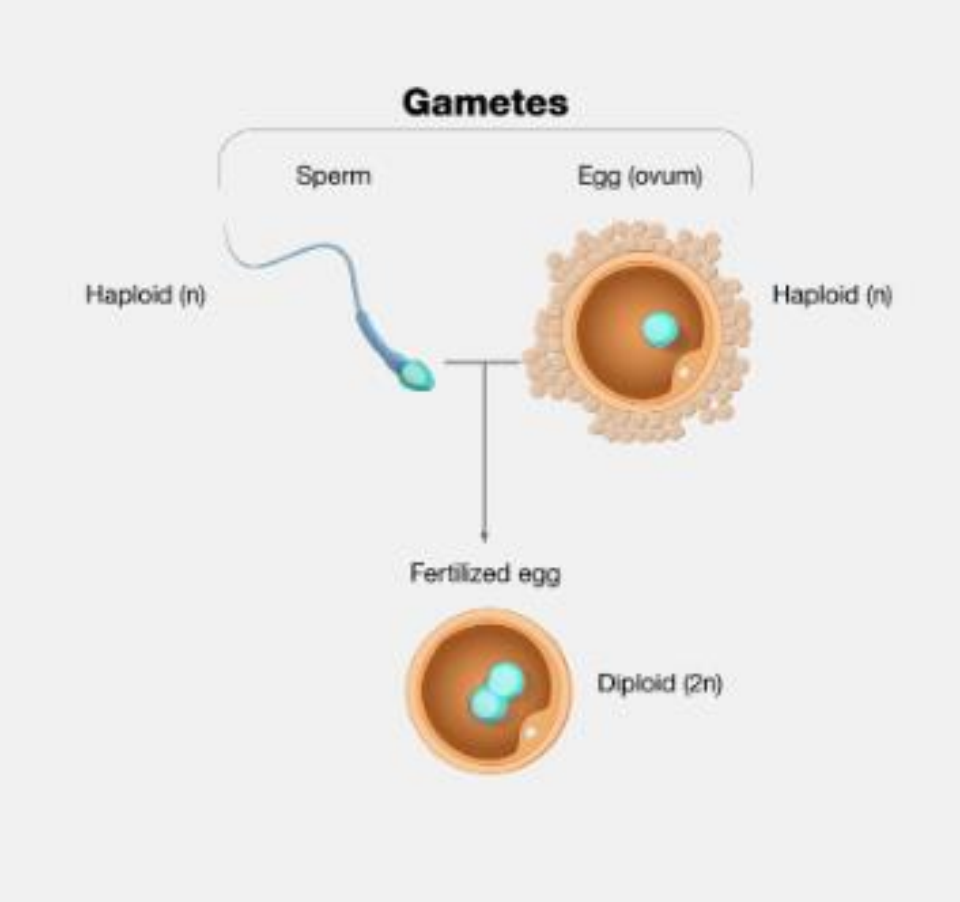
A gamete (sperm or egg) contains a **single set** of chromosomes, and is **haploid** (n)

For humans, the haploid number is 23 (n = 23)

Each set of 23 consists of 22 autosomes and a single sex chromosome

In an unfertilized egg (ovum), the sex chromosome is X

In a sperm cell, the sex chromosome may be either X or Y



The Variety of Sexual Life Cycles

• **life cycle**: generation-to-generation sequence of stages in the reproductive history of an organism from conception to production of its own offspring.

Fertilization and meiosis alternate in sexual life cycles, maintaining a constant number of chromosomes in a species from one generation to the next.

Although the alternation of meiosis and fertilization is common to all organisms that reproduce sexually, the timing of these two events in the life cycle varies, depending on the species.

The Variety of Sexual Life Cycles

- 3 types of life cycles in different organisms are-
    - Animal Sexual Life cycle,
    - Plants and Algae life cycle
    - Fungi and Protista life cycle.

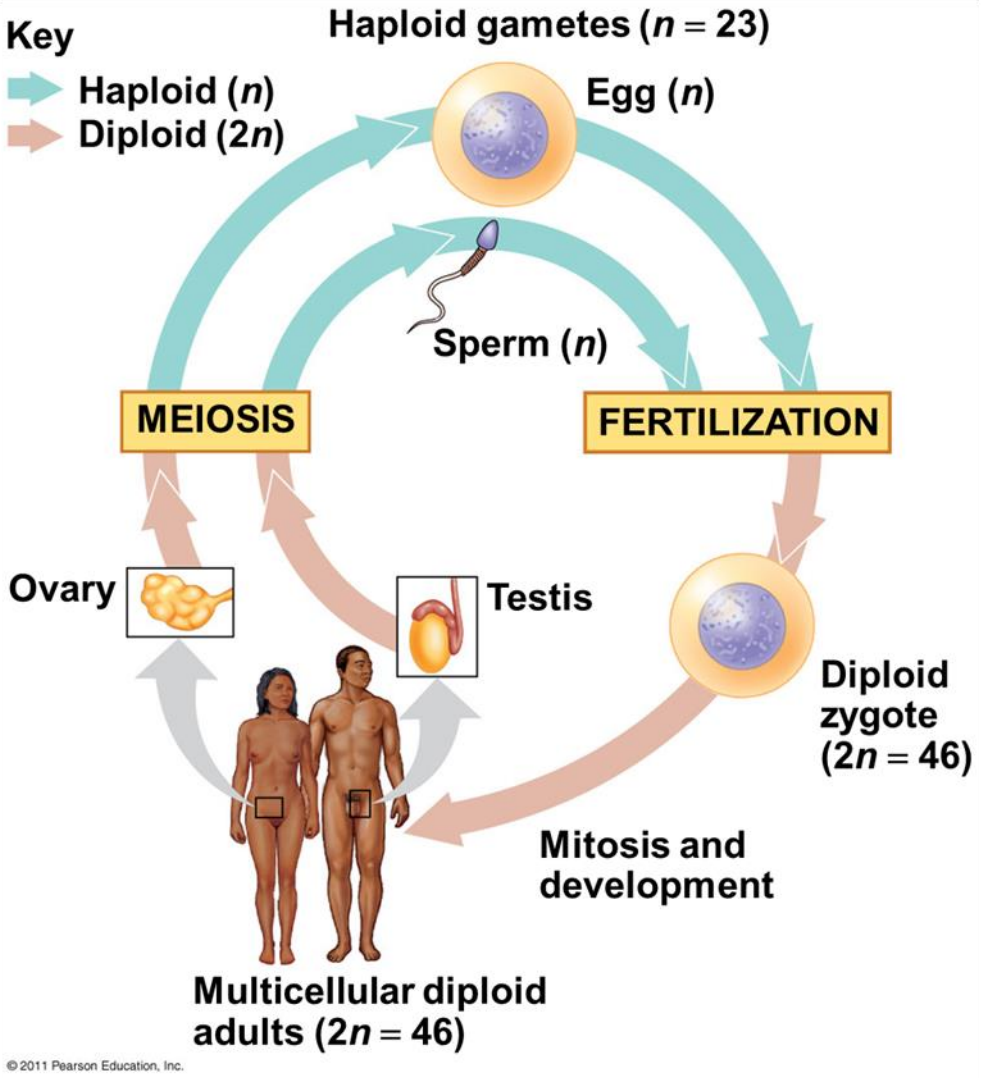
- Depending on the type of life cycle, either haploid or diploid cells can divide by mitosis

- only diploid cells can undergo meiosis

- In all three life cycles, the halving and doubling of chromosomes contributes to genetic variation in offspring

1- Animal Sexual Life Cycle

- In animals, at sexual maturity, the ovaries and testes produce haploid gametes through meiosis, which undergo no further cell division before fertilization
- Gametes are the only types of animal cells produced by meiosis. Each has 1 set of chromosomes (Gametes are the only haploid cells in animals)
- **Fertilization** is the union of gametes (the sperm and the egg)
- The fertilized egg is called a zygote and has one set of chromosomes from each parent (diploid)
- The zygote produces somatic cells by mitosis and develops into a multicellular organism.
- Fertilization and meiosis alternate in sexual life cycles to maintain chromosome number



2-Plants and Algae Life Cycle

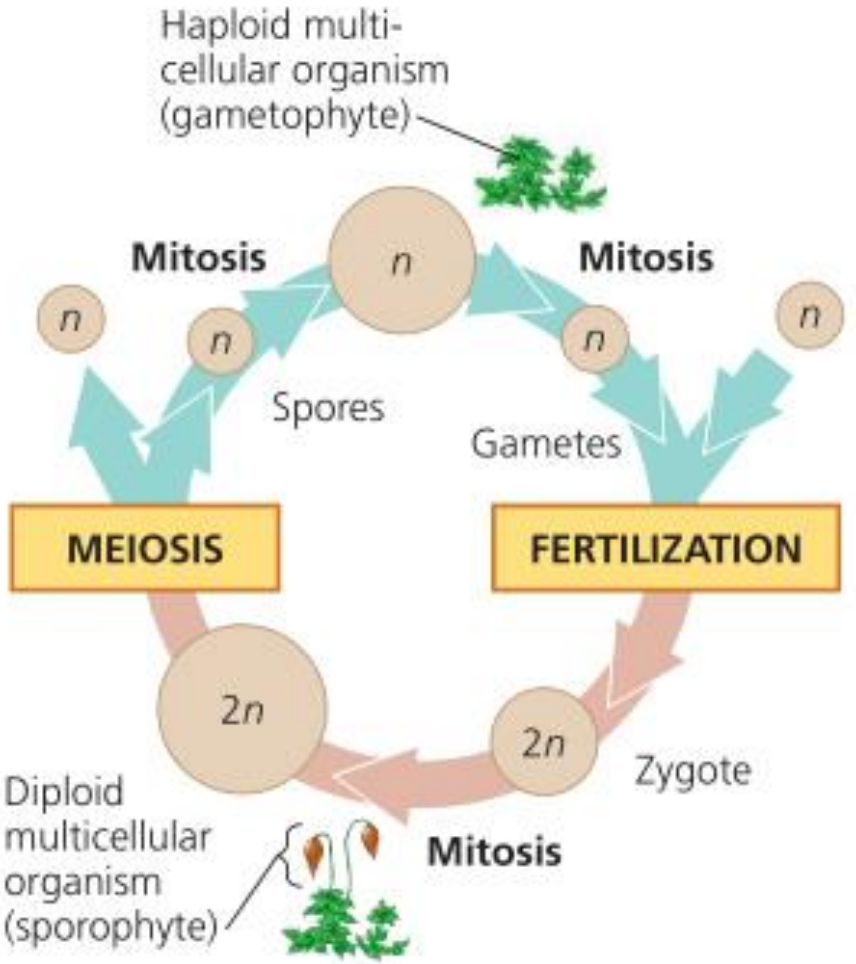
Plants and some algae exhibit an **alternation of generations**
This life cycle includes both a diploid and haploid multicellular stage
The diploid zygote, called the sporophyte, makes haploid spores by meiosis

Each spore grows by mitosis into a haploid organism called a gametophyte

A gametophyte makes haploid gametes by mitosis

Fusion of 2 haploid gametes at fertilization=a diploid zygote

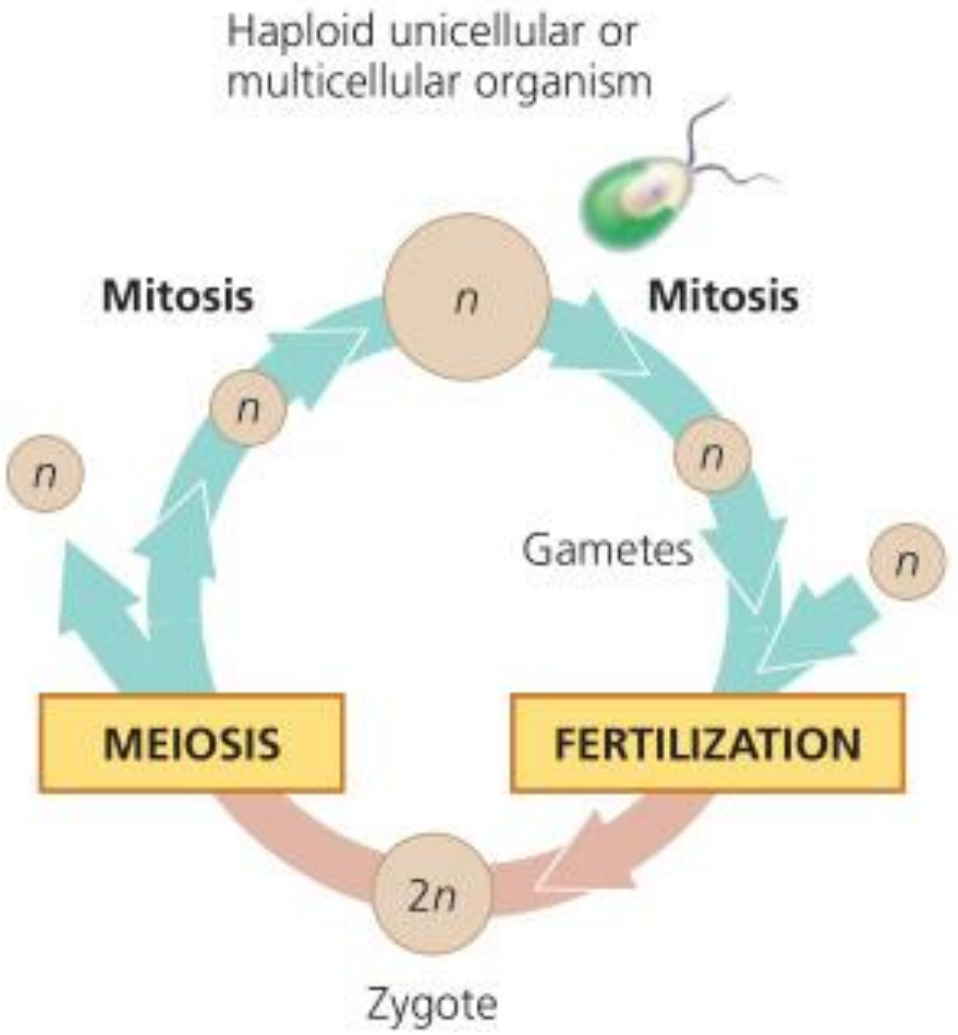
The sporophyte generation produces a gametophyte offspring, which produces the next sporophyte generation.



(b) Plants and some algae

3-Fungi and Protists Life Cycle

- In most fungi and some protists, the only diploid stage is the single-celled zygote; there is no multicellular diploid stage
- The zygote produces haploid cells by meiosis
- Each haploid cell grows by mitosis into a haploid multicellular organism
- The haploid adult produces gametes by mitosis



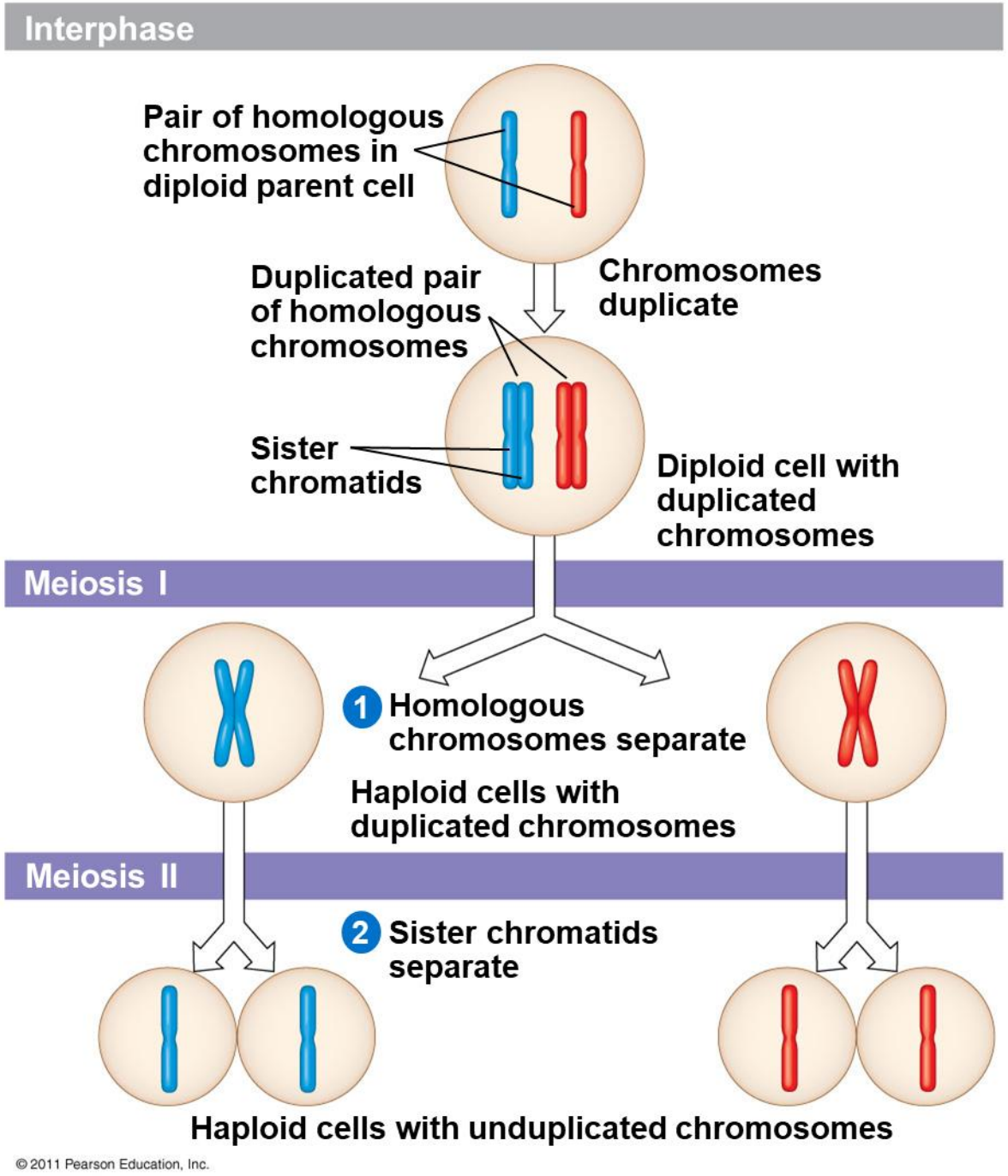
Meiosis reduces the number of chromosome sets from diploid to haploid

Like mitosis, Meiosis is preceded by interphase, which includes S phase (the duplication of chromosomes).

- Meiosis involves two sets of cell divisions, called meiosis I and meiosis II

Meiosis I: starts with 1 cell that is 2n and makes 2 cells that are 1n
Meiosis II: the cells resulted from Meiosis I divide again forming 2 cells each, so a total of 4 haploid cells, rather than the two daughter cells in Mitosis

- Each daughter cell has only half as many chromosomes as the parent cell

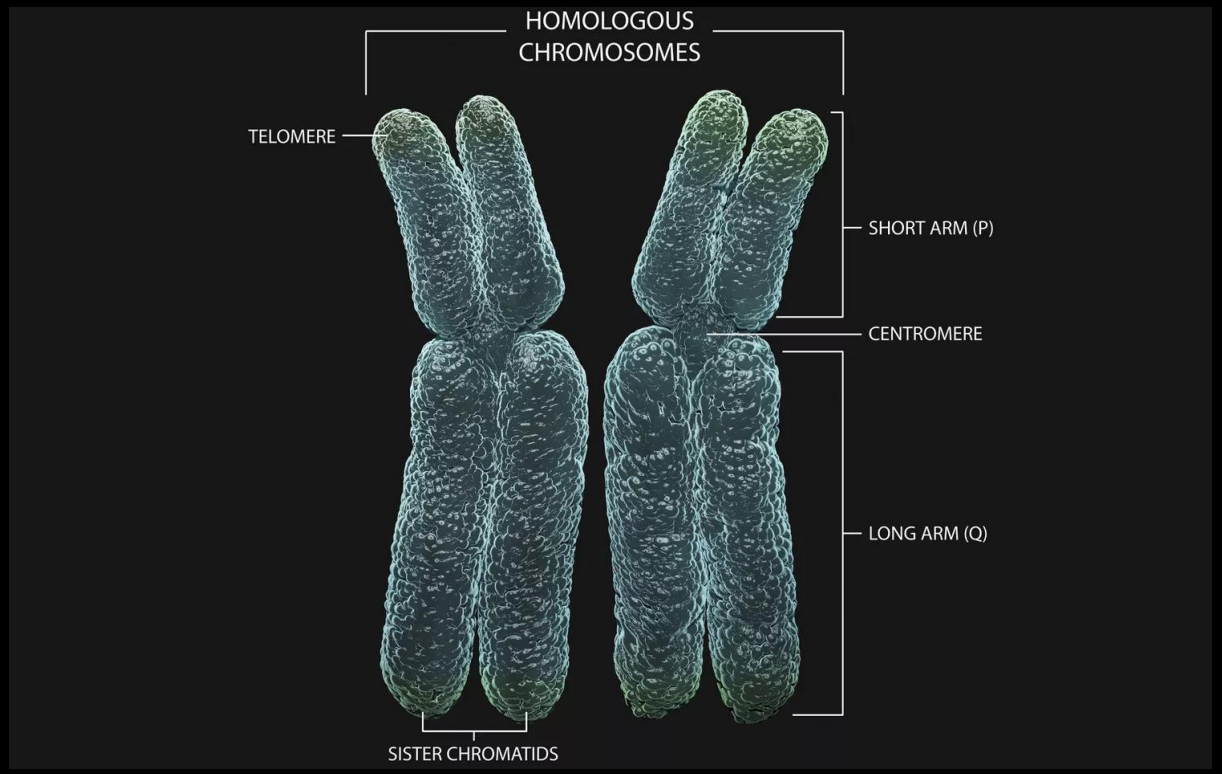


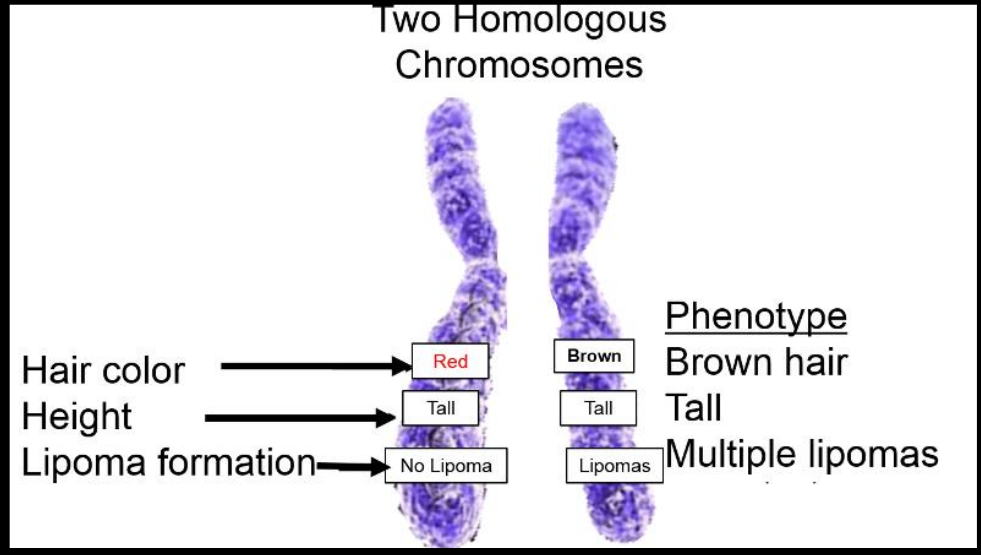
- During Interphase a homologous pair of chromosomes in a diploid parent cell replicate, forming sister chromatids
- The resulting sister chromatids are closely associated along their lengths
- This is called **sister chromatid cohesion**
- In the first cell division (meiosis I), homologous chromosomes pair up and separate
- Meiosis I results in two haploid daughter cells with replicated chromosomes; it is called the **reductional division**
- In the second cell division (meiosis II), sister chromatids separate
- Meiosis II results in four haploid daughter cells with unreplicated chromosomes; it is called the **equational division**

Homologs appear alike in the microscope, but they may have different versions of genes at corresponding loci; each version is called an **allele** of that gene.

For example, one chromosome might have an allele for freckles, but the homologous chromosome may have an allele for the absence of freckles at the same locus.

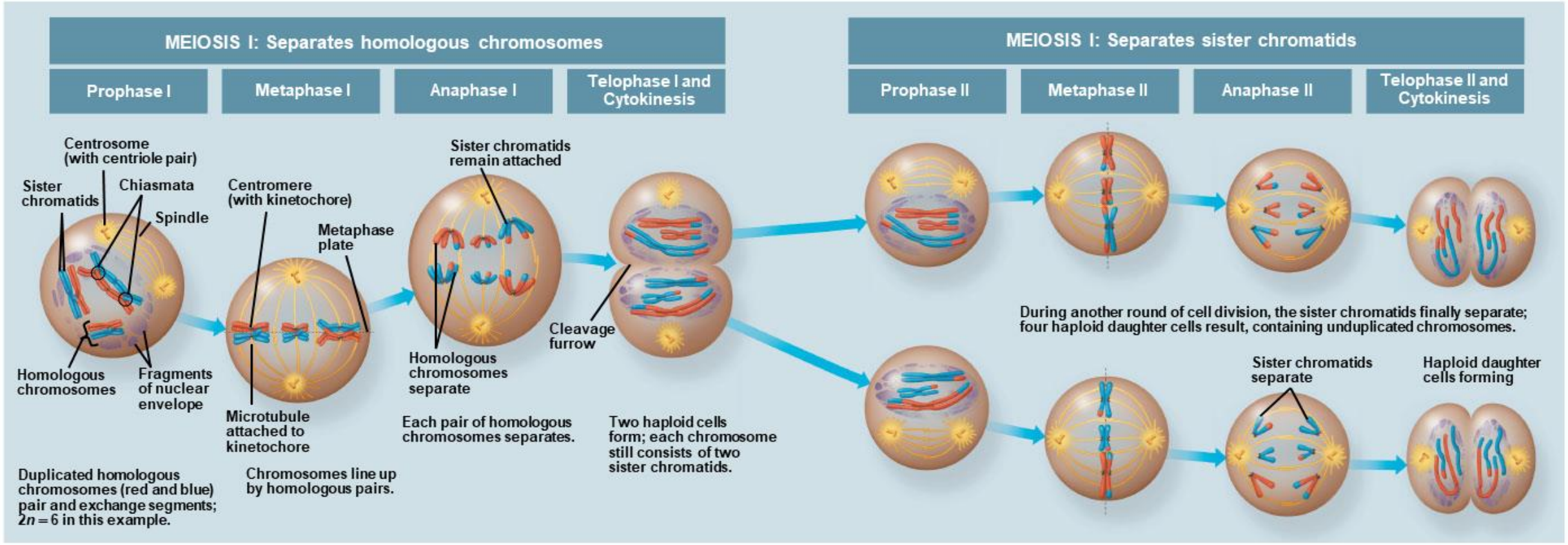
Homologs are not associated with each other in any obvious way except during meiosis.





Exploring meiosis in an animal cell- A closer look

- Meiosis I is preceded by interphase, in which chromosomes are replicated to form sister chromatids
- The sister chromatids are genetically identical and joined at the centromere
- The single centrosome replicates, forming two centrosomes



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Meiosis I separates homologous chromosomes:

Division in meiosis I occurs in four phases:

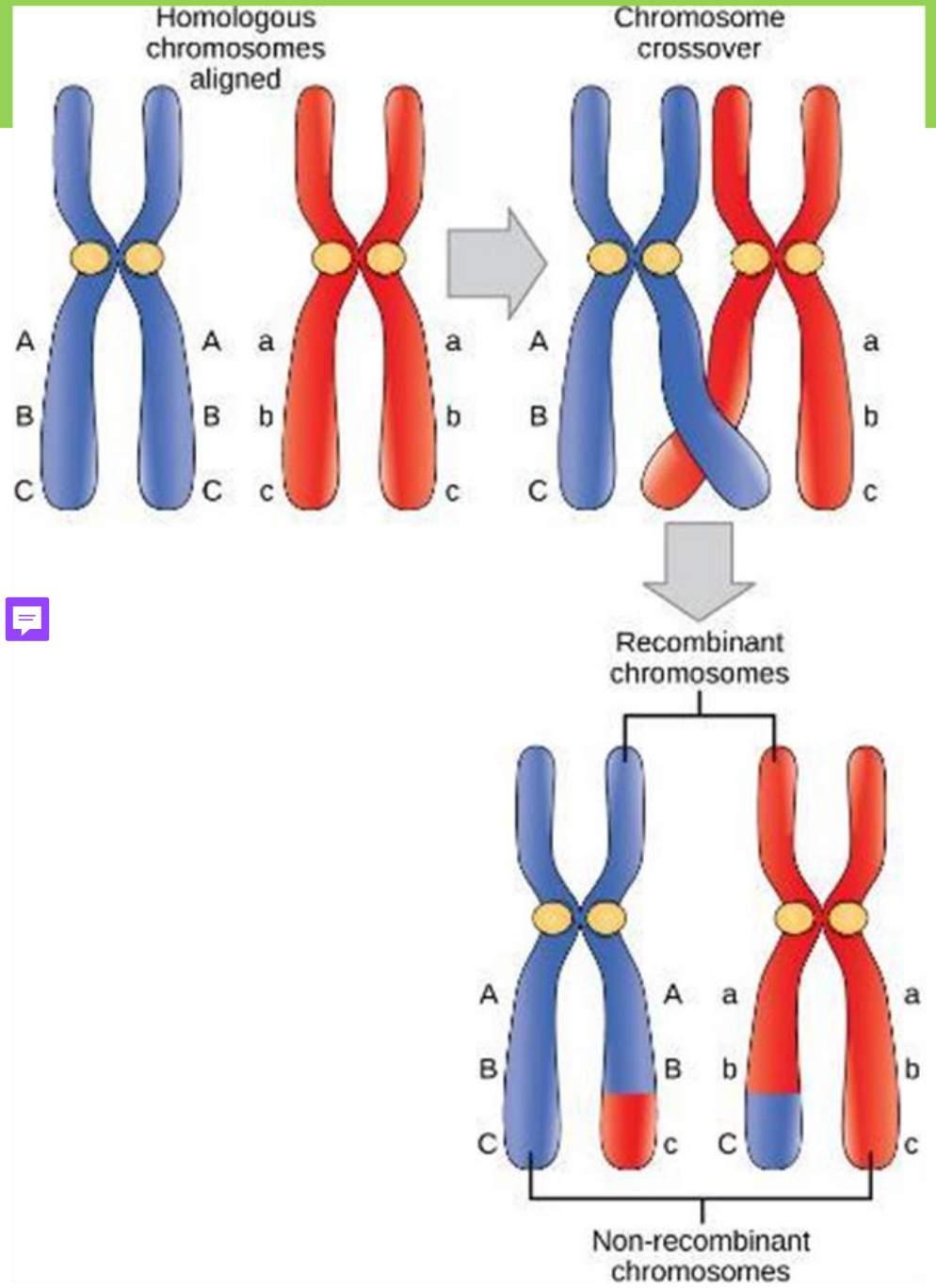
– Prophase I
– Metaphase I
– Anaphase I
– Telophase I and cytokinesis

Prophase I:

- Prophase I typically occupies more than 90% of the time required for meiosis
  - The nucleolus and nuclear membrane completely disappear, centrosomes move and kinetic spindle becomes apparent. Chromosomes begin to condense (as in prophase of mitosis).
- In prophase I (unlike in the prophase of mitosis) the homologous chromosomes:
  Find each other and go through **synapsis** = aligned with each other
  Exchange DNA pieces through **crossing-over** at places called **chiasmata** resulting in **recombinant** sister chromatids (containing a mixture of maternal and paternal DNA)

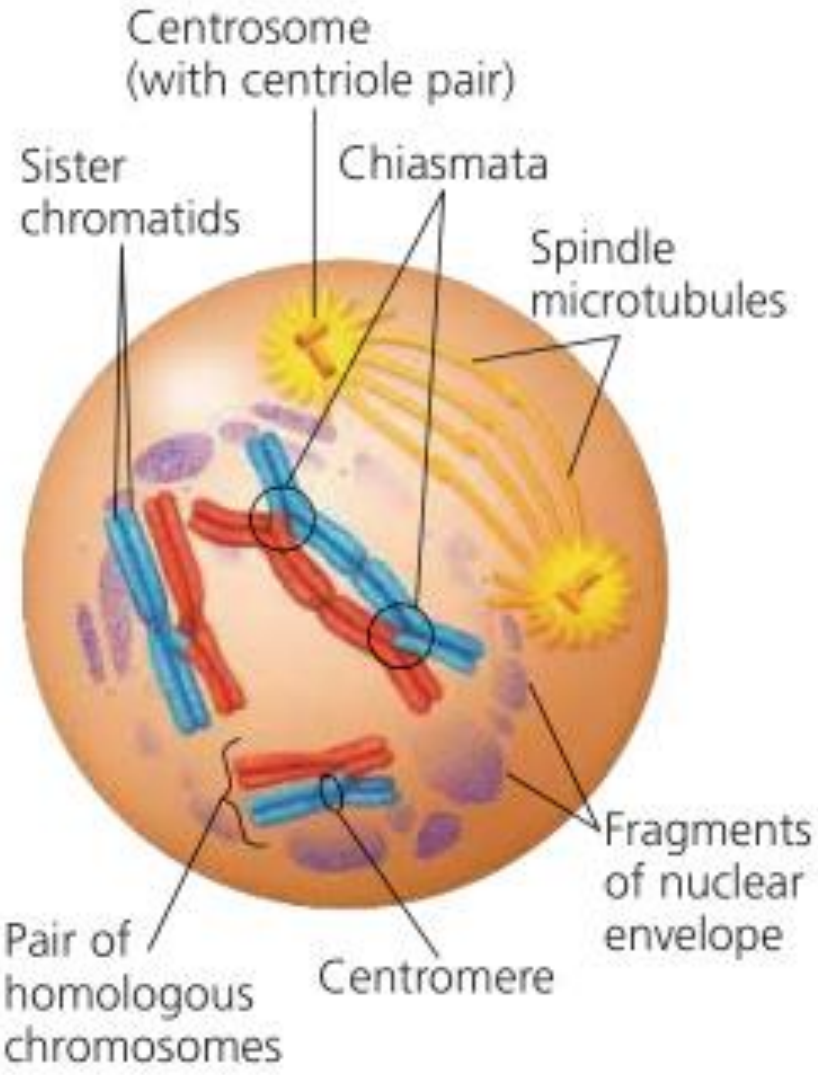
Prophase I:

- In synapsis, homologous chromosomes loosely pair up, aligned gene by gene and crossing over occurs.
- In crossing over, the DNA molecules of nonsister chromatids are broken (by proteins) at precisely corresponding positions.
- A zipper-like structure called the **synaptonemal complex** holds the homologs together tightly.
- DNA breaks are repaired, joining DNA from one nonsister chromatid to the corresponding segment of another.
- Each pair of chromosomes forms a tetrad, a group of four chromatids
- Each tetrad usually has one or more **chiasmata**, X-shaped regions where crossing over occurred



Prophase I:

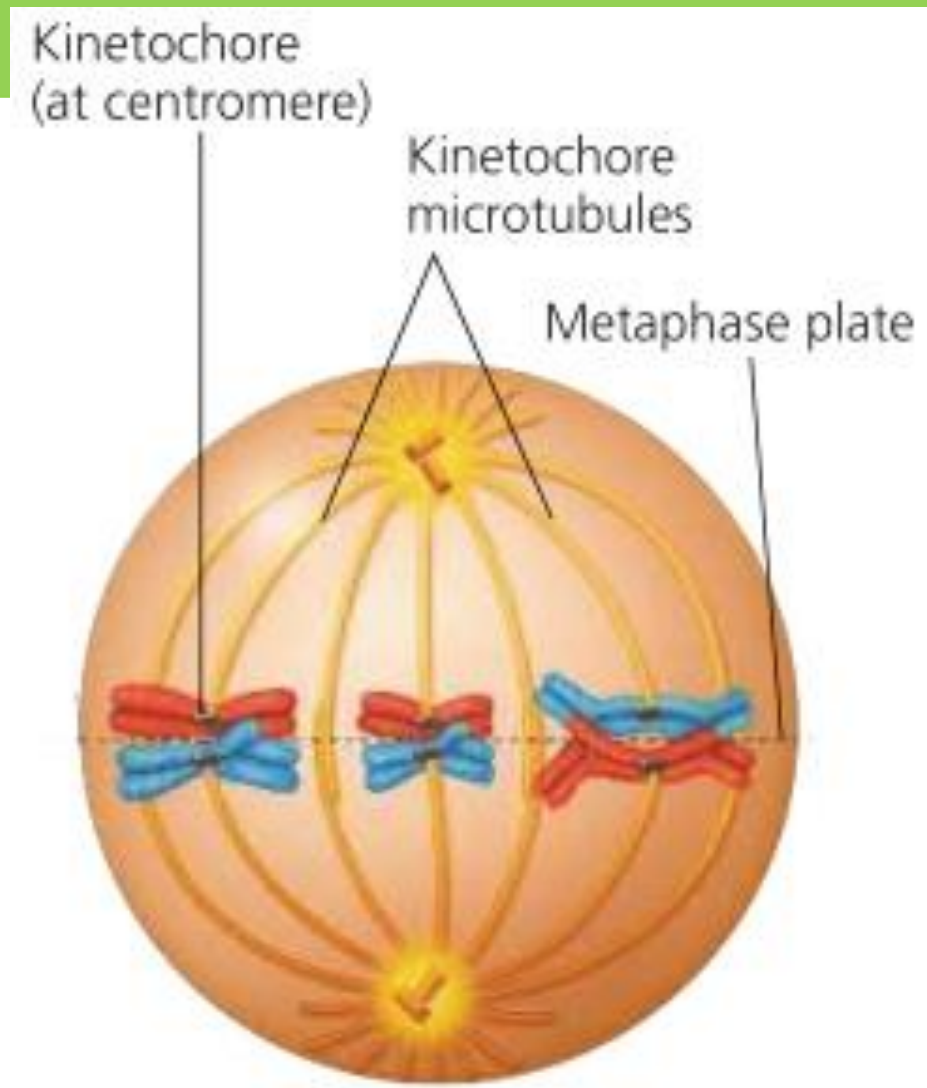
Later in prophase I, microtubules from one pole or the other attach to the kinetochores, one at the centromere of each homolog. (The two kinetochores on the sister chromatids of a homolog are linked together by proteins and act as a single kinetochore.) Microtubules move the homologous pairs toward the metaphase plate.



Duplicated homologous chromosomes (red and blue) pair up and exchange segments; $2n = 6$ in this example.

Metaphase I

- Pairs of homologous chromosomes are now arranged at the metaphase plate, with one chromosome of each pair facing each pole.
- Each pair has lined up independently of other pairs. (This arrangement is **called independent assortment**.)
- Both chromatids of one homolog are attached to kinetochore microtubules from one pole; the chromatids of the other homolog are attached to microtubules from the opposite pole.



Chromosomes line up by homologous pairs.

Anaphase I

- Breakdown of cohesins (proteins that are responsible for sister chromatid cohesion along chromatid arms) allows homologs to separate.
- The homologs move toward opposite poles, guided by the spindle apparatus.
- Sister chromatid cohesion persists at the centromere, causing the two chromatids of each chromosome to move as a unit toward the same pole.

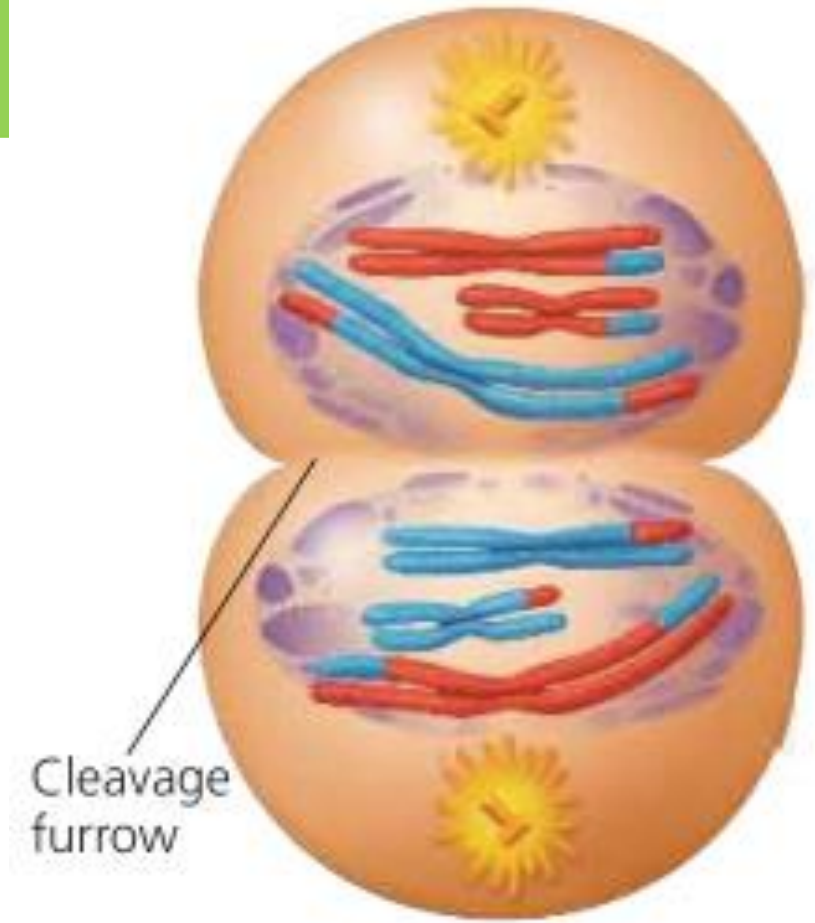


The two homologous chromosomes of each pair separate.

Telophase I and Cytokinesis

• When telophase I begins, each half of the cell has a complete haploid set of duplicated chromosomes. Each chromosome is composed of two sister chromatids; one or both chromatids include regions of nonsister chromatid DNA.

• Cytokinesis (division of the cytoplasm) usually occurs simultaneously with telophase I, forming two haploid daughter cells.
• In animal cells, a cleavage furrow forms. (In plant cells, a cell plate forms.)
• In some species, chromosomes decondense and nuclear envelopes form.
• No chromosome duplication occurs between meiosis I and meiosis II.



Two haploid cells form; each chromosome still consists of two sister chromatids.

Meiosis II separates sister chromatids:

- Division in meiosis II also occurs in four phases:
  - Prophase II
  - Metaphase II
  - Anaphase II
  - Telophase II and cytokinesis

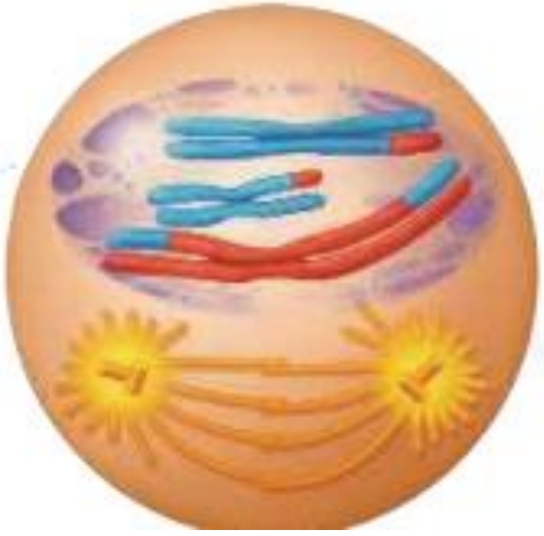
- Meiosis II is very similar to mitosis

Prophase II



- A spindle apparatus forms.

- In late prophase II (not shown here), chromosomes, each still composed of two chromatids associated at the centromere, are moved by microtubules toward the metaphase II plate.



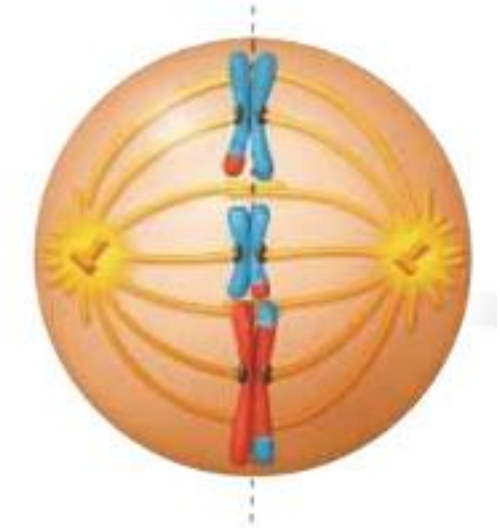
Metaphase II

- In metaphase II, the sister chromatids are arranged at the metaphase plate

- Because of crossing over in meiosis I, the two sister chromatids of each chromosome are no longer genetically identical.

- The kinetochores of sister chromatids are attached to microtubules extending from opposite poles.

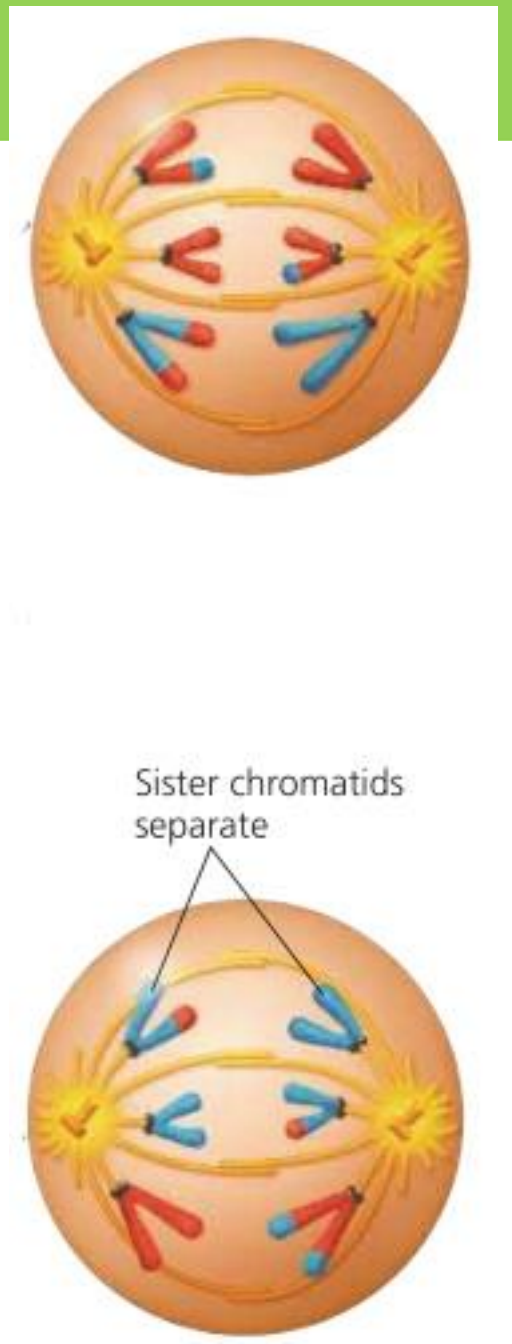




Anaphase II

- Breakdown of proteins holding the sister chromatids together at the centromere allows the chromatids to separate.

- The sister chromatids of each chromosome now move as two newly individual chromosomes toward opposite poles



Telophase II and Cytokinesis



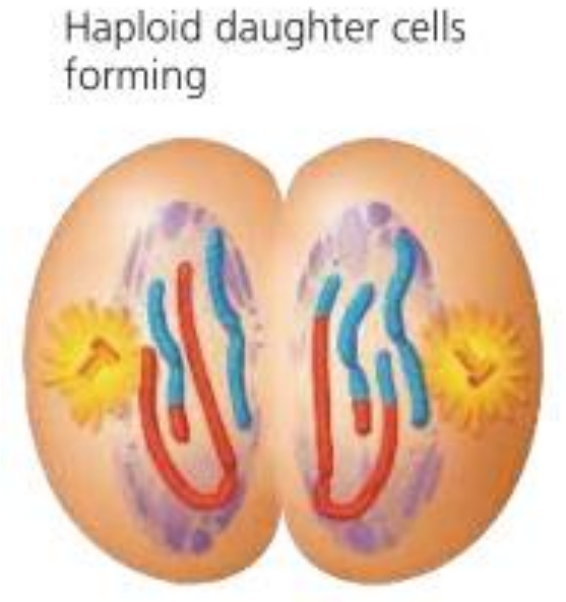
- In telophase II, the chromosomes arrive at opposite poles.

- Nuclei form, the chromosomes begin decondensing, and cytokinesis occurs.

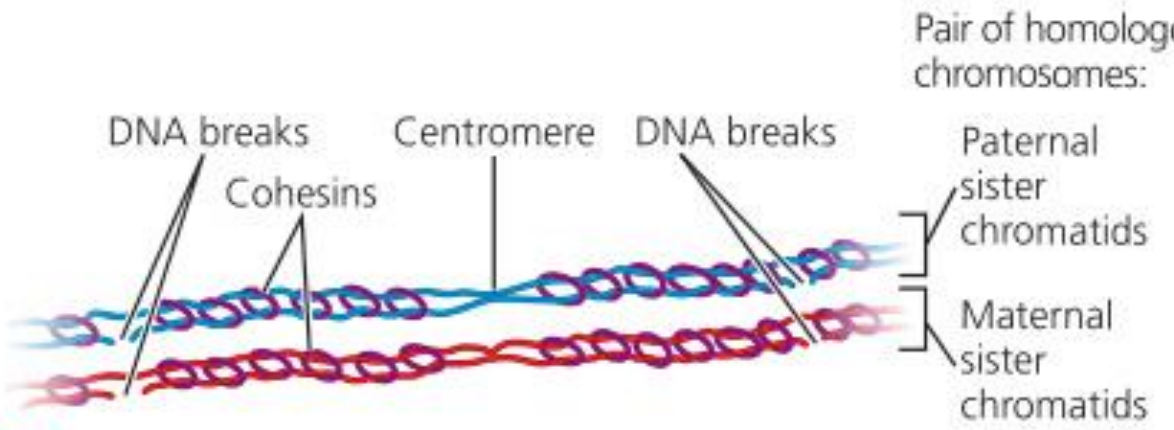
- Cytokinesis separates the cytoplasm

- The meiotic division of one parent cell produces four daughter cells, each with a haploid set of (unduplicated) chromosomes.

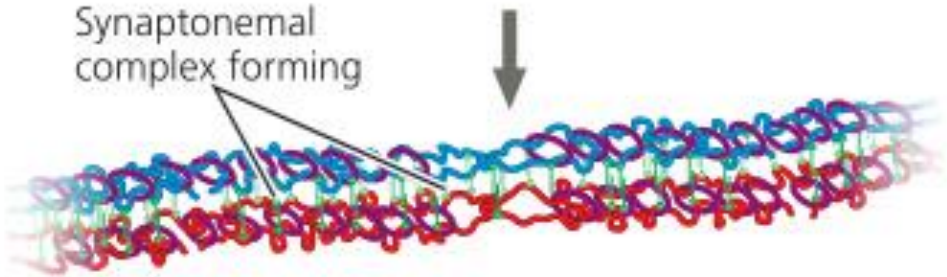
- The four daughter cells are genetically distinct from one another and from the parent cell.



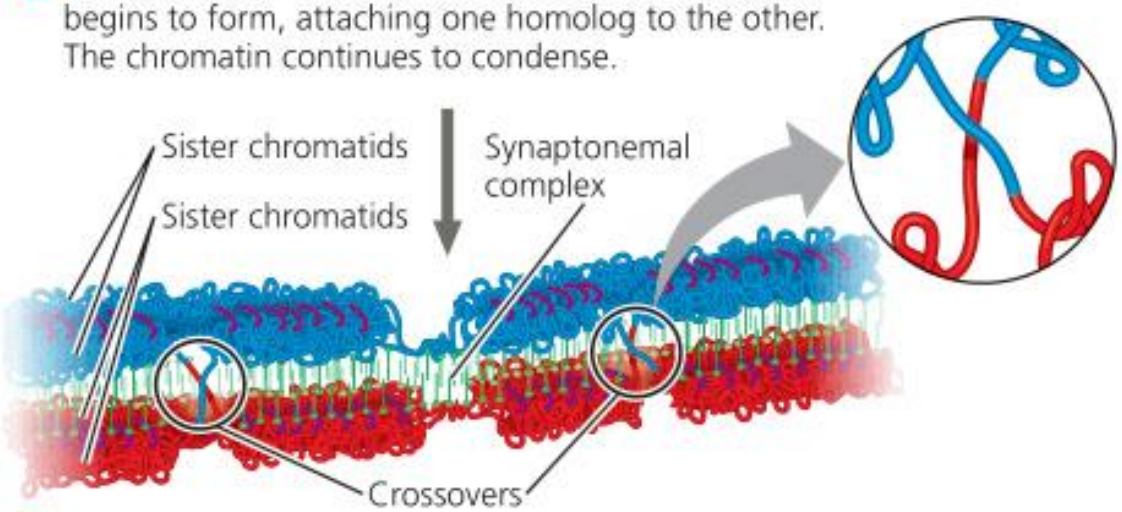
Crossing over and synapsis in prophase I: a closer look.



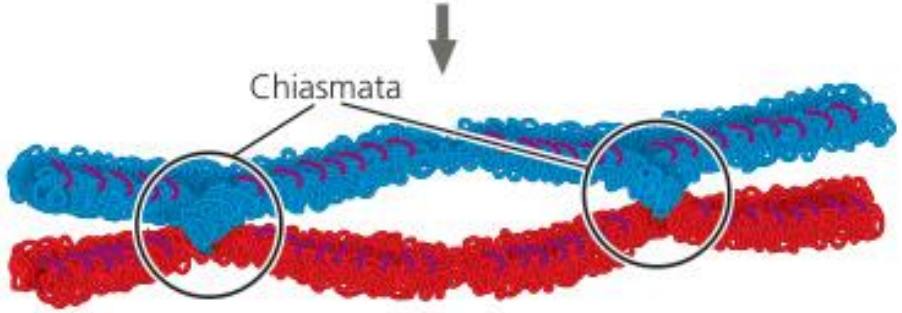
1 After interphase, the chromosomes have been duplicated, and sister chromatids are held together by proteins called cohesins (purple). Each pair of homologs associate along their length. The DNA molecules of two nonsister chromatids are broken at precisely corresponding points. The chromatin of the chromosomes starts to condense.



2 A zipper-like protein complex, the synaptonemal complex (green), begins to form, attaching one homolog to the other. The chromatin continues to condense.



3 The synaptonemal complex is fully formed; the two homologs are said to be in synapsis. During synapsis, the DNA breaks are closed up when each broken end is joined to the corresponding segment of the nonsister chromatid, producing crossovers.



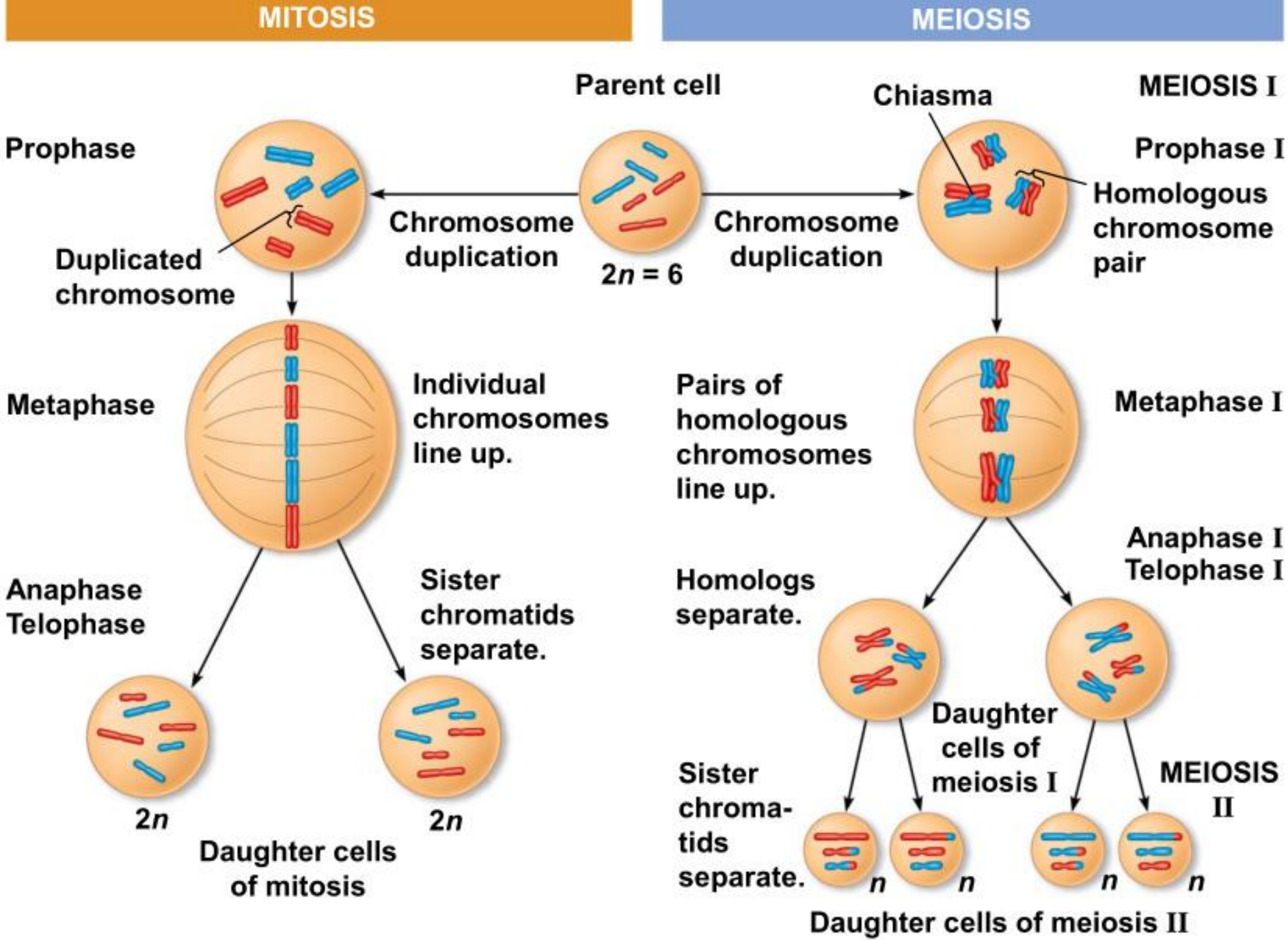
4 After the synaptonemal complex disassembles, the homologs move slightly apart from each other but remain attached because of sister chromatid cohesion, even though some of the DNA may no longer be attached to its original chromosome. The points of attachment where crossovers have occurred show up as chiasmata. The chromosomes continue to condense as they move toward the metaphase plate.

A Comparison of Mitosis and Meiosis

- Mitosis conserves the number of chromosome sets, producing cells that are genetically identical to the parent cell

- Meiosis reduces the number of chromosome sets from two (diploid) to one (haploid), producing cells that differ genetically from each other and from the parent cell

- The mechanism for separating sister chromatids is virtually identical in meiosis II and mitosis


MITOSIS
MEIOSIS
Parent cell
Prophase
Duplicated chromosome
Chromosome duplication
2n = 6
Chromosome duplication
Chiasma
MEIOSIS I
Prophase I
Homologous chromosome pair
Metaphase
Individual chromosomes line up.
Pairs of homologous chromosomes line up.
Metaphase I
Anaphase
Telophase
Sister chromatids separate.
Homologs separate.
Anaphase I
Telophase I
Daughter cells of meiosis I
Sister chroma-tids separate.
MEIOSIS II
2n
2n
Daughter cells of mitosis
n
n
n
n
Daughter cells of meiosis II

Property	Mitosis (occurs in both diploid and haploid cells)	Meiosis (can only occur in diploid cells)
DNA replication	Occurs during interphase before mitosis begins	Occurs during interphase before meiosis I begins
Number of divisions	One, including prophase, prometaphase, metaphase, anaphase, and telophase	Two, each including prophase, metaphase, anaphase, and telophase
Synapsis of homologous chromosomes	Does not occur	Occurs during prophase I along with crossing over between nonsister chromatids; resulting chiasmata hold pairs together due to sister chromatid cohesion
Number of daughter cells and genetic composition	Two, each genetically identical to the parent cell, with the same number of chromosomes	Four, each haploid (*n*); genetically different from the parent cell and from each other
Role in the animal or plant body	Enables multicellular animal or plant (gametophyte or sporophyte) to arise from a single cell; produces cells for growth, repair, and, in some species, asexual reproduction; produces gametes in the gametophyte plant	Produces gametes (in animals) or spores (in the sporophyte plant); reduces number of chromosomes sets by half and introduces genetic variability among the gametes or spores

Three events are unique to meiosis, and all three occur in meiosis I:

1. Synapsis and crossing over in prophase I:
Homologous chromosomes physically connect and exchange genetic information

2. Homologous chromosomes pair at the metaphase plate (tetrads), instead of individual replicated chromosomes

3. Separation of homologs chromosomes at anaphase I, instead of sister chromatids, that separate.

Cohesions

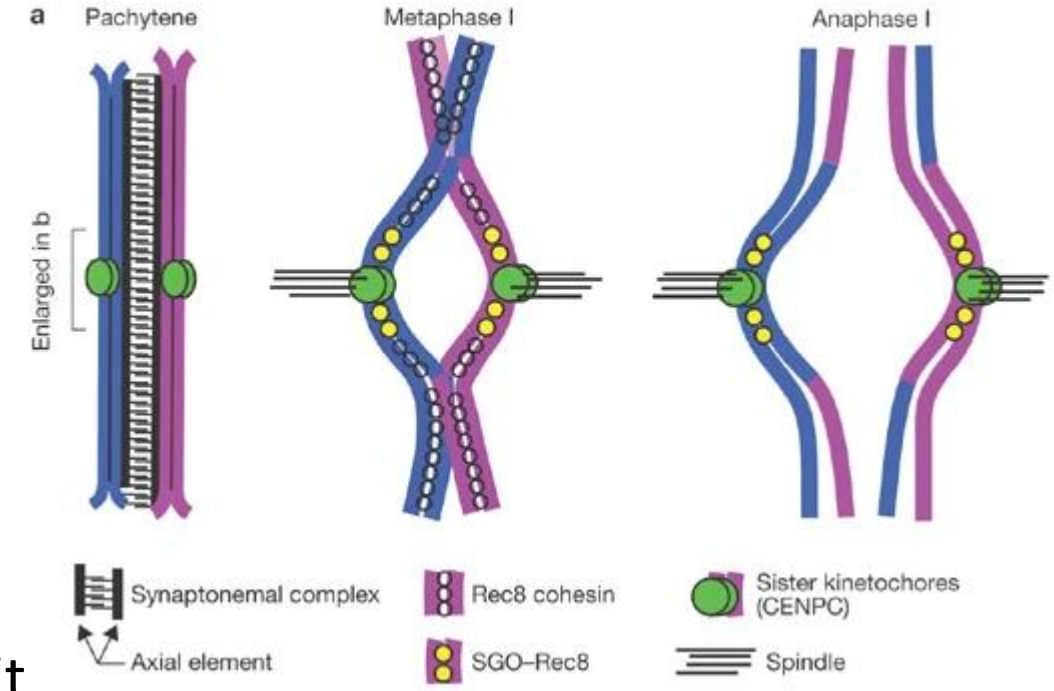
Sister chromatid cohesion allows sister chromatids of a single chromosome to stay together through meiosis I

- Protein complexes called cohesins are responsible for this cohesion

- In mitosis, cohesins are cleaved at the end of metaphase

- In meiosis, cohesins are cleaved along the chromosome arms in anaphase I (separation of homologs) and at the centromeres in anaphase II (separation of sister chromatids)

- Meiosis I is called the reductional division because it reduces the number of chromosomes per cell

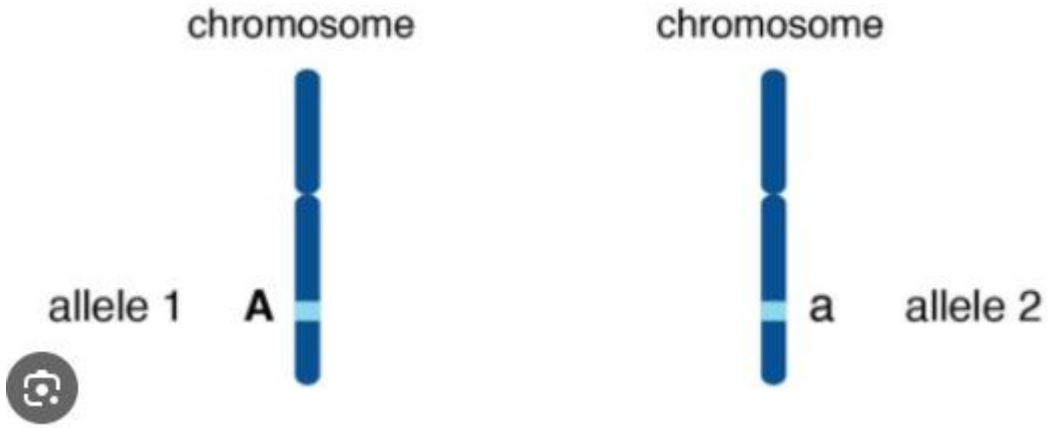


Genetic variation produced in sexual life cycles contributes to evolution

- **Mutations** (changes in an organism's DNA) are the original source of genetic diversity

- Mutations create different versions of genes called **alleles**

- <u>Reshuffling of the alleles during sexual reproduction produces the variation</u> that results in each member of a sexually reproducing population having a unique combination of traits.



Origins of Genetic Variation Among Offspring

• The behavior of chromosomes during meiosis and fertilization is responsible for most of the variation that arises in each generation

• Three mechanisms contribute to genetic variation among offsprings:

– Independent assortment of chromosomes
– Crossing over
– Random fertilization

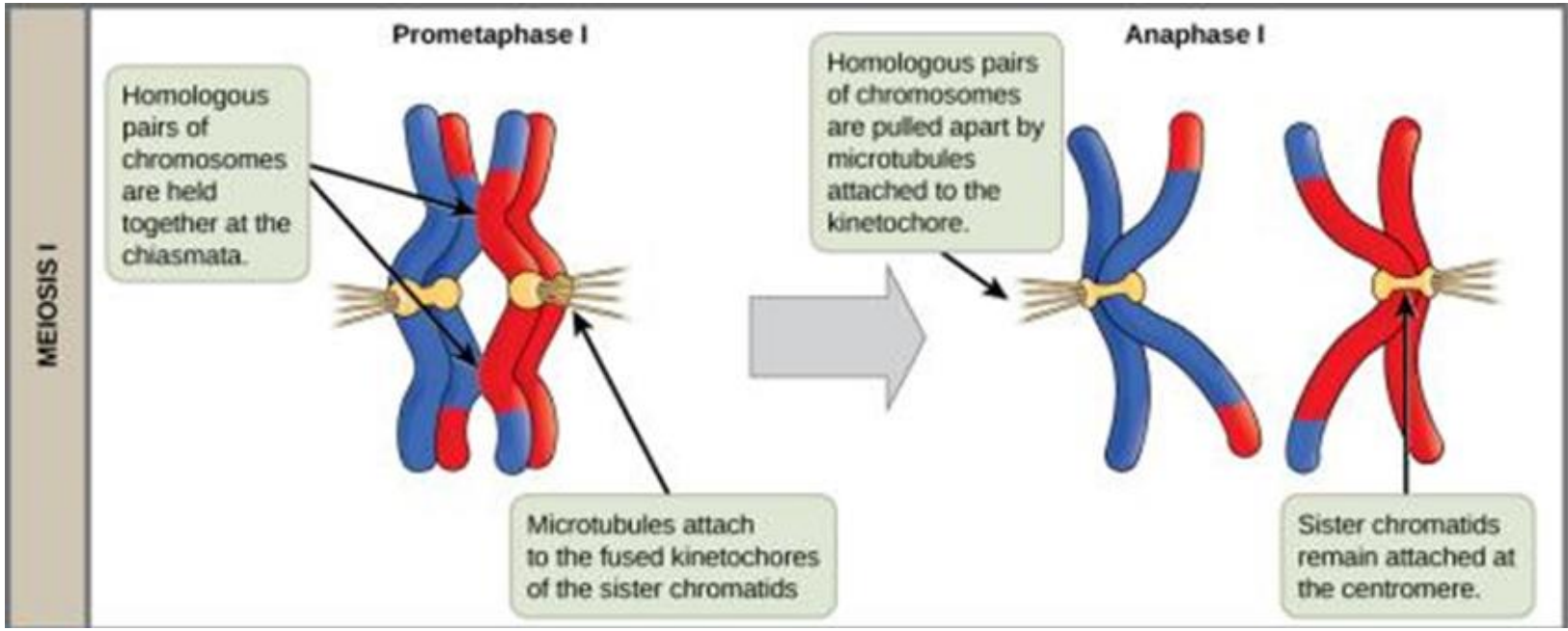
Genetic Variation Due To Independent Assortment of Chromosomes

- Homologous pairs of chromosomes orient randomly at metaphase I of meiosis

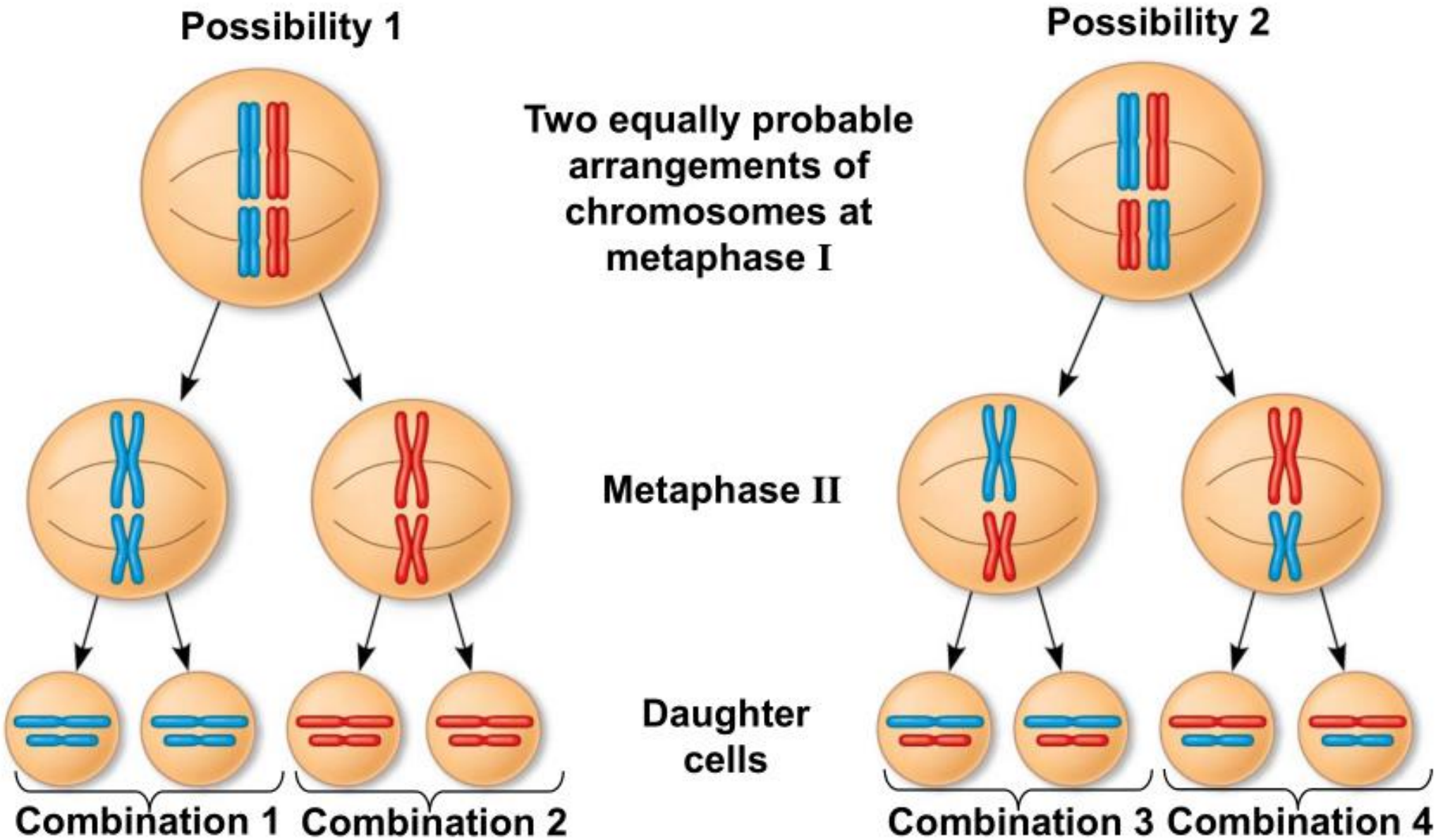
- In independent assortment, each pair of chromosomes sorts maternal and paternal homologues into daughter cells independently of the other pairs

- The number of combinations possible when chromosomes assort independently into gametes is 2^n, where n is the haploid number

- For humans (n = 23), there are more than 8 million (2^{23}) possible combinations of chromosomes!



Independent Assortment



Warning!
This figure assumes that no crossing over occurs between chromosome pairs for simplicity.
In reality this is not the case.

Genetic Variation Due To Crossing Over

Crossing over produces recombinant chromosomes, individual chromosomes that carry genes (DNA) from two different parents

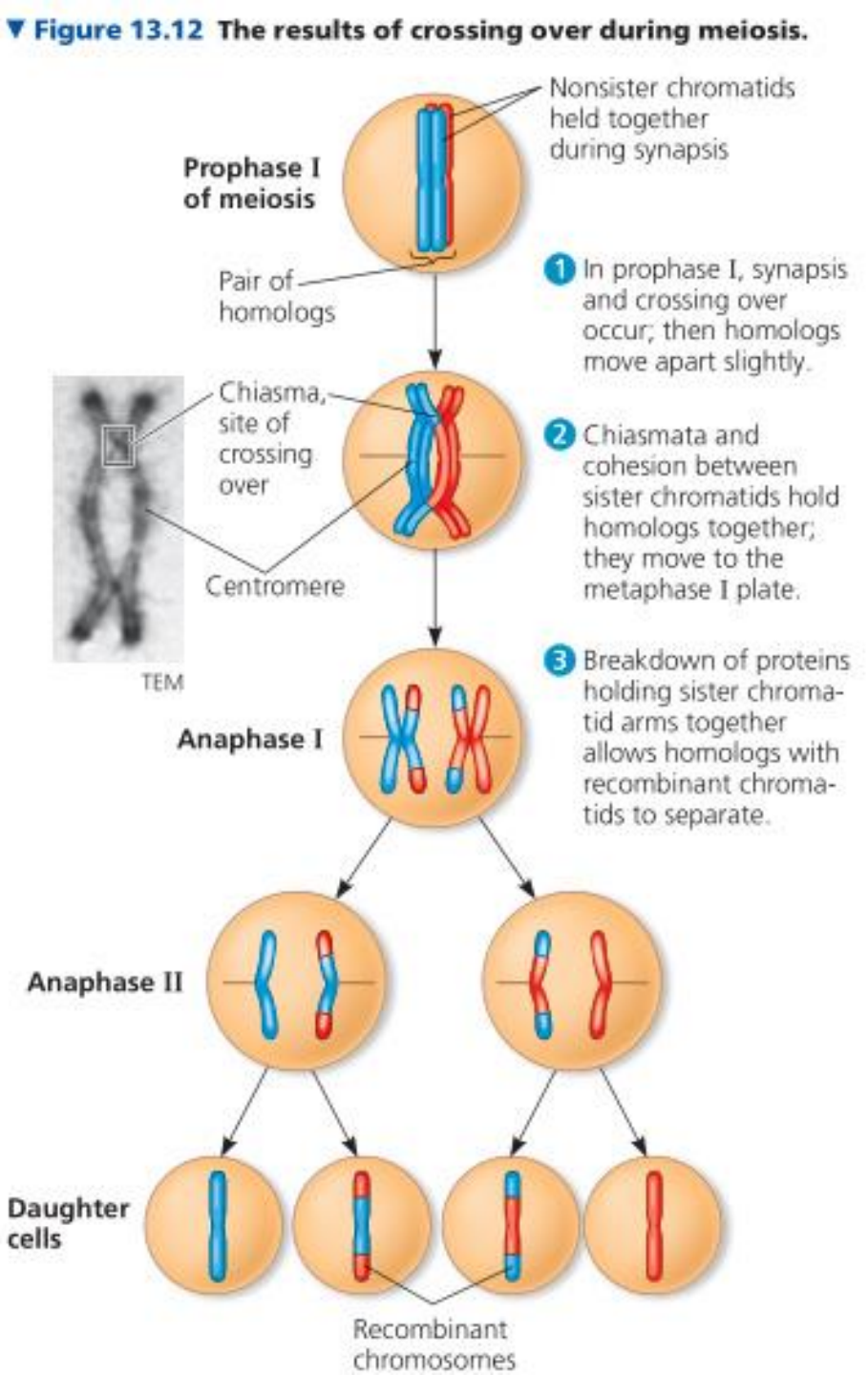
- An average of 1-3 crossing over events per chromosome pair occur during human meiosis depending on the size of the chromosomes and the position of their centromeres.

- Crossing over begins very early in prophase I, as homologous chromosomes pair up gene
by gene

- In crossing over, homologous portions of two nonsister chromatids trade places

- Crossing over contributes to genetic variation by combining DNA from two parents into a
single chromosome

Crossing over produces chromosomes with new combinations of maternal and paternal alleles.



Genetic Variation Due To Random Fertilization

- Random fertilization adds to genetic variation because any sperm can fuse with any ovum (unfertilized egg)

- In humans, each male and female gamete represents one of about 8.4 million (2^{23}) possible chromosome combinations due to independent assortment. The fusion of a male gamete with a female gamete during fertilization will produce a zygote with any of about 70 trillion ($2^{23} * 2^{23)}$ diploid combinations.

- Crossing over adds even more variation

- This variation gives each zygote a unique genetic identity.

The Evolutionary Significance of Genetic Variation Within Populations

Darwin recognized that a population evolves through the differential reproductive success of its variant members. On average, the individuals best suited to the local environment leave the most offspring, thereby transmitting their genes. Thus, natural selection results in the accumulation of genetic variations favored in a certain environment. As the environment changes, the population may survive if, in each generation, at least some of its members can cope effectively with the new conditions.

Mutations are the original source of different alleles, which are then mixed and matched during meiosis. New and different combinations of alleles may work better than those that previously prevailed.

In a stable environment, sexual reproduction seems as if it would be less advantageous than asexual reproduction, which ensures continuation of successful combinations of alleles. Furthermore, sexual reproduction is more expensive energetically than asexual reproduction. In spite of these apparent disadvantages, sexual reproduction is almost universal among animals. Why is this?

The ability of sexual reproduction to generate genetic diversity is the most commonly proposed explanation for the evolutionary persistence of this process.

The Evolutionary Significance of Genetic Variation Within Populations

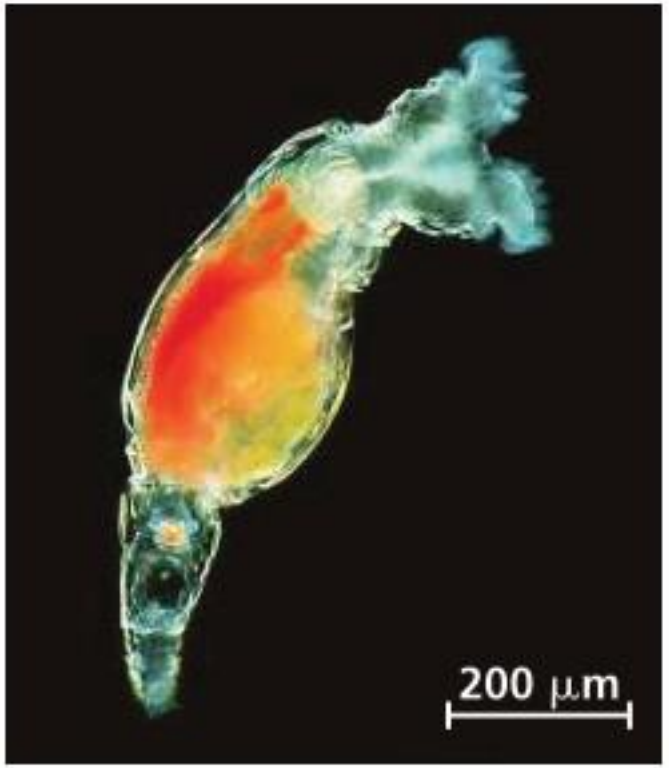
• Natural selection results in the accumulation of genetic variations favored by the environment

• Sexual reproduction contributes to the genetic variation in a population, which originates from mutations

• A bdelloid rotifer is an animal reproduces only asexually. Bdelloids have mechanisms other than sexual reproduction for generating genetic diversity. They take up foreign DNA from their environments and that DNA can become incorporated into their genome leading to increased genetic diversity.

- Genetic diversity is advantageous, but sexual reproduction is not the only way of generating such diversity.

▼ Figure 13.13 A bdelloid rotifer, an animal that reproduces only asexually.



Mendel and the Gene Idea

Character: A heritable feature that varies among individuals (e.g.flower color).

Trait: Each variant for a character, such as purple or white color for flowers.

True-breeding: the plants that produce only the same variety as the parent plant over many generations of self-pollination.

Hybridization: mating, or crossing, of two true-breeding varieties.

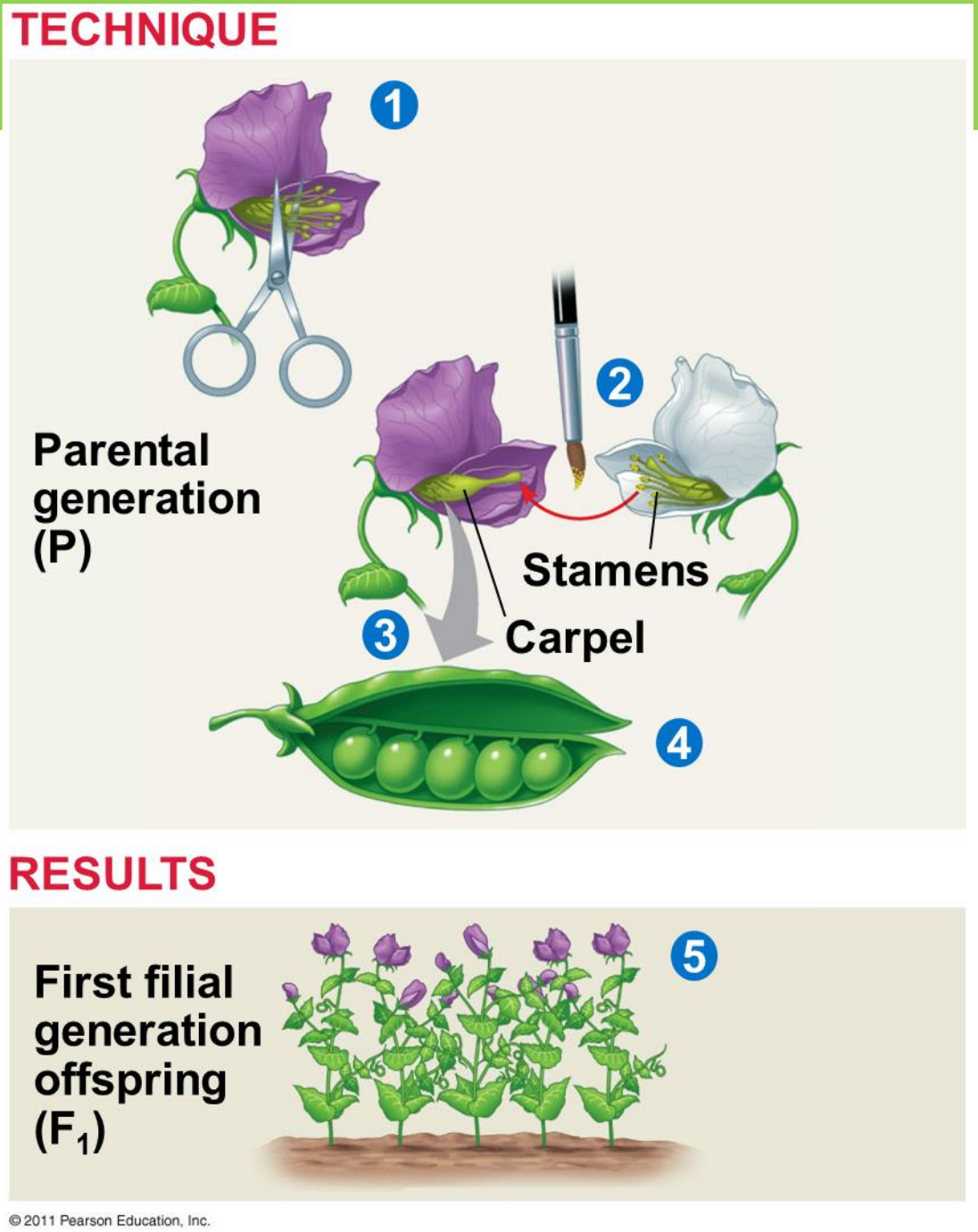
P generation (parental generation): The true-breeding parents.

F1 generation (first filial generation): First offsprings of the P gereration

F2 generation (second filial generation): produced by allowing F1 hybrids to self-pollinate.

Mendel's Experiment

1.Removed stamen's (male fertilizing organ of a flower)from true- breeding purple flower.
2. Transferred pollen (sperm) from stamens of true breeding white flower to carpel of purple flowers.
3. The pollinated carpel matured into a pod.
4. Planted the pod seeds.
Results:
The F1 offspring were all purple.
When the purple flower's pollen was transferred to white flower's carpel all of the F1 were also purple.

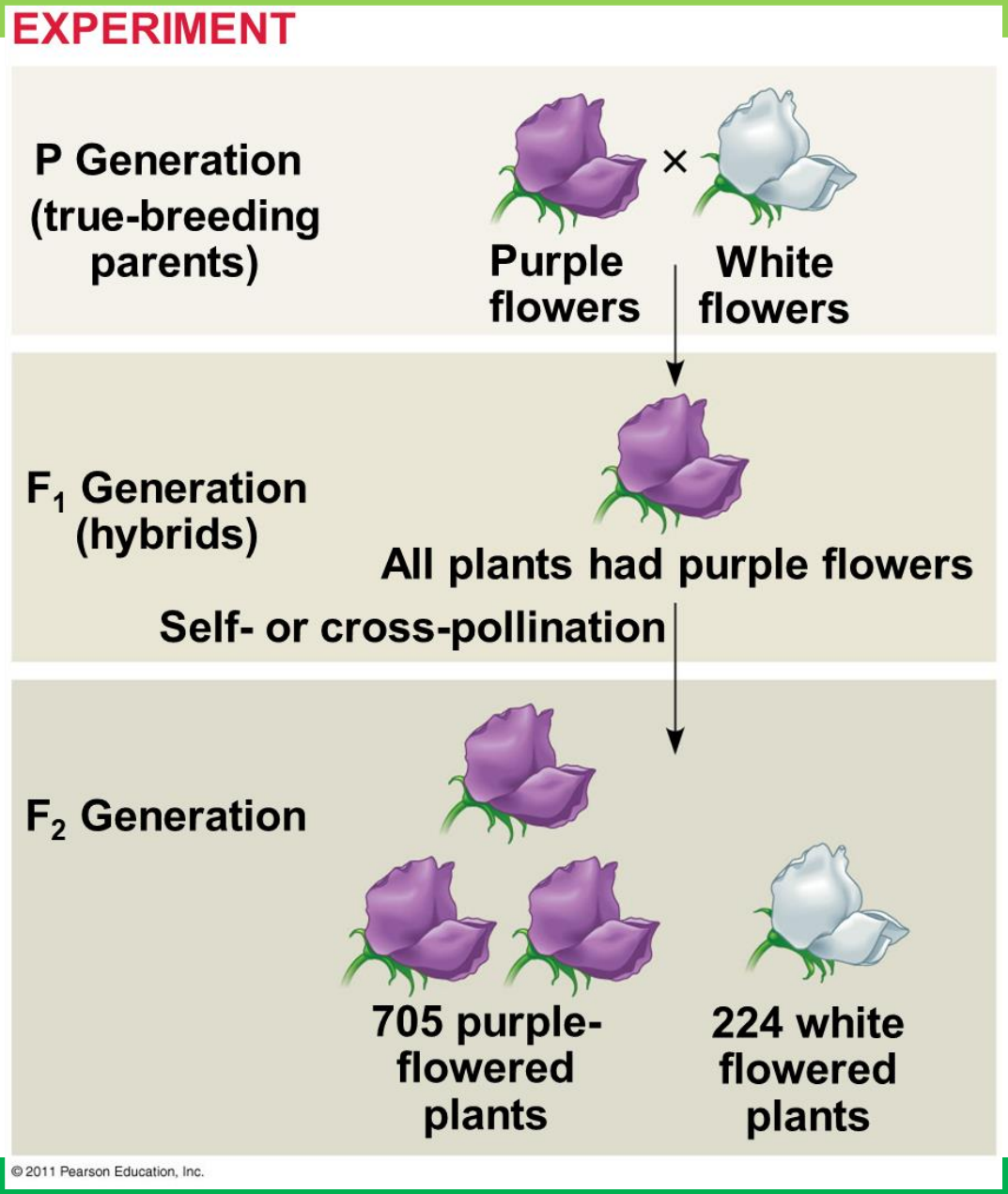


F1 hybrid plants are allowed to self-pollinate what traits do have in F2 generation

1. Repeated experiment as before. All F1 are purple.
2. Allowed F1 to cross-pollinate itself.

Results:

F2 plants both purple and white flowered were produced in a ratio of 3:1.



Dominant and Recessive Traits

- Mendel reasoned that only the purple flower factor was affecting flower color in the F1 hybrids
- Mendel called the purple flower color a **dominant trait** and the white flower color a **recessive trait**
- Mendel observed the same pattern of inheritance in six other characters in pea plant, each represented by two trait
- What Mendel called a “heritable factor” is what we now call a gene

Table 14.1 The Results of Mendel’s F_1 Crosses for Seven Characters in Pea Plants

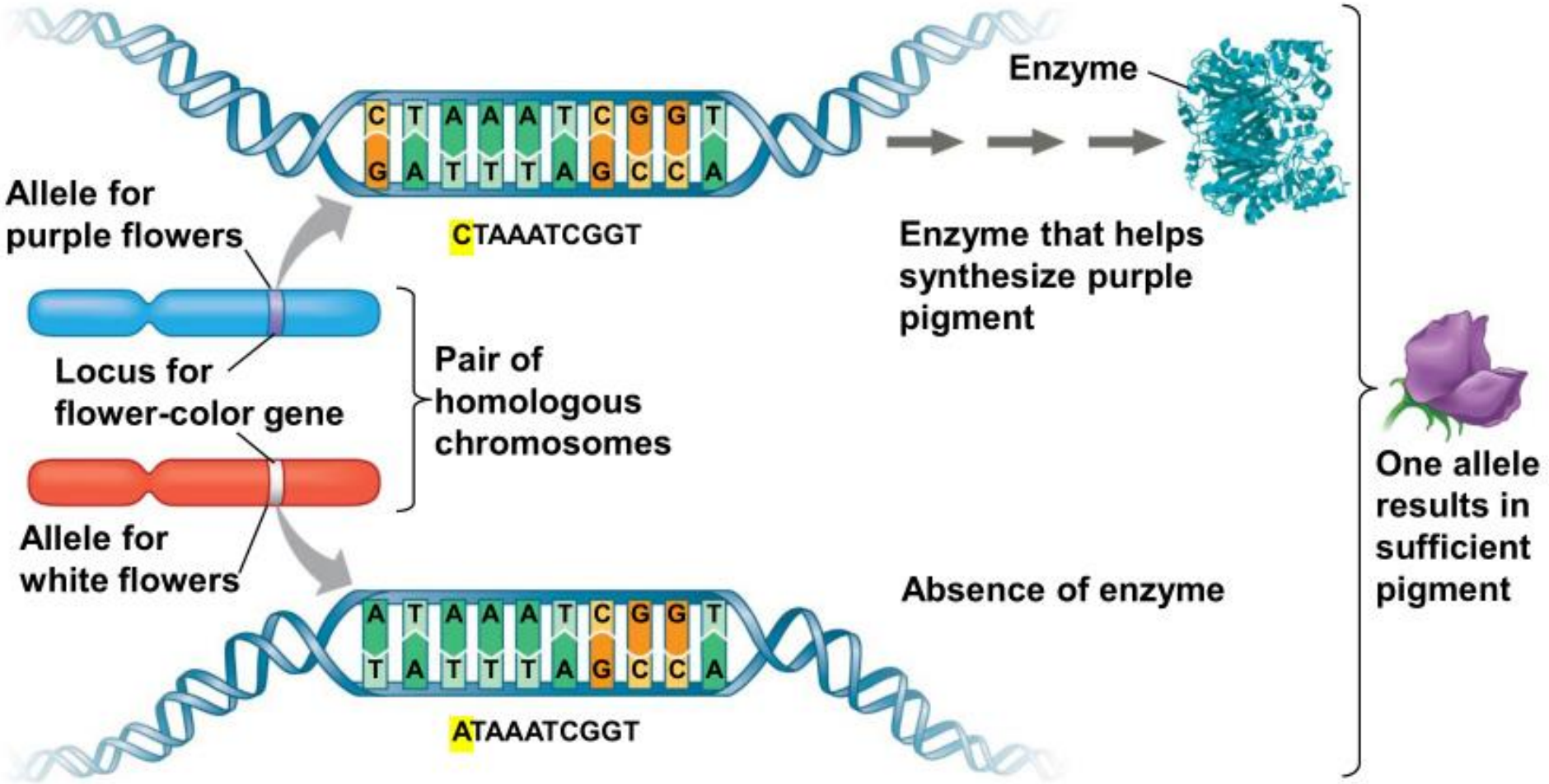
Character	Dominant Trait	×	Recessive Trait	F_2 Generation Dominant: Recessive	Ratio
Flower color	Purple 	×	White 	705:224	3.15:1
Flower position	Axial 	×	Terminal 	651:207	3.14:1
Seed color	Yellow 	×	Green 	6,022:2,001	3.01:1
Seed shape	Round 	×	Wrinkled 	5,474:1,850	2.96:1
Pod shape	Inflated 	×	Constricted 	882:299	2.95:1
Pod color	Green 	×	Yellow 	428:152	2.82:1
Stem length	Tall 	×	Dwarf 	787:277	2.84:1

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Mendel's Model

- The first concept is **that alternative versions of heritable factors (genes) account for variations in inherited characters**
- For example, the gene for flower color in pea plants exists in two versions, one for purple flowers and the other for white flowers
- These alternative versions of a gene are now called **alleles**
- Each gene resides at a specific locus on a specific chromosome
- The DNA at that locus can vary in its sequence of nucleotides.
- The purple-flower and white-flower alleles are two DNA variations at the flower-color locus.

Alleles, alternative versions of a gene



The paternally inherited chromosome (blue) has an allele for purple flowers, which codes for a protein that indirectly controls synthesis of purple pigment.

The maternally inherited chromosome (red) has an allele for white flowers, which results in no functional protein being made.

Mendel's Model

- The second concept is that **for each character an organism inherits two alleles (alternative forms of the same gene), one from each parent**

- Mendel made this deduction without knowing about the role of chromosomes

- A diploid organism inherits one set of chromosomes from each parent.

- Each diploid organism has a pair of homologous chromosomes and, therefore, two copies of each gene.

- 2 alleles may be identical (true-breeding & P1)

- 2 alleles may differ (F1 hybrids)

Mendel's Model

• The third concept is that if the two alleles at a locus differ, then one (the **dominant allele**) determines the organism's appearance, and the other (the **recessive allele**) has no noticeable effect on appearance

• In the flower-color example, the F1 plants inherited a purple-flower allele from one parent and a white-flower allele from the other.

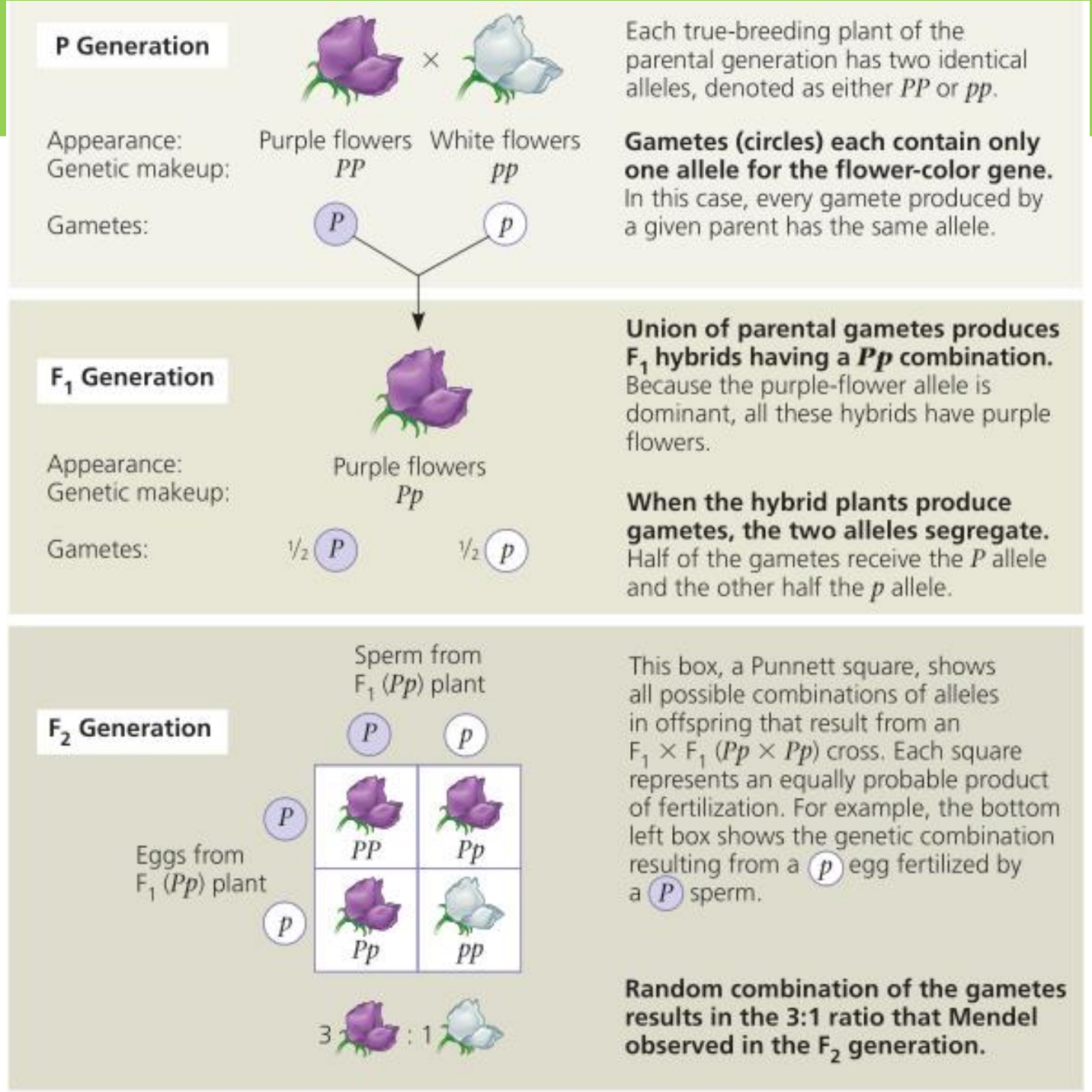
• In the flower-color example, the F1 plants had purple flowers because the allele for that trait is dominant

Mendel's Model

• The fourth concept, now known as **the law of segregation, states that the two alleles for a heritable character separate (segregate) during gamete formation and end up in different gametes**

• Thus, an egg or a sperm gets only one of the two alleles that are present in the somatic cells of an organism

• This segregation of alleles corresponds to the distribution of homologous chromosomes to different gametes in meiosis.



- The possible combinations of sperm and egg can be shown using a **Punnett square**, a diagram for predicting the results of a genetic cross between individuals of known genetic makeup

- A capital letter represents a dominant allele, and a lowercase letter represents a recessive allele

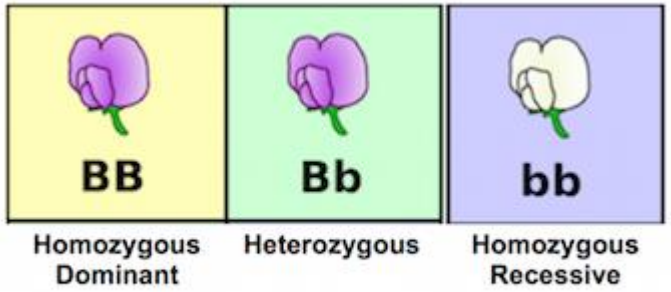
Mendel's law of segregation accounts for the 3:1 ratio that he observed in the F2 generation

Useful Genetic Vocabulary

• An organism with two identical alleles for a character is said to be **homozygous** for the gene controlling that character
- For flower color in peas- white flowers are homozygous recessive (pp) and purple flower are homozygous dominant (PP) for the flower-color gene.

• An organism that has two different alleles for a gene is said to be **heterozygous** for the gene controlling that character
- For flower color in peas- purple flower are heterozygous (Pp)

• Unlike homozygotes, heterozygotes are not by true-breeding



Useful Genetic Vocabulary

• Because of the different effects of dominant and recessive alleles, an organism's traits do not always reveal its genetic composition

• Therefore, we distinguish between an organism's **phenotype**=physical appearance, and its **genotype**=genetic makeup

Genotype	Phenotype
BB Homozygous dominant	
Bb Heterozygous	
bb Homozygous recessive	

• Two organisms can have the same phenotype but different genotypes if one is homozygous dominant and the other is heterozygous.

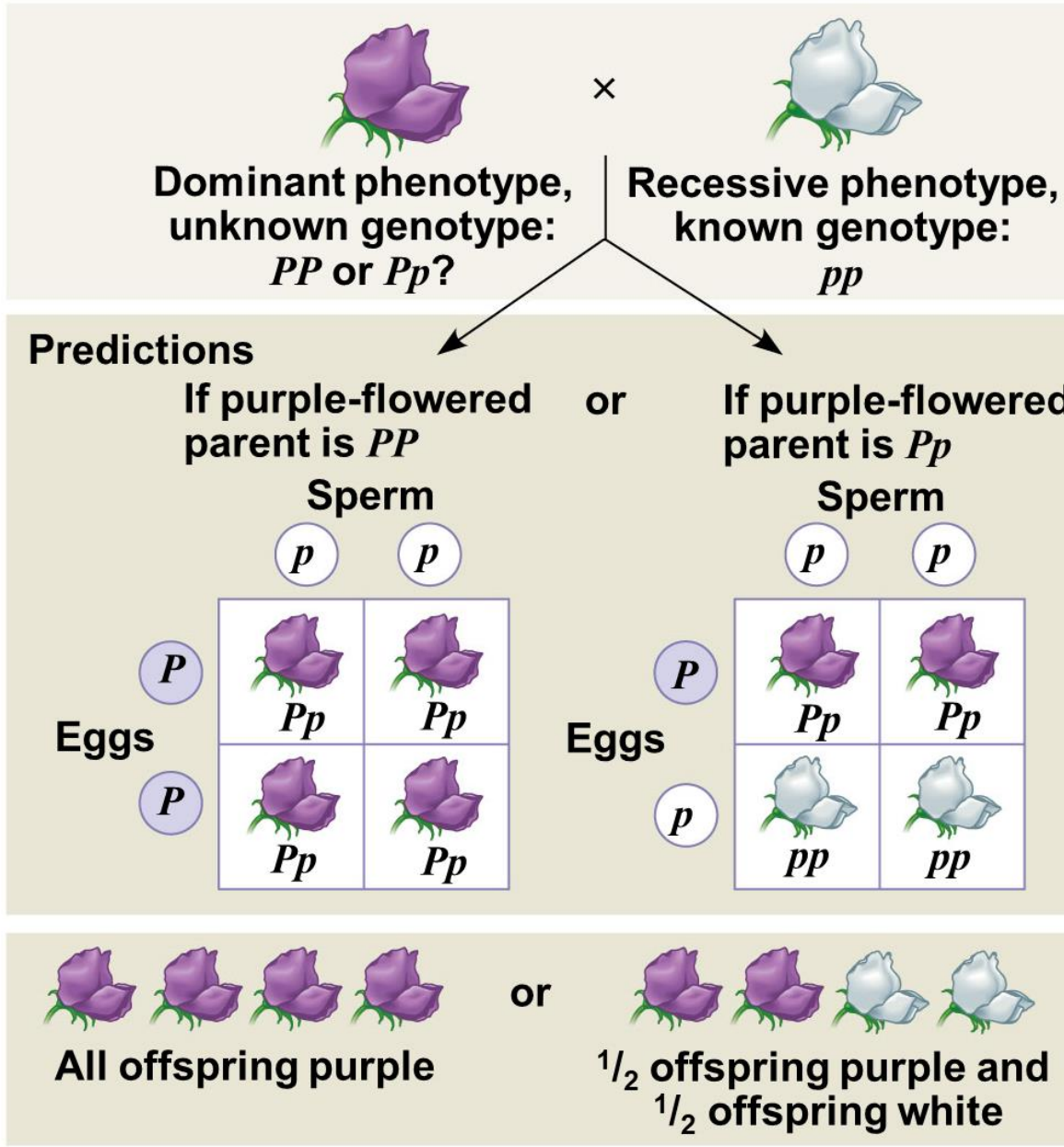
• In the example of flower color in pea plants, PP anc Pp plants have the same phenotype (purple) but different genotypes (homozygous dominant and heterozygous).

The Testcross is used to determine genotype

TECHNIQUE

- How can we tell the genotype of an individual with the dominant phenotype?
- Such an individual must have one dominant allele, but the individual could be either homozygous dominant or heterozygous
- The answer is to carry out a testcross:

breeding the mystery individual with a homozygous recessive individual



- If an organism has two identical alleles for a particular character, then that allele is present as a single copy in all gametes. All offsprings are puple.

RESULTS

- If different alleles are present, then 50% of the gametes will receive one allele and 50% will receive the other (half puple and half white flower.)

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The Law of Independent Assortment

- Mendel's first testcross experiments followed only a single character, such as flower color.
- All the F1 progeny produced in these crosses were **monohybrids**, heterozygous for one character.

- A cross between two heterozygotes is a **monohybrid cross**.

- Mendel identified the second law of inheritance by following two characters at the same time.
- Crossing two true-breeding parents differing in two characters produces **dihybrids** in the F1 generation, heterozygous for both characters

- In one such **dihybrid cross**, Mendel studied the inheritance of seed color and seed shape.

- A dihybrid cross, a cross between F1 dihybrids, can determine whether two characters are transmitted to offspring as a package or independently

• Mendel crossed two true-breeding pea varieties that differ in both color and seed shape: **EXPERIMENT**

The F1 will be dihybrids, heterozygous for 2 characters (YyRr)

Are these 2 characters transmitted from parents to offspring as a package (Y and R alleles always stay together and y and r always stay together) or are seed color and seed shape inherited independently?

• YyRr F1's will have the yellow round phenyotype no matter which hypothesis is correct.

• What happens when the F1's self pollinate and produce F2 offspring?

• Dependent Assortment hypothesis-

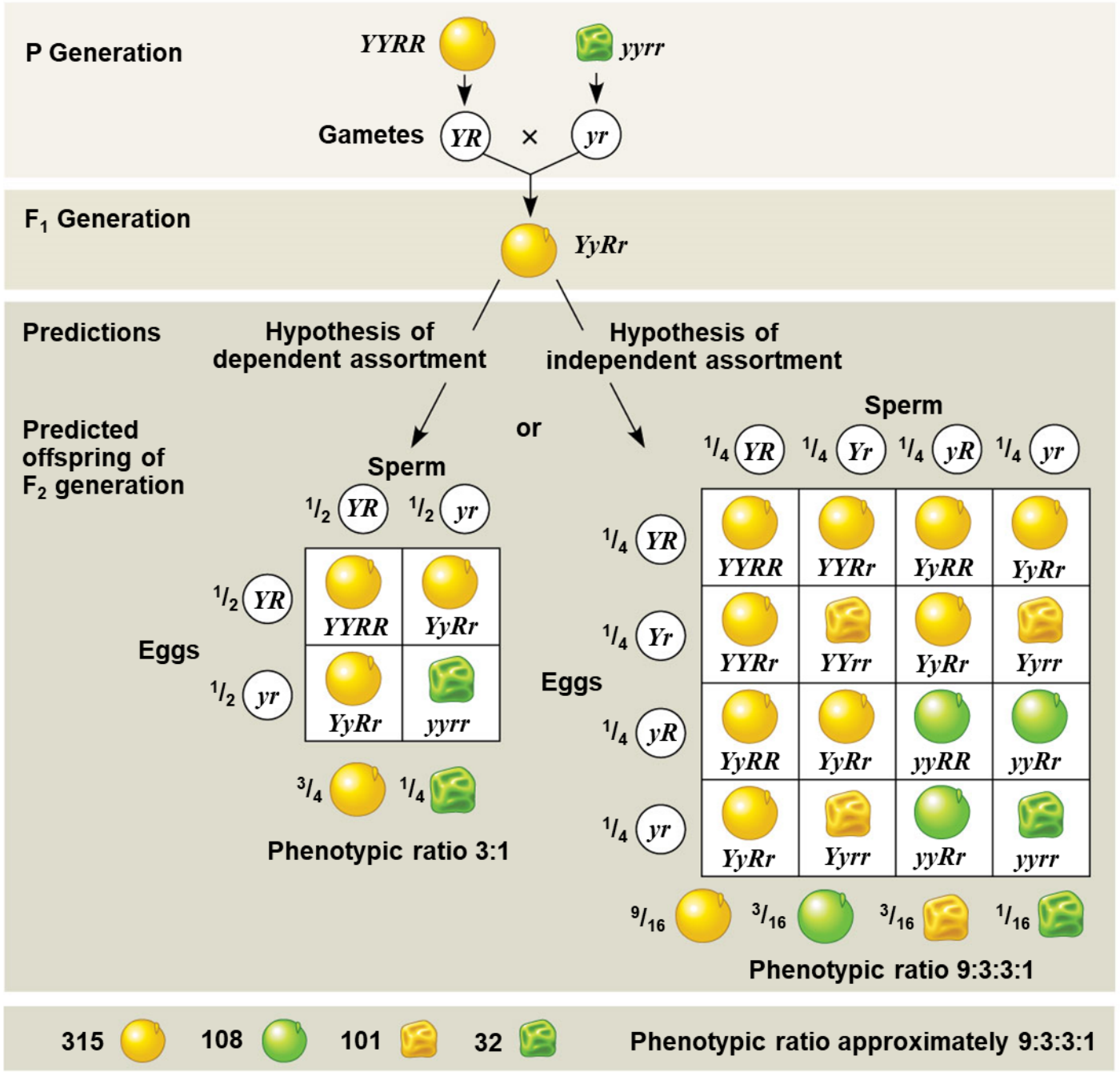
2 classes of gametes produced: YR and yr so F2 phenotype will be in 3:1 (3 yellow round:1 green wrinkled)

Independent Assortment hypothesis-

4 classes of gametes produced: YR, Yr, yR and yr

16 equally probable ways Y and R can combine so F1 phenotype will be in 9:3:3:1

9 yellow round:3 green round:3 yellow wrinkled:1 green wrinkled **RESULTS**



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The law of independent assortment

- Mendel's law of independent assortment states that the alleles of two (or more) different genes get sorted into gametes independently of one another. In other words, the allele a gamete receives for one gene does not influence the allele received for another gene.

- Strictly speaking, this law applies only to genes on different, nonhomologous chromosomes

- Genes located near each other on the same chromosome tend to be inherited together.

Inheritance patterns are often more complex than predicted by simple Mendelian genetics

• The relationship between genotype and phenotype is rarely as simple as in the pea plant characters Mendel studied

• Many heritable characters are not determined by only one gene with two alleles

• However, the basic principles of segregation and independent assortment apply even to more complex patterns of inheritance

Inheritance of characters by a single gene may deviate from simple Mendelian patterns in the following situations:

– When alleles are not completely dominant or recessive

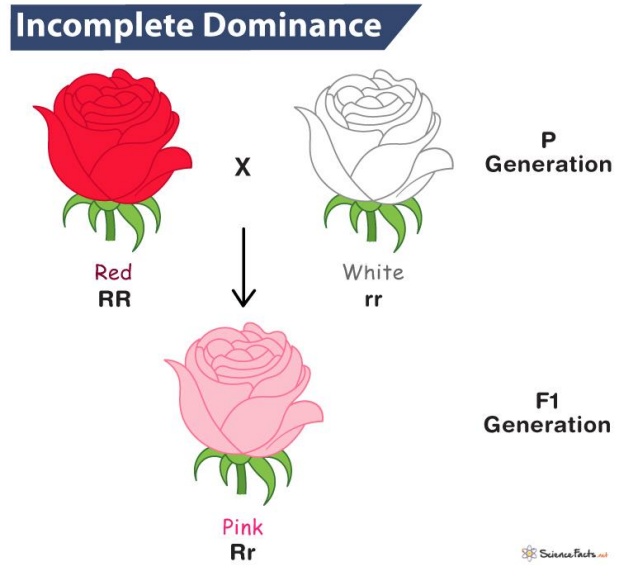
– When a gene has more than two alleles

– When a gene produces multiple phenotypes

Degrees of Dominance

• **Complete dominance** occurs when phenotypes of the heterozygote and dominant homozygote are identical
e.g. Mendel's pea experiments.

• In **incomplete dominance** the phenotype of F1 hybrids is somewhere between the phenotypes of the two parental varieties.



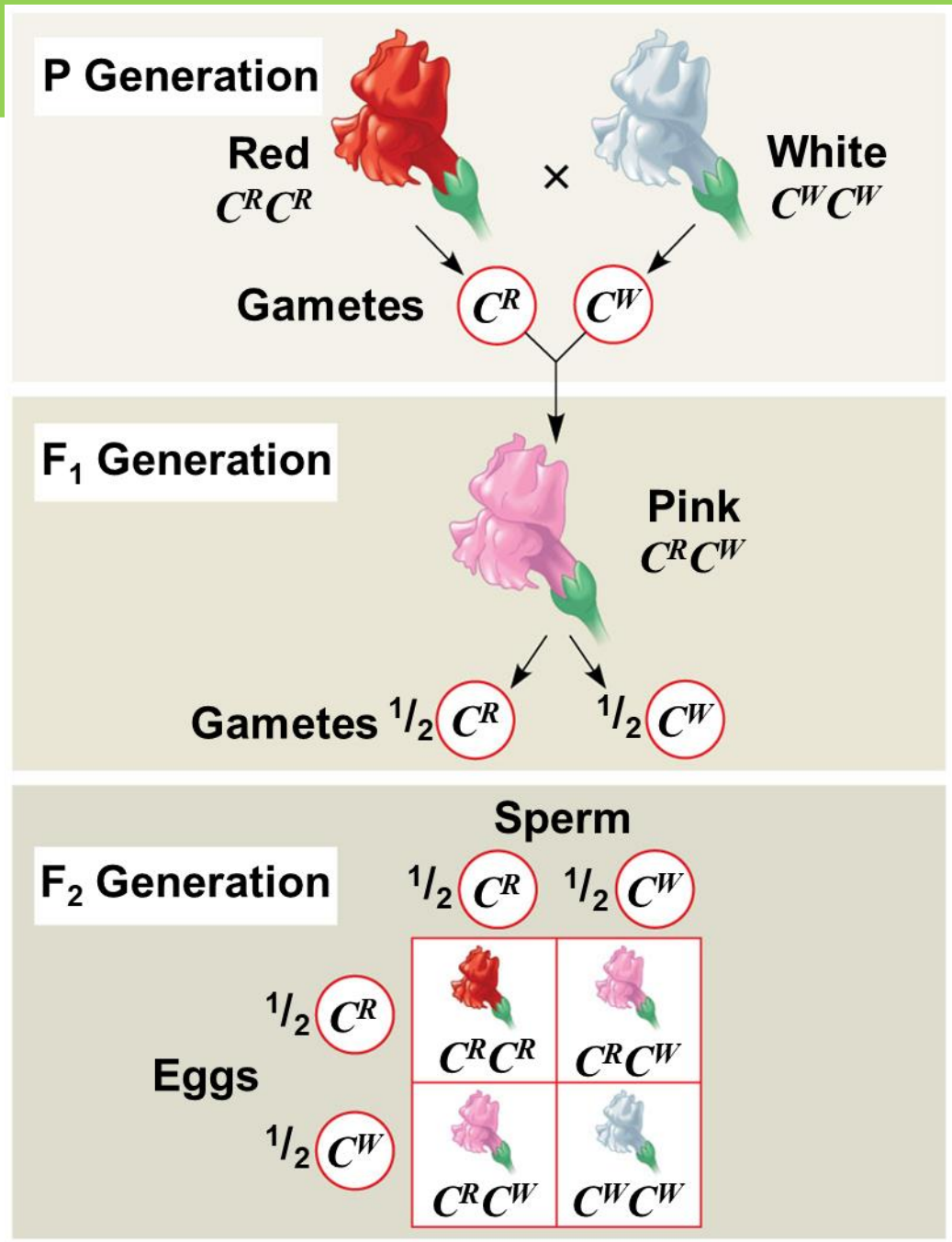
• In **codominance** two dominant alleles affect the phenotype in separate, distinguishable ways e.g. ABO type human blood cell.

Blood type	Genotype	
A	I^A, I^O	AO
	I^A, I^A	AA
B	I^B, I^O	BO
	I^B, I^B	BB
AB	I^A, I^B	AB
O	$I^O I^O$	OO

Incomplete Dominance

- Incomplete dominance occurs when neither trait is truly dominant over the other.

- This is not blending inheritance because the traits are separable (particulate), as shown in further crosses.
- Offspring of a cross between heterozygotes show three phenotypes: each parental phenotype and the heterozygous phenotype.
- The phenotypic and genotypic ratios are identical: 1:2:1.
- A clear example of incomplete dominance is the flower color of snapdragons.
- A cross between a white-flowered plant and a red-flowered plant produces all pink F1 offspring.
- Self-pollination of the F1 offspring produces 25% white, 25% red, and 50% pink F2 offspring.



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Codominance

• For example, the M, N, and MN blood groups of humans are due to the presence of two specific molecules on the surface of red blood cells.

• People of group M (genotype MM) have one type of molecule on their red blood cells, people of group N (genotype NN) have the other type, and people of group MN (genotype MN) have both molecules present.

• The MN phenotype is not intermediate between M and N phenotypes but rather exhibits both the M and the N phenotype.

MN Blood Group System

genotype	phenotype	antigen present on red bc
$L^M L^M$	M	M
$L^M L^N$	MN	M and N
$L^N L^N$	N	N

The Relationship Between Dominance and Phenotype

- An allele is called dominant because it is seen in the phenotype, not because it somehow subdues a recessive allele.

- Alleles are simply variations in a gene's nucleotide sequence.

- When a dominant allele coexists with a recessive allele in a heterozygote, they do not interact at all.

It is in the pathway from genotype to phenotype that dominance and recessiveness come into play.

Tay-Sachs disease, an inherited disorder in humans

For any character, the observed dominant/recessive relationship of alleles depends on the level at which we examine phenotype.

• Tay-Sachs disease is fatal; a dysfunctional enzyme causes an accumulation of lipids in the brain

– Only children who inherit two copies of the Tay-Sachs allele (homozygotes) have the disease. Thus, at the organismal level, the Tay-Sachs allele qualifies as recessive.

– At the biochemical level, the phenotype (i.e., the enzyme activity level) is incompletely dominant. The activity level of the lipid-metabolizing enzyme in heterozygotes is intermediate between the activity level in individuals homozygous for the normal allele and the activity level in individuals with Tay-Sachs disease.

– At the molecular level, the alleles are codominant. Heterozygous individuals produce equal numbers of normal and dysfunctional enzyme molecules.

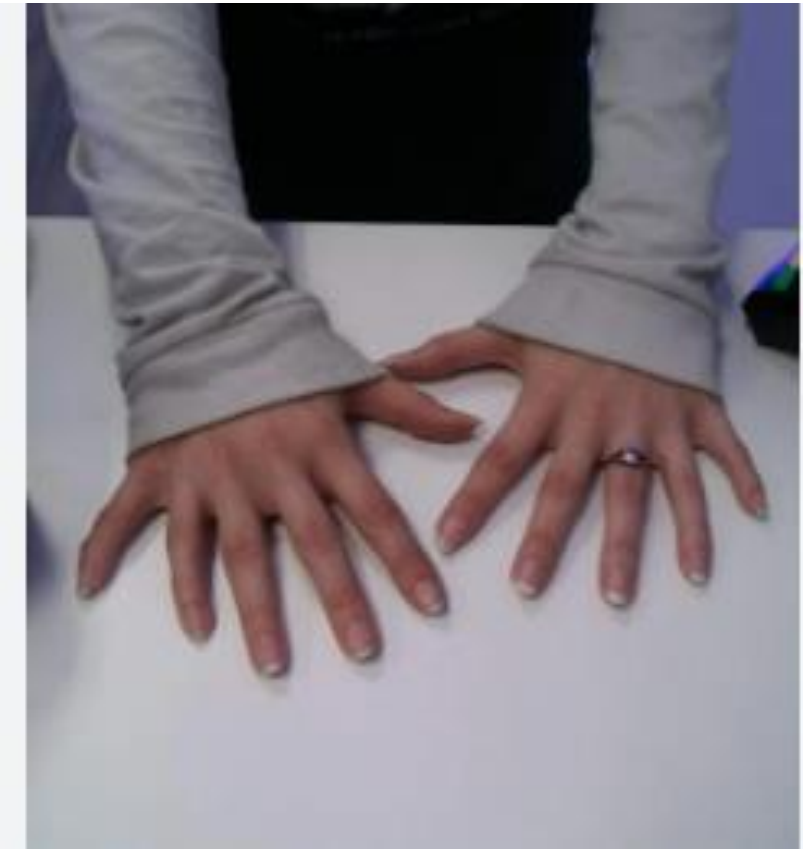
Frequency of Dominant Alleles

- Dominant alleles are not necessarily more common in populations than recessive alleles

- For example, one baby out of 400 in the United States is born with extra fingers or toes

- The allele for this unusual trait is dominant to the allele for the more common trait of five digits per appendage

- In this example, the recessive allele is far more prevalent than the population's dominant allele



Multiple Alleles

• Most genes exist in populations in more than two allelic forms

• For example, the four phenotypes of the ABO blood group in humans are determined by three alleles for the enzyme (I) that attaches A or B carbohydrates to red blood cells: IA, IB, and i.

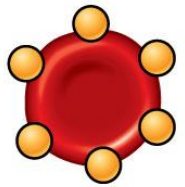
• The enzyme encoded by the IA allele adds the A carbohydrate, whereas the enzyme encoded by the IB allele adds the B carbohydrate; the enzyme encoded by the i allele adds neither

Both the IA and IB alleles are dominant to the i allele.

(a) The three alleles for the ABO blood groups and their carbohydrates

Allele	I^A	I^B	i
Carbohydrate	A	B	none

(b) Blood group genotypes and phenotypes

Genotype	I^AI^A or I^Ai	I^BI^B or I^Bi	I^AI^B	ii
Red blood cell appearance				
Phenotype (blood group)	A	B	AB	O

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Extending Mendelian Genetics for Two or More Genes

- Most genes have multiple phenotypic effects, a property called **pleiotropy**

- For example, pleiotropic alleles are responsible for the multiple symptoms of certain hereditary diseases, such as cystic fibrosis and sickle-cell disease

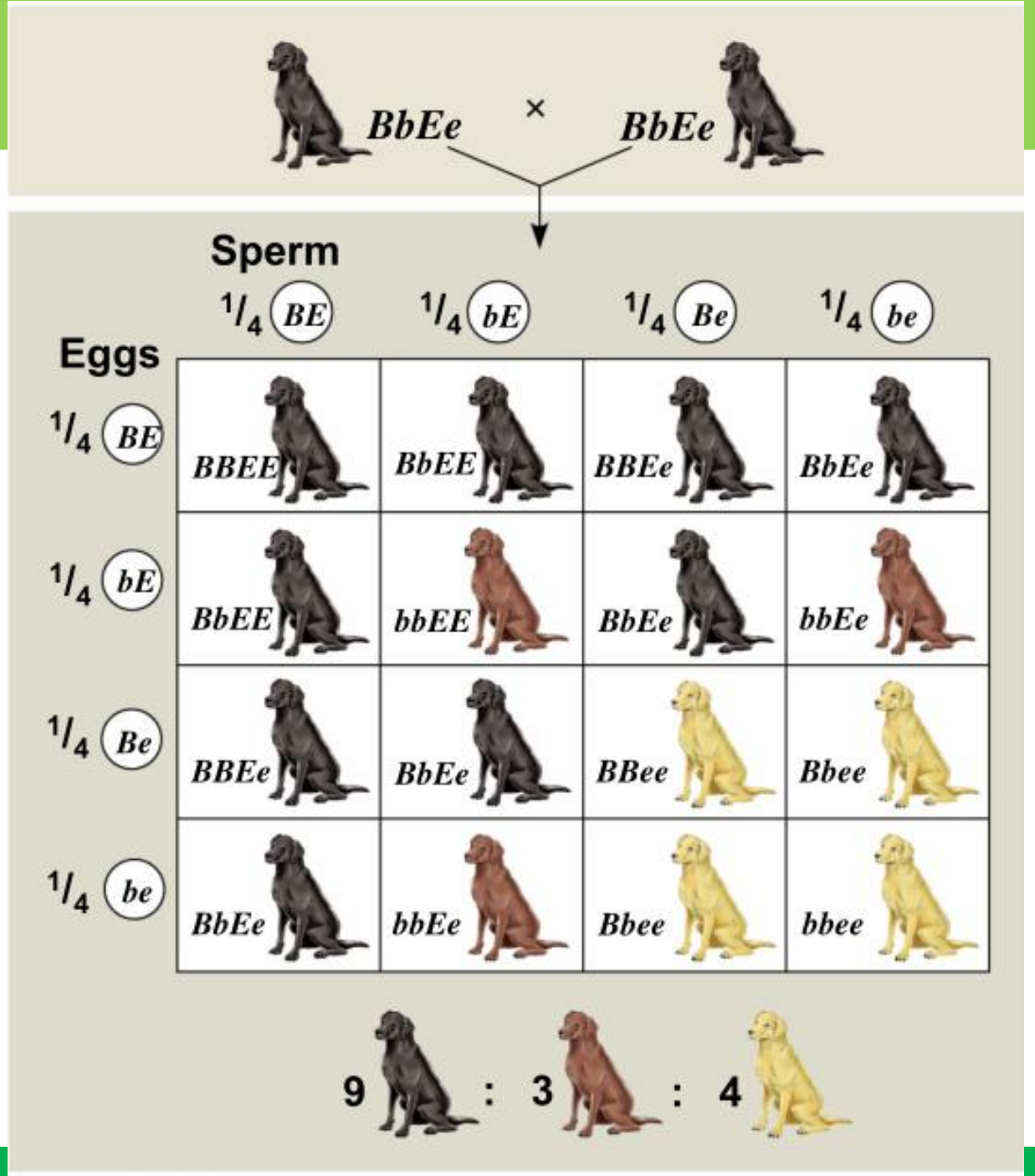
- Some traits may be determined by two or more genes

- In **epistasis**, a gene at one locus alters the phenotypic expression of a gene at a second locus.

e.g. Coat color in Labrador retrievers depends on two genes.
1 gene determines pigment color (B for black, and b for brown)
1 gene determines (C: color; c: no color) whether the pigment will be deposited in the hair

Epistasis

An example of epistasis. This Punnett square illustrates the genotypes and phenotypes predicted for offspring of matings between two black Labrador retrievers of genotype BbEe. The E/e gene, which is epistatic to the B/b gene coding for hair pigment, controls whether or not pigment of any color will be deposited in the hair.



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Polygenic Inheritance

•Polygenic inheritance : quantitative inheritance, where multiple independent genes have an additive or similar effect on a single quantitative trait.

Quantitative characters are those that vary in the population in a gradual fashion
e.g. height

• Quantitative variation usually indicates **polygenic inheritance** an additive effect of two or more genes on a single phenotype

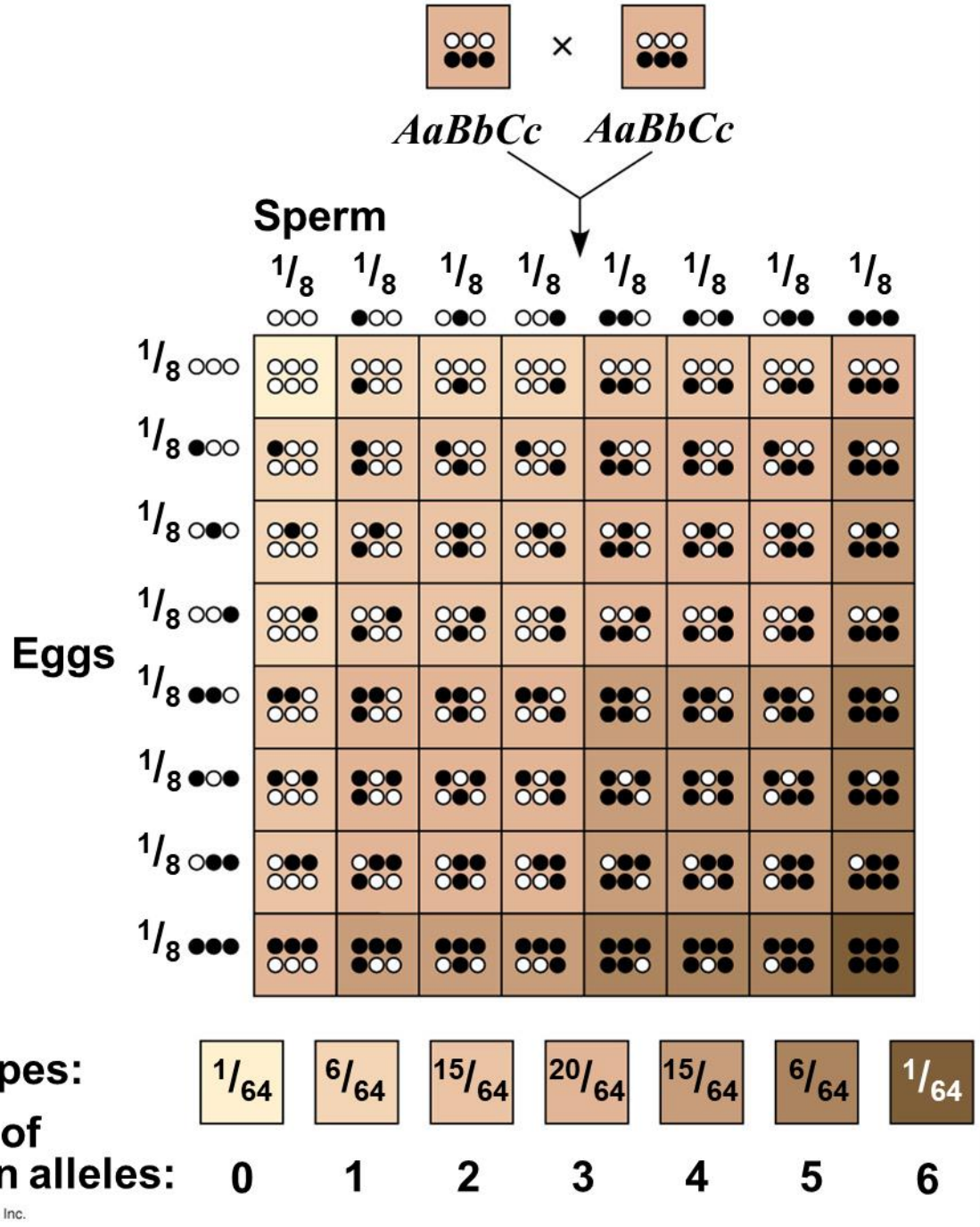
• Skin color in humans is an example of polygenic inheritance

• A cross between two AaBbCc individuals (with intermediate skin shade) produces offspring with a wide range of shades.

In 2014, a genomic study of over 250,000 people found almost 700 genetic variations in over 180 genes that affect height.

Polygenic Inheritance

A simplified model for polygenic inheritance of skin color. In this model, three separately inherited genes affect skin color. (The reported number is actually 378 genes.) The heterozygous individuals (AaBbCc) represented by the two rectangles at the top of this figure each carry three dark-skin alleles (black circles, which represent A, B, or C) and three light- skin alleles (white circles, which represent a, b, or c). The Punnett square shows all possible genetic combinations in gametes and offspring of hypothetical matings between these two. The results are summarized by phenotypic frequencies (fractions) under the Punnett square. (The phenotypic ratio is 1:6:15:20:15:6:1.)



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The Environmental Impact on Phenotype

• Another departure from Mendelian genetics arises when the phenotype for a character depends on environment as well as genotype.

• Traits that depend on multiple genes combined with environmental influences are called **multifactorial.** Such characters are called multifactorial because genetic and environmental factors collectively influence phenotype

• The **norm of reaction** is the phenotypic range of a genotype influenced by the environment.
Norms of reaction are generally broadest for polygenic characters
• For example, hydrangea flowers of the same genotype range from blue-violet to pink, depending on soil acidity



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Integrating a Mendelian View of Heredity and Variation

- An organism's phenotype includes its physical appearance, internal anatomy, physiology, and behavior

- An organism's phenotype reflects its overall genotype and unique environmental history

Many human traits follow Mendelian patterns of inheritance

- Humans are not good subjects for genetic research

– Generation time is too long

– Parents produce relatively few offspring

– Breeding experiments are unacceptable

- However, basic Mendelian genetics endures as the foundation of human genetics

Pedigree Analysis

• A **pedigree** is a family tree that describes the interrelationships of parents and children across generations

•Inheritance patterns of particular traits can be traced and described using pedigrees

• Information about the presence or absence of a particular phenotypic trait is collected from as many individuals in a family as possible, across generations.

• The distribution of these characters is then mapped on the family tree.

An important application of a pedigree is to help us calculate the probability that a future child will have a particular genotype and phenotype.

Pedigree Analysis

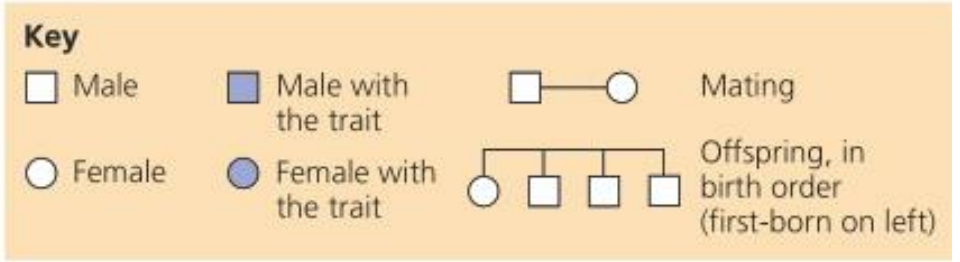
– For example, the occurrence of a widow's peak (W) is dominant to a straight hairline (w).

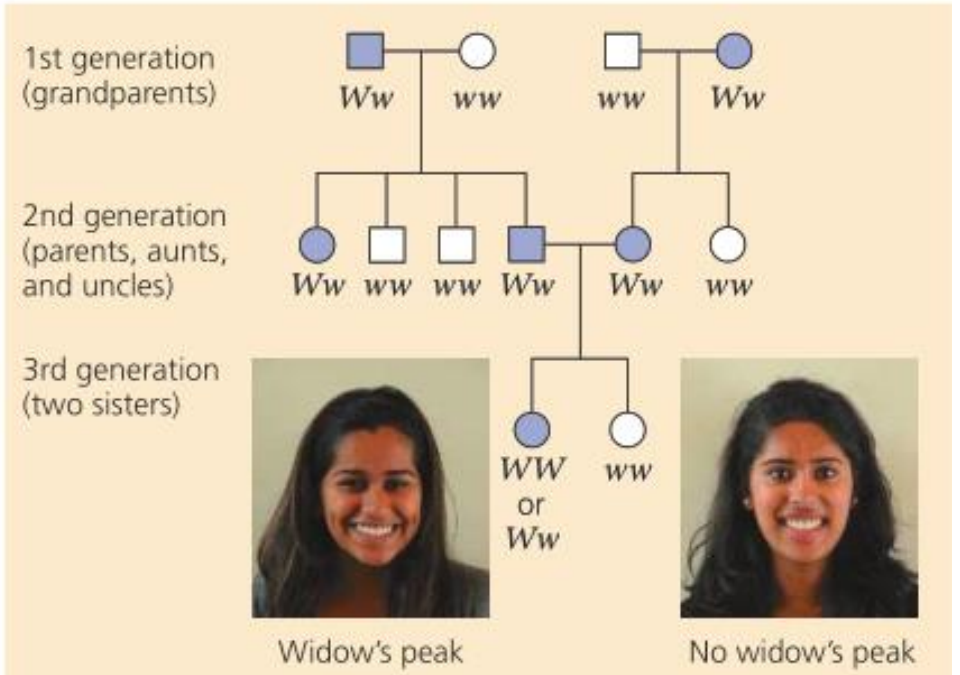
• Phenotypes of family members and knowledge of dominance/recessiveness relationships between alleles allow researchers to predict the genotypes of members of this family.

– For example, if an individual in the third generation lacks a widow's peak but both her parents have widow's peaks, then her parents must be heterozygous for that gene.

– If some siblings in the second generation lack a widow's peak and one of the grandparents (first generation) also lacks one, then we know the other grandparent must be heterozygous, and we can determine the genotype of many other individuals.

▼ Figure 14.15 Pedigree analysis. Each pedigree traces a trait through three generations of the same family; the two traits have different inheritance patterns. For the sake of understanding genetic principles, each trait is shown as being determined by two alleles of a single gene—a simplification, since other genes likely also affect these two characters.





(a) Is a widow's peak a dominant or recessive trait?
Tips for pedigree analysis: Notice in the third generation that the second-born daughter lacks a widow's peak, although both of her parents had the trait. Such a pattern indicates that the trait is due to a dominant allele. If it were due to a *recessive* allele, and both parents had the recessive phenotype (straight hairline), *all* of their offspring would also have the recessive phenotype.

Recessively Inherited Disorders

- Many genetic disorders are inherited in a recessive manner.
- These range from relatively mild to life-threatening
- Recessively inherited disorders show up only in individuals homozygous for the recessive allele
- **Carriers** are heterozygous individuals who carry the recessive allele but are phenotypically normal

- If a recessive allele that causes a disease is rare, then the chance of two carriers meeting and mating is low

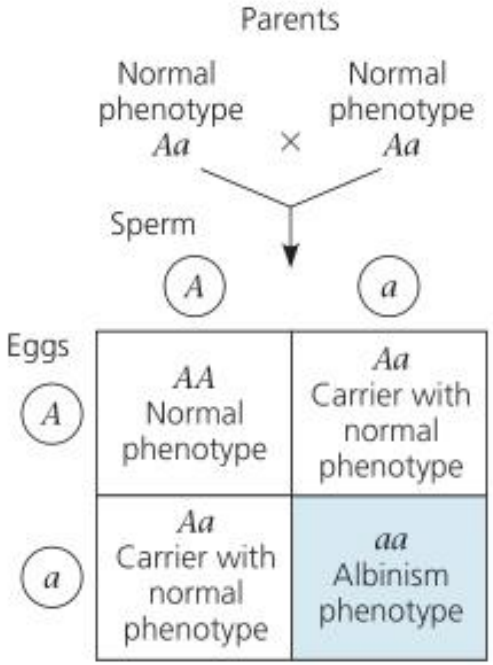
- Consanguineous matings (i.e., matings between close relatives) increase the chance of mating between two carriers of the same rare allele

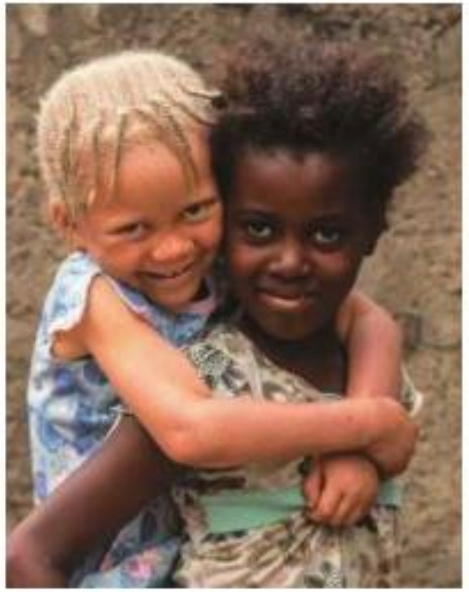
- Most societies and cultures have laws or taboos against marriages between close relatives

Recessively Inherited Disorders

Genes code for proteins of specific function. An allele that causes a genetic disorder (let's call it allele a) codes for either a malfunctioning protein or no protein at all. In recessive disorders, heterozygotes (Aa) typically have the normal phenotype because one copy of the normal allele (A) produces a sufficient amount of the specific protein. Thus, a recessively inherited disorder shows up only in the homozygous individuals (aa) who inherit a recessive allele from each parent. Although phenotypically normal with regard to the disorder, heterozygotes may transmit the recessive allele to their offspring and thus are called carriers.

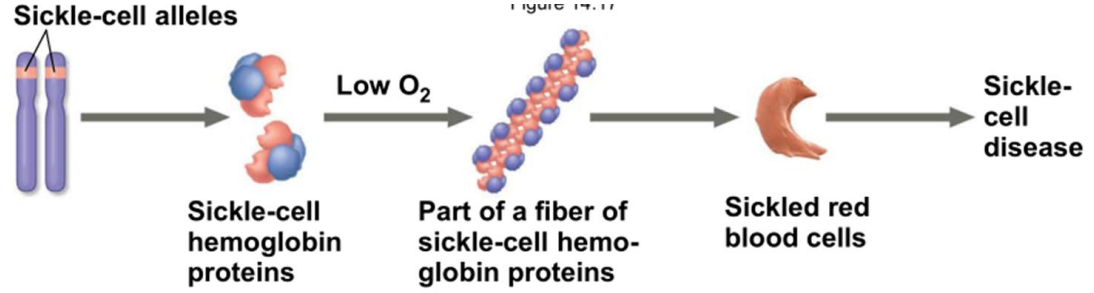
- Albinism is a recessive condition characterized by a lack of pigmentation in skin and hair
- Most recessive homozygotes are born to parents who are carriers of the disorder, but themselves lack the disorder



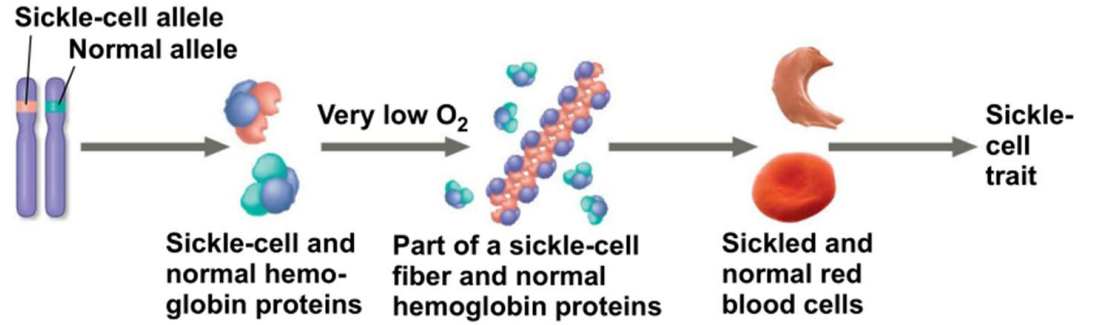


Sickle-Cell Disease

- The disease is caused by the substitution of a single amino acid in the hemoglobin protein in red blood cells. Glutamine is replaced by Valine.
- The non sickle allele is incompletely dominant to the sickle-cell allele.
- Heterozygotes (said to have sickle-cell trait) are usually healthy but may suffer symptoms of sickle-cell disease under blood-oxygen stress.
- About one out of ten African Americans has sickle cell trait, an unusually high frequency of an allele with detrimental effects in homozygotes
- Heterozygotes are less susceptible to the malaria parasite, so there is an advantage to being Heterozygous.



(a) Homozygote with sickle-cell disease: Weakness, anemia, pain and fever, organ damage



(b) Heterozygote with sickle-cell trait: Some symptoms when blood oxygen is very low; reduction of malaria symptoms

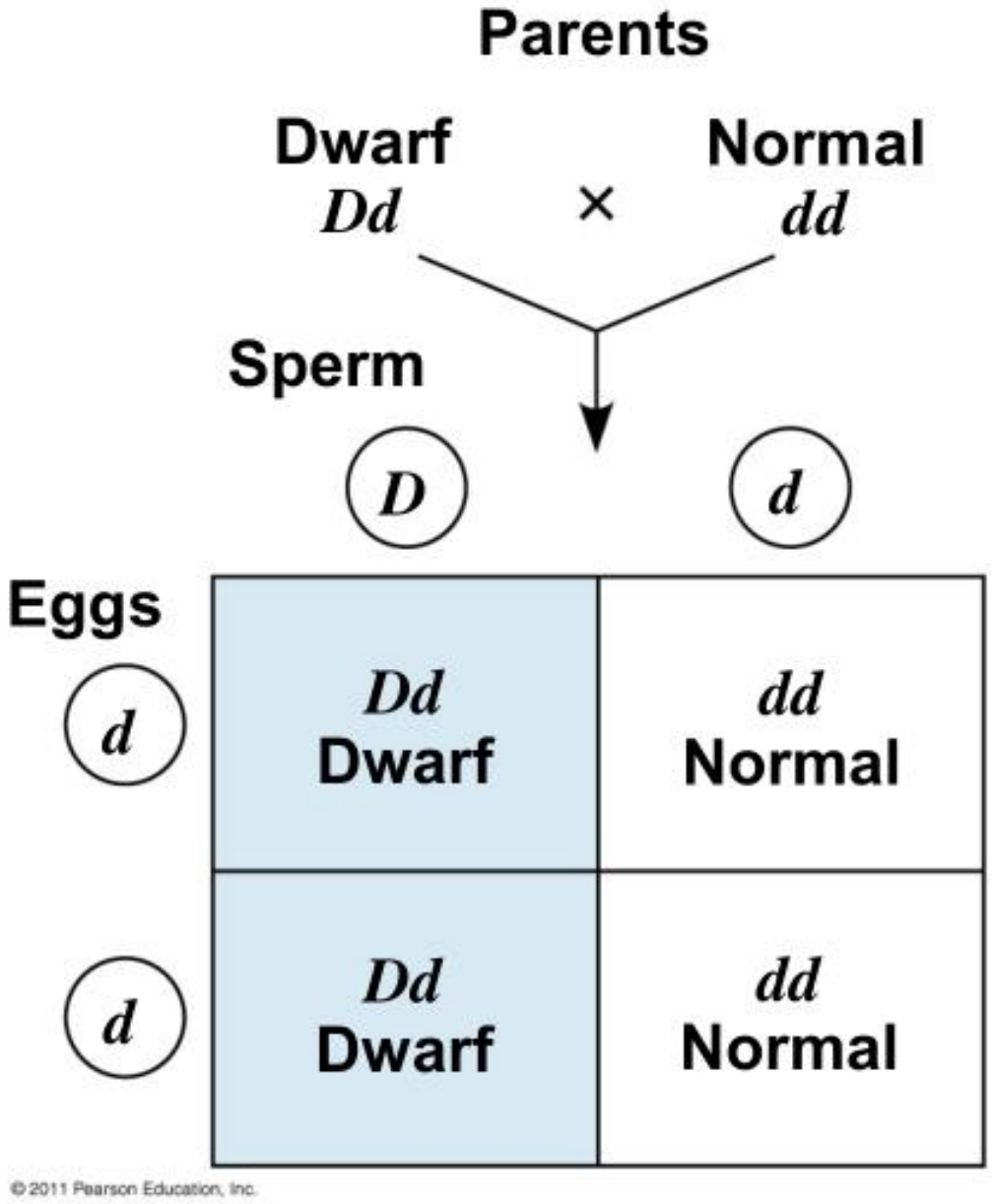
Dominantly Inherited Disorders

- Some human disorders are caused by dominant alleles
- Dominant alleles that cause a lethal disease are rare and arise by mutation
- Achondroplasia is a form of dwarfism caused by a rare dominant allele

– Heterozygous individuals have the dwarf phenotype.

– The 99.99% of the population who are not achondroplastic dwarfs are homozygous recessive for this trait.

– Achondroplasia is another example of a trait for which the recessive allele is far more prevalent than the dominant allele.



Dominantly Inherited Disorders

- A lethal dominant allele can escape elimination if it causes death at a relatively advanced age, after the individual has already passed on the lethal allele to his or her children.
- One example is Huntington's disease, a degenerative disease of the nervous system (1:10,000 people)
  - The dominant lethal allele has no obvious phenotypic effect until the individual is about 35 to 45 years old.
  - Then the deterioration of the nervous system is irreversible and inevitably fatal.
- Any child born to a parent who has the allele for Huntington's disease has a 50% chance of inheriting the disease and the disorder.
- Recently, molecular geneticists have used pedigree analysis of affected families to track the Huntington's allele to a locus near the tip of chromosome 4.
- The gene has now been sequenced.
- This has led to the development of a test that can detect the presence of the Huntington's allele in an individual's genome.

Multifactorial Disorders

- Many diseases, such as heart disease, diabetes, alcoholism, mental illnesses, and cancer have both genetic and environmental components
- No matter what our genotype, our lifestyle has a tremendous effect on phenotype
- Little is understood about the genetic contribution to most multifactorial diseases

Counseling Based on Mendelian Genetics and Probability Rules

- Genetic counselors can provide information to prospective parents concerned about a family history for a specific disease
- Using family histories, genetic counselors help couples determine the odds that their children will have genetic disorders
- It is important to remember that each child represents an independent event in the sense that its genotype is unaffected by the genotypes of older siblings
- For a growing number of diseases, tests are available that identify carriers and help define the odds more accurately
- The tests enable people to make more informed decisions about having children
- However, they raise other issues, such as whether affected individuals fully understand their genetic test results
- Probabilities are predicted on the most accurate information at the time; predicted probabilities may change as new information is available

In Utero Fetal Testing

- In amniocentesis, the liquid that bathes the fetus is removed and tested usually between 14-16 weeks
  – Amniocentesis can be used to assess whether the fetus has a specific disease.
- Fetal cells extracted from amniotic fluid are cultured and karyotyped to identify some disorders.
- Other disorders can be identified from chemicals in the amniotic fluids.
- In chorionic villus sampling (CVS), a sample of the placenta is removed and tested
- Other techniques, such as ultrasound and fetoscopy, allow fetal health to be assessed visually in utero

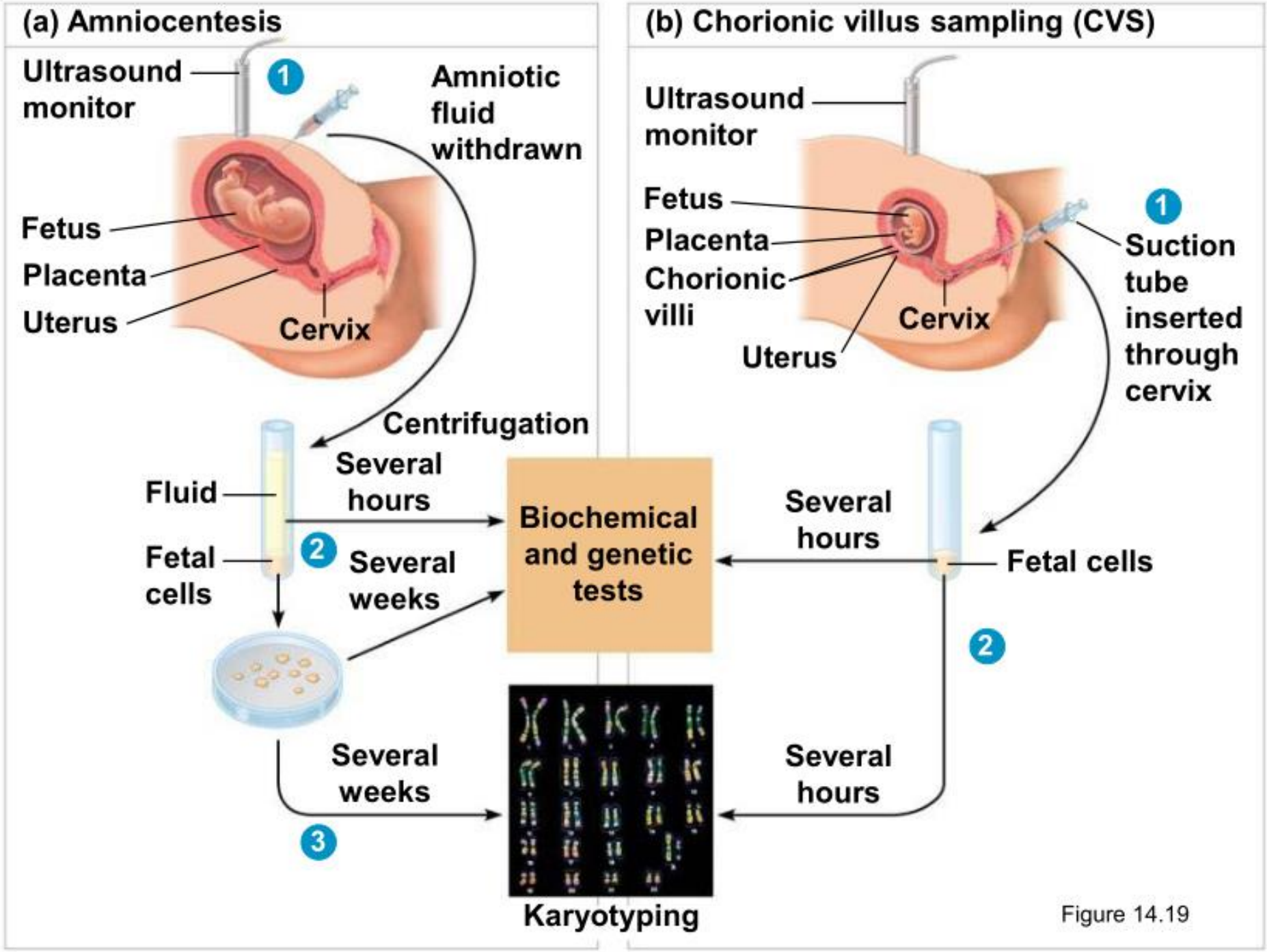

(a) Amniocentesis
Ultrasound monitor
1
Amniotic fluid withdrawn
Fetus
Placenta
Uterus
Cervix
Centrifugation
Fluid
Several hours
Fetal cells
2
Several weeks
Biochemical and genetic tests
Several weeks
3
Karyotyping
(b) Chorionic villus sampling (CVS)
Ultrasound monitor
Fetus
Placenta
Chorionic villi
Uterus
Cervix
1
Suction tube inserted through cervix
Several hours
Fetal cells
2
Several hours

Figure 14.19

Newborn Screening

- Some genetic disorders can be detected at birth by simple

- One common test is for phenylketonuria (PKU), a recessively inherited disorder that occurs in one of every 10,000–15,000 births in the United States

Relationship among alleles of a single gene	Description	Example
Complete dominance of one allele	Heterozygous phenotype same as that of homozygous dominant	
Incomplete dominance of either allele	Heterozygous phenotype intermediate between the two homozygous phenotypes	
Codominance	Both phenotypes expressed in heterozygotes	
Multiple alleles	In the whole population, some genes have more than two alleles	ABO blood group alleles I^A, I^B, i
Pleiotropy	One gene is able to affect multiple phenotypic characters	Sickle-cell disease

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Chromosome Theory of Inheritance (~1902)

- The chromosome theory of inheritance states:

–Mendelian genes have specific loci (positions) on chromosomes

–Chromosomes undergo segregation and independent assortment

- The behavior of chromosomes during meiosis was said to account for Mendel's laws of segregation and independent assortment

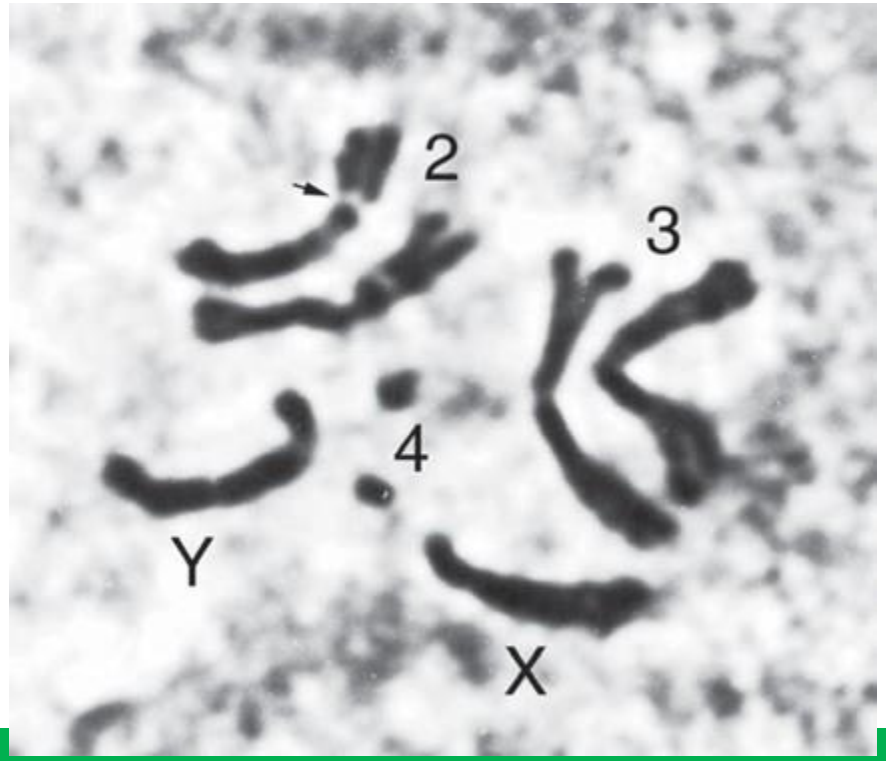
Morgan's Experimental Evidence

• The first solid evidence associating a specific gene with a specific chromosome came from Thomas Hunt Morgan, an experimental embryologist at Columbia University.

• Morgan's experiments with Drosophila melanogaster (fruit flies) provided convincing evidence that chromosomes are the location of Mendel's heritable factors.

- Several characteristics make fruit flies a convenient
- organism for genetic studies:

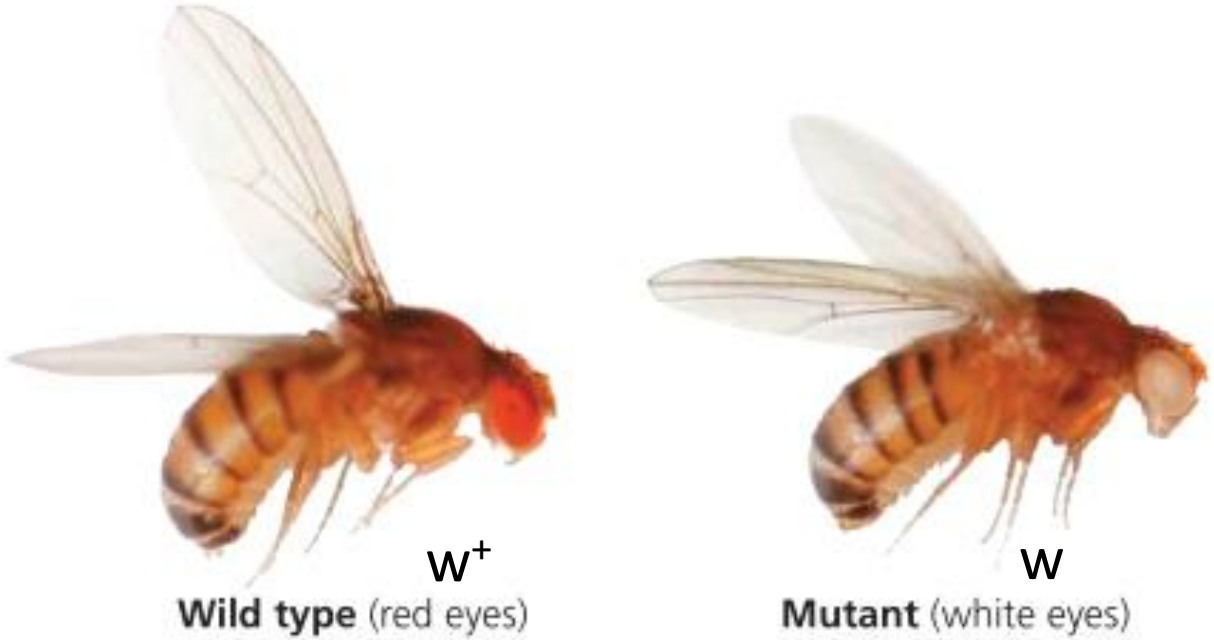
– a single mating will produce hundreds of offspring
– a new generation can be bred every two weeks
–have only four pairs of chromosomes which are easily distinguishable with a light microscope.



Morgan's Experimental Evidence

Morgan noted **wild type, or normal, phenotypes** that were common in the fly populations
Traits alternative to the wild type are called **mutant phenotypes** (because they are due to alleles assumed to have originated as changes, or mutations, in the wild-type allele).

– Morgan spent a year looking for variant individuals among the flies he was breeding!
– Finally he discovered a single male fly with white eyes instead of the usual red.



For a given character in flies, the gene takes its symbol from the first mutant (non–wild type) discovered. Thus, the allele for white eyes in Drosophila is symbolized by w. A super-script + identifies the allele for the wild-type trait: w^+ for the allele for red eyes, for example. Over the years, a variety of gene notation systems have been developed for different organisms.

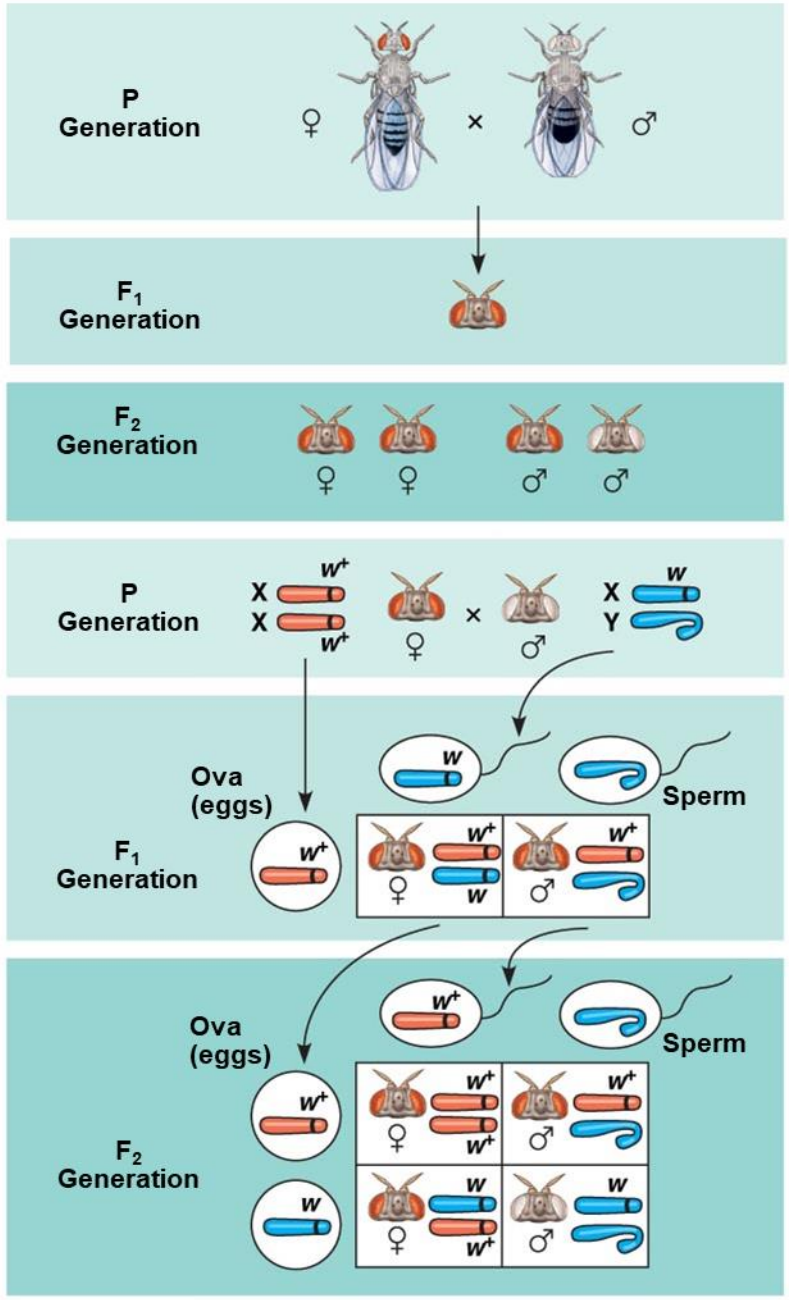
Correlating Behavior of a Gene's Alleles with Behavior of a Chromosome Pair

- Morgan mated his white-eyed male fly with a red-eyed female.
- F1 offspring had red eyes, suggesting that the wild-type allele is dominant.
- The F2 generation showed the 3:1 red:white eye ratio, but only males had white eyes

Conclusion: Somehow a fly's eye color was linked to its sex. The gene with the white-eyed mutation is on the X chromosome, with no corresponding allele present on the Y chromosome.

Females homozygous (XX) or heterozygous (Xx) have red eyes.

Males with (XY) have red eyes but (xY) have white eyes. Single X chromosome is enough to determine eye color for male flies.



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The relationship between the chromosome theory of inheritance and Mendel's laws: Chromosomes during F1 Meiosis

The two alleles for each gene separate during gamete formation.

R and r alleles segregate at anaphase I, yielding two types of daughter cells.

Each gamete gets either the R or r allele.

During fertilization, the R and r alleles recombine randomly.

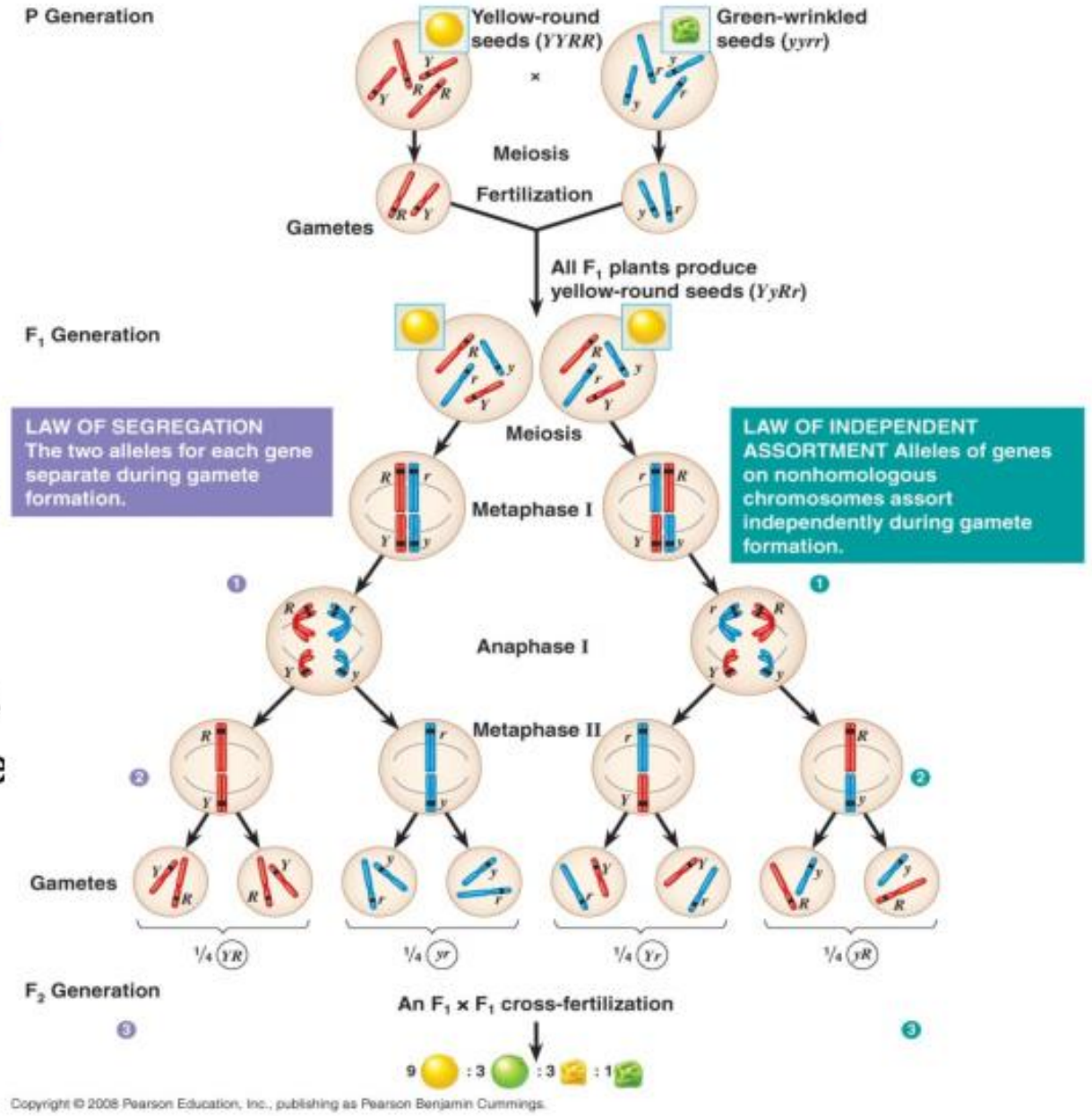


Figure15.2

Alleles on nonhomologous chromosomes assort independently during gamete formation.

1.Alleles at both loci segregate during anaphase I giving rise to 4 different daughter cells, depending on how they arranged during metaphase I.

2. Each gamete gets a chromosome in one of 4 allele combinations.

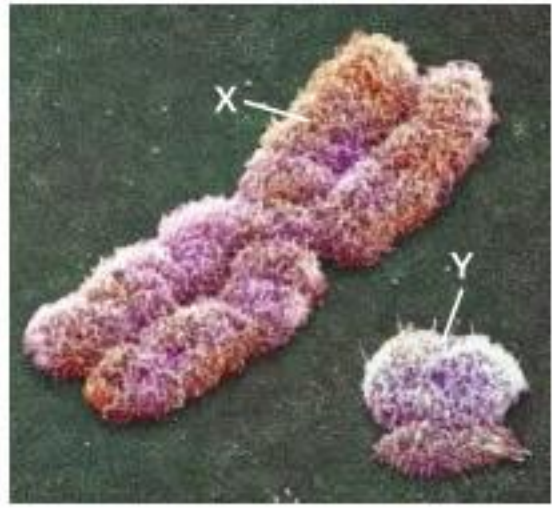
3.Fertilization results in a F2 9:3:3:1 phenotypic ratio.

Sex-linked genes exhibit unique patterns of inheritance

• In humans and some other animals, there is a chromosomal basis of sex determination. Sex in many species is determined largely by inheritance of sex chromosomes.

• Humans and other mammals have 2 types of sex chromosomes, X and Y.
– An individual who inherits two X chromosomes usually develops as a female.
– An individual who inherits an X and a Y chromosome usually develops as a male.
• Short segments at either end of the Y chromosome are the only regions that are homologous with the corresponding regions of the X. These homologous regions allow the X and Y chromosomes in males to pair and behave like homologs during meiosis.
• The SRY gene on the Y chromosome codes for a protein that directs the development of male anatomical features.
– Gonads develop into ovaries in individuals lacking SRY.

▼ **Figure 15.5** Human sex chromosomes (duplicated).



Each conception has about a fifty-fifty chance of producing a particular sex

• In both testes (XY) and ovaries (XX), the two sex chromosomes segregate during meiosis, and each gamete receives one.

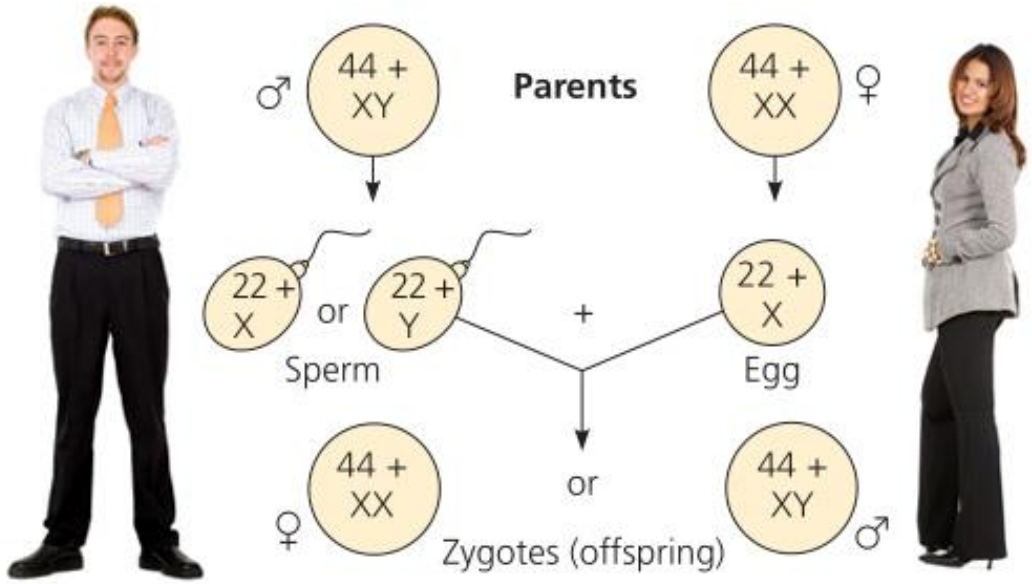
• Each ovum receives an X chromosome.

• Half the sperm cells receive an X chromosome, and half receive a Y chromosome.

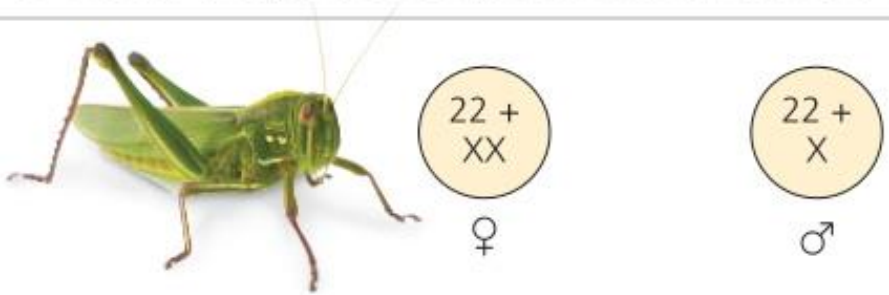
– If a sperm cell bearing an X chromosome fertilizes an ovum, the resulting zygote is female (XX).
– If a sperm cell bearing a Y chromosome fertilizes an ovum, the resulting zygote is male (XY).

--the mammalian X-Y system isn’t the only chromosomal system for determining sex.

Some chromosomal systems of sex determination



(a) The X-Y system. In mammals, the sex of an offspring depends on whether the sperm cell contains an X chromosome or a Y.



(b) The X-0 system. In grasshoppers, cockroaches, and some other insects, there is only one type of sex chromosome, the X. Females are XX; males have only one sex chromosome (X0). Sex of the offspring is determined by whether the sperm cell contains an X chromosome or no sex chromosome.



(c) The Z-W system. In birds, some fishes, and some insects, the sex chromosomes present in the egg (not the sperm) differ, and thus determine the sex of offspring. The sex chromosomes are designated Z and W. Females are ZW and males are ZZ.



(d) The haplo-diploid system. There are no sex chromosomes in most species of bees and ants. Females develop from fertilized eggs and are thus diploid. Males develop from unfertilized eggs and are haploid; they have no fathers.

Inheritance of Sex-Linked Genes

- The sex chromosomes have genes for many characters unrelated to sex.

- A gene located on either sex chromosome is called a sex-linked gene

- The human X chromosome contains approximately 1,100 genes, which are called **X-linked Genes.**

- Genes on the Y chromosome are called **Y-linked genes.** ~78 genes that code for about 25 proteins. About half of these genes are expressed only in the testis, and some are required for normal testicular functioning and the production of normal sperm.

- X chromosome have genes for many characters unrelated to sex, whereas the Y chromosome mainly encodes genes related to sex determination.

- X-linked genes in humans follow the same pattern of inheritance that Morgan observed for the eye color locus he studied in Drosophila

Inheritance of Sex-Linked Genes

- Sex-linked genes follow specific patterns of inheritance

– Fathers pass sex-linked alleles to all their daughters but none of their sons

– Mothers pass sex-linked alleles to both sons and daughters

- For a recessive sex-linked trait to be expressed

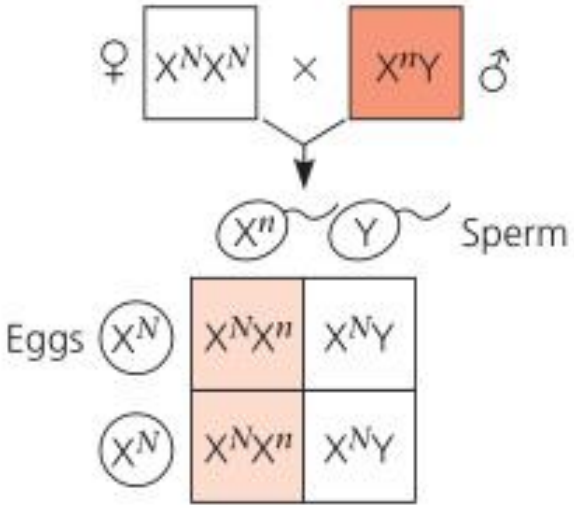
–A female needs two copies of the allele
Heterozygous females are carriers for the recessive trait.

–A male needs only one copy of the allele

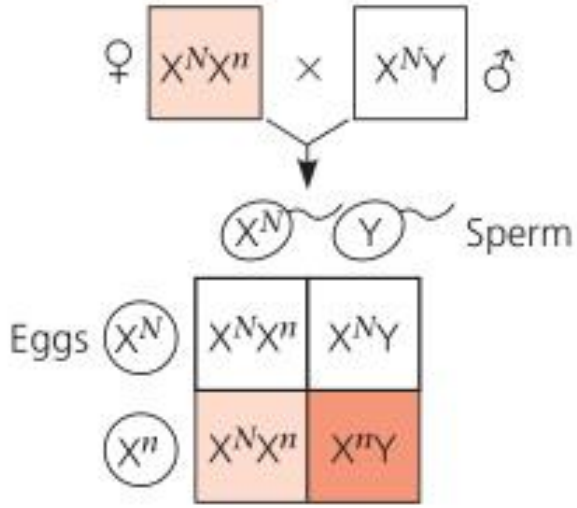
- Sex-linked recessive disorders are much more common in males than in females

Inheritance of Sex-Linked Genes

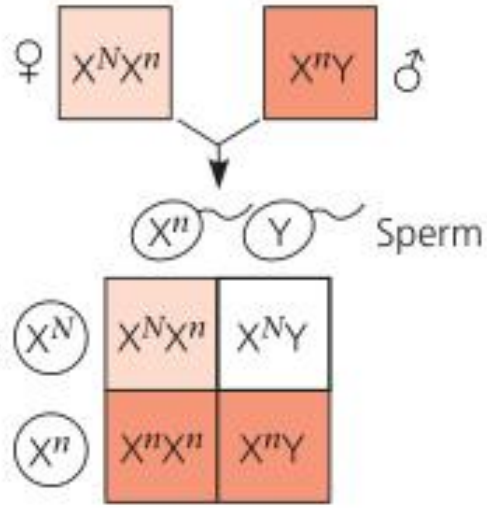
▼ Figure 15.7 The transmission of X-linked recessive traits. In this diagram, red-green color blindness is used as an example. The superscript *N* represents the dominant allele for normal color vision carried on the X chromosome, while *n* represents the recessive allele, which has a mutation causing color blindness. White boxes indicate unaffected individuals, light orange boxes indicate carriers, and dark orange boxes indicate color-blind individuals.



(a) A father with color-blindness will transmit the mutant allele to all daughters but to no sons. When the mother is a dominant homozygote, the daughters will have the normal phenotype but will be carriers of the mutation.



(b) If a carrier mates with a male who has normal color vision, there is a 50% chance that each daughter will be a carrier like her mother and a 50% chance that each son will have the disorder.



(c) If a carrier mates with a color-blind male, there is a 50% chance that each child born to them will have the disorder, regardless of sex. Daughters who have normal color vision will be carriers, whereas males who have normal color vision will be free of the recessive allele.

Inheritance of Sex-Linked Genes

- Some disorders caused by recessive alleles on the X chromosome in humans:

– **Color blindness** (mostly X-linked)

– **Duchenne muscular dystrophy**: a lethal muscular disorder
Affected individuals rarely live past their early 20s. This disorder is due to the absence of an X-linked gene for a key muscle protein called dystrophin. The disease is characterized by a progressive weakening of the muscles and a loss of coordination

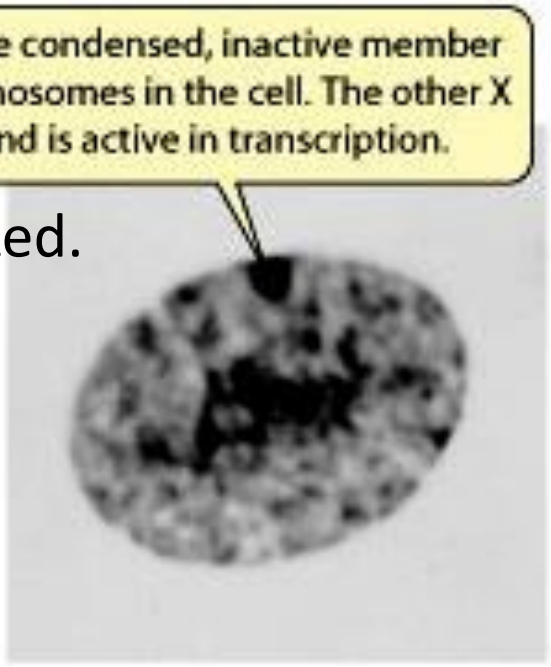
– Hemophilia: absence of one or more proteins required for blood clotting.
These proteins normally slow and then stop bleeding. Individuals with hemophilia have prolonged bleeding because a firm clot forms slowly. Bleeding in muscles and joints can be painful and can lead to serious damage. Today, people with hemophilia can be treated with intravenous injections of the missing protein.

X Inactivation in Female Mammals

- Female mammals inherit two X chromosomes—twice the number inherited by males. But females don't make twice as many of the proteins encoded by X-linked genes as males.

- One of the two X chromosomes in each cell is randomly inactivated during embryonic development. As a result, the cells of females and males have the same effective dose (one active copy) of most X-linked genes.

- The inactive X condenses into a compact object called a Barr body. During interphase an inactive X-chromosome can be visualized.



- Most of the genes on the Barr-body chromosome are not expressed.
- In the ovaries, however, Barr body chromosomes are reactivated in the cells that give rise to eggs, resulting in every egg having an active X chromosome after meiosis.

X Inactivation in Female Mammals

Selection of which X chromosome will form the Barr body occurs randomly and independently in embryonic cells at the time of X inactivation.

As a consequence, females consist of a mosaic of two types of cells: those with the active X derived from the father and those with the active X derived from the mother.

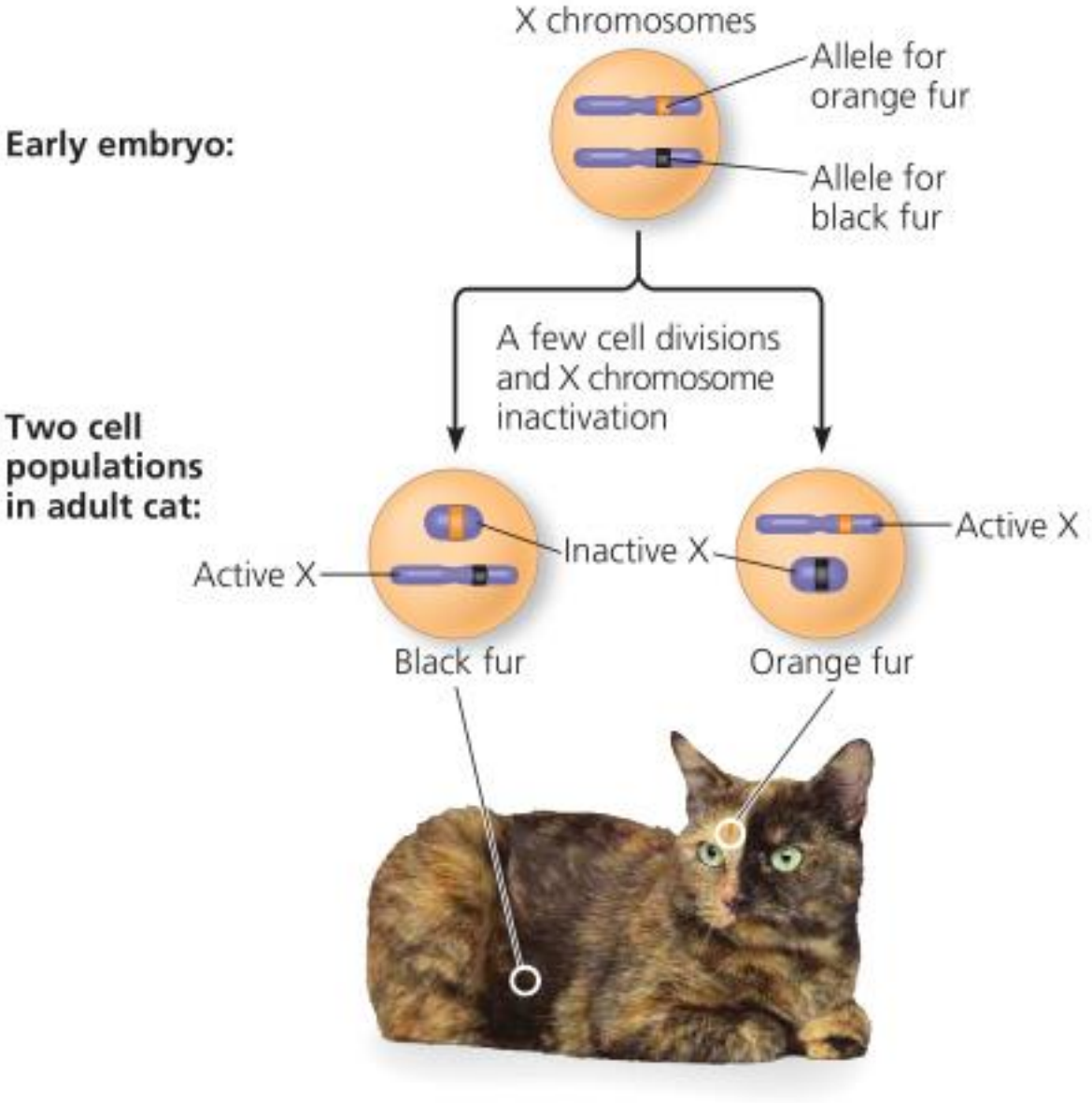
After an X chromosome is inactivated in a particular cell, all mitotic descendants of that cell have the same inactive X.

If a female is heterozygous for a sex-linked trait, approximately half her cells will express one allele, and the other half will express the other allele.

In humans, mosaicism can be observed in a recessive X-linked mutation that prevents the development of sweat glands. A woman who is heterozygous for this trait has patches of normal skin and patches of skin lacking sweat glands.

X Inactivation in the tortoiseshell cat

The tortoise shell gene is located on the X chromosome.
The tortoise-shell phenotype requires both the orange fur and black fur alleles.
Only females can have both alleles (XX).
Females heterozygous for the tortoiseshell gene have orange patches of fur where the orange allele is active as well as patches of black fur where the black allele is active.



What causes X chromosome inactivation?

1. Methylation of DNA nucleotides of the X chromosome.

2. XIST (X-inactive specific transcript) gene.

–XIST gene is active only on the Barr-body chromosome and produces multiple copies of an RNA molecule that attach to the X chromosome on which they were made.
–This initiates X inactivation.

Linked genes tend to be inherited together

Linked genes tend to be inherited together because they are located near each other on the same chromosome.

- Each chromosome has hundreds or thousands of genes.

- Genes located near each other on the same chromosome tend to be inherited together in genetic crosses. Such genes are said to be **genetically linked** and are called **linked genes**.

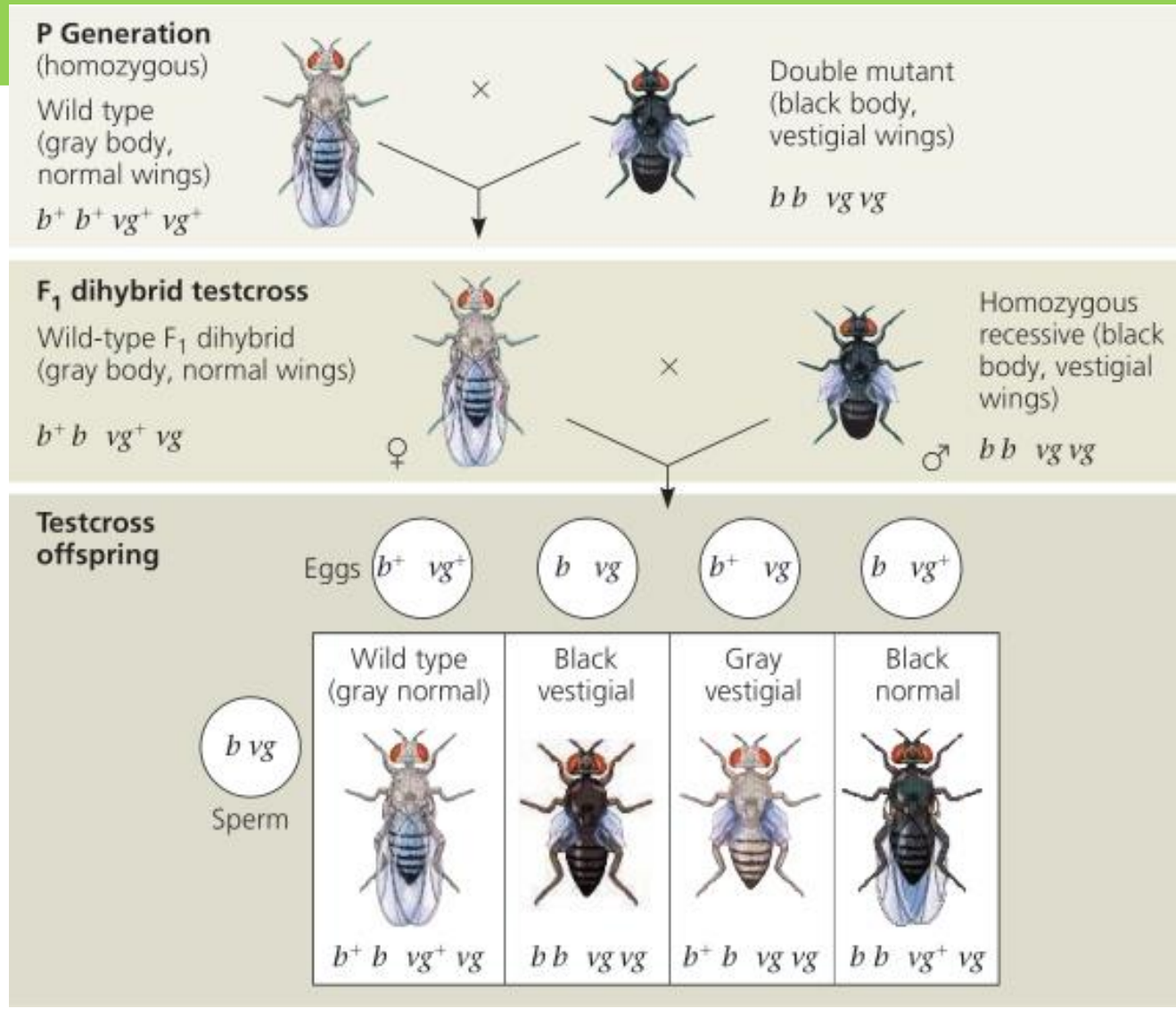
- The results of crosses with linked genes differ from the results expected according to the Mendel's law of independent assortment.

- Morgan observed this linkage and its deviations when he followed the inheritance of characters for body color and wing size in Drosophila.

How does linkage between two genes affect inheritance of characters?

Morgan wanted to know whether the genes for body color and wing size are genetically linked and, if so, how this affects their inheritance.

- Body color: wild type is gray (b+), and the mutant is black (b).
- Wing size: Wild type is (vg+), and the mutant is vestigial (vg).
  – The mutant alleles are recessive to the wild-type alleles.
  – Neither gene is on a sex chromosome.
- Morgan crossed F1 heterozygous females (b^+bvg^+vg) with homozygous recessive males (bbvgvg).
- According to independent assortment, this should produce four phenotypes in a 1:1:1:1 ratio.
- Morgan observed that most F1 offspring resembled parents



if the genes were on seperate chromosomes	1:	1:	1:	1
if genes are located on the same chromosome	1:	1:	0:	0:
Morgan's data	965:	944:	206:	185

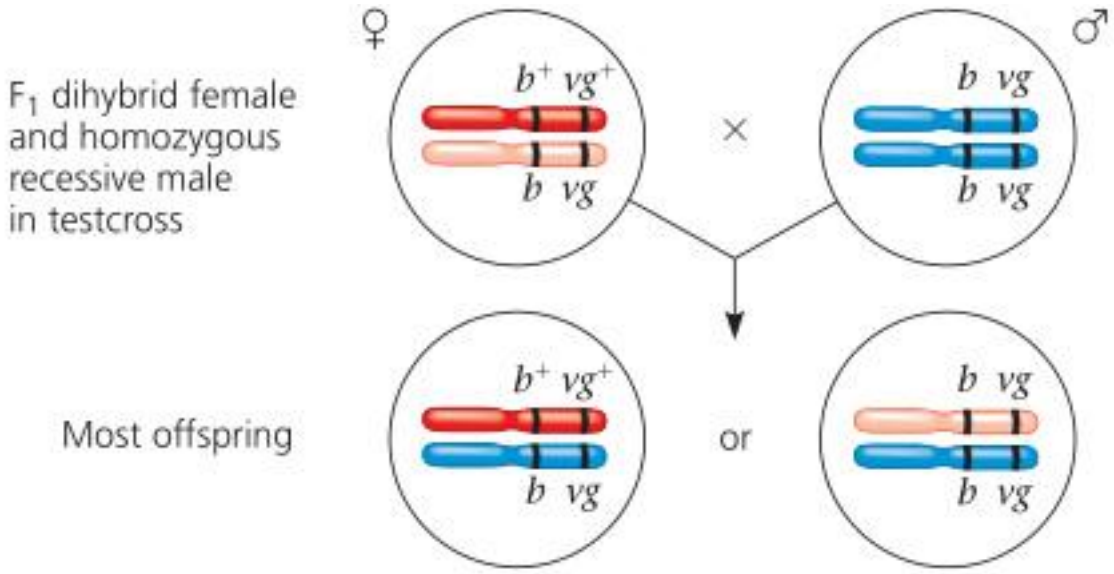
How does linkage between two genes affect inheritance of characters?

How Linkage Affects Inheritance

• Morgan found that body color and wing size are <u>usually</u> inherited together in specific combinations (parental phenotypes)

• He noted that these genes do not assort independently, and reasoned that they were on the same chromosome

• However, nonparental phenotypes were also produced. Understanding this result involves exploring genetic **recombination**, the production of offspring with combinations of traits differing from either parent.



Genetic Recombination

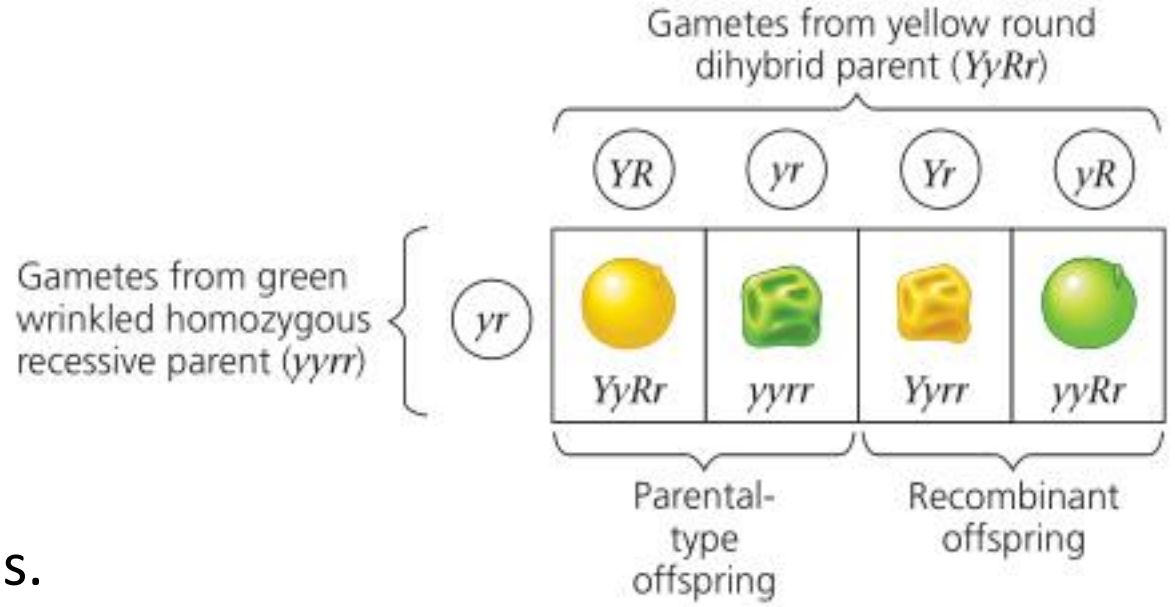
• **Genetic recombination:** The production of offspring with combinations of traits that differ from those found in either parent.
– Genetic recombination can result from independent assortment of genes located on nonhomologous chromosomes.

• Offspring with a phenotype matching one of the parental phenotypes are called **parental types**
• Offspring with nonparental phenotypes (new combinations of traits) are called **recombinant types**, or recombinants

• A 50% frequency of recombination (50% of all offspring are recombinants) is observed for any two genes on different chromosomes.

Recombination of Unlinked Genes: Independent Assortment of Chromosomes

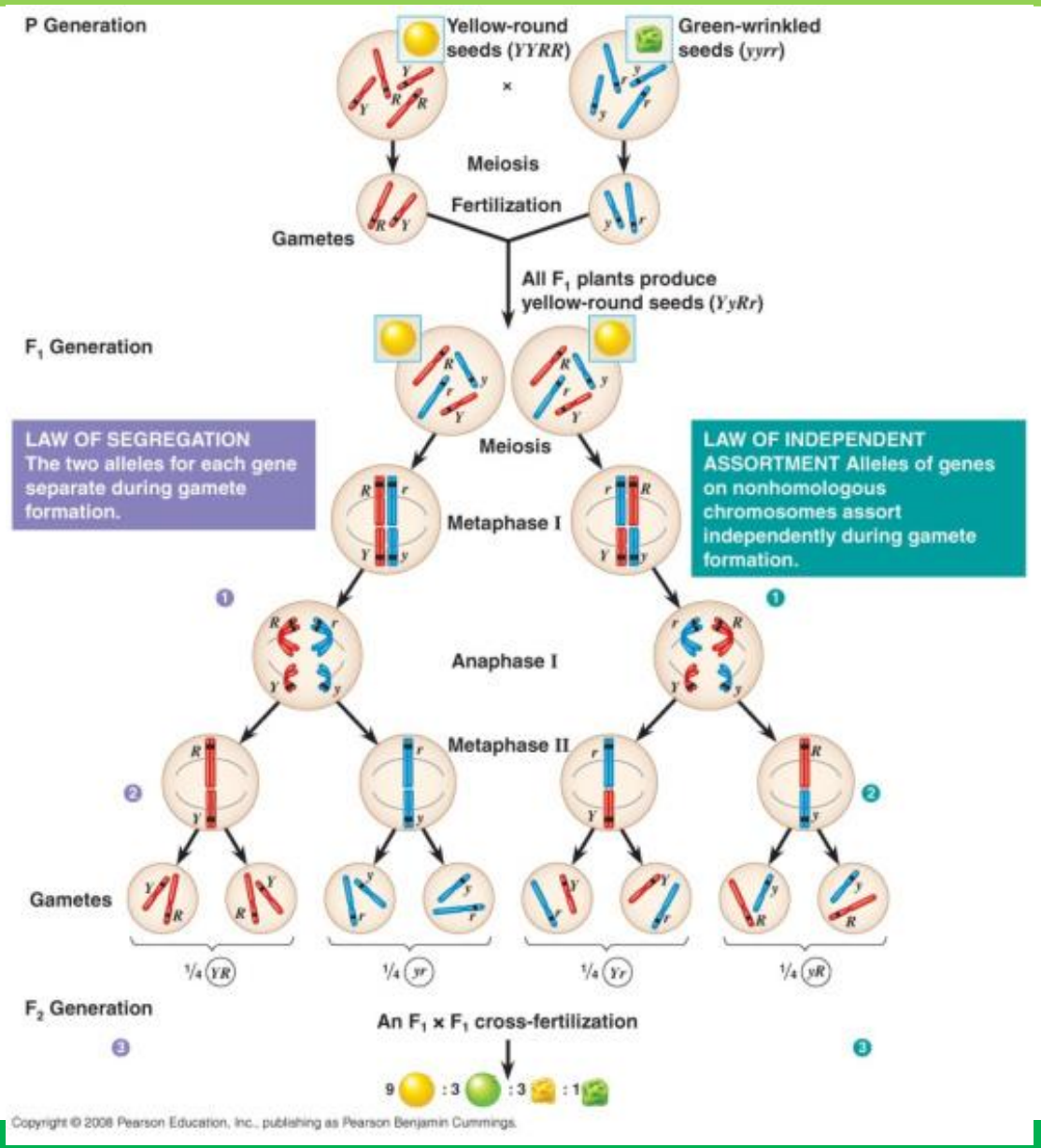
- Mendel's dihybrid cross experiments produced offspring that had a combination of traits that did not match either parent in the P generation.
- If the P generation consists of a yellow-round seed parent (YYRR) crossed with a green-wrinkled seed parent (yyrr), all the F1 plants have yellow-round seeds (YyRr).
- A cross between an F1 plant and a homozygous recessive plant (a testcross) produces four phenotypes.
- Half are the parental types, with phenotypes that match the original P parents, with either yellow-round seeds or green-wrinkled seeds.
- Half are recombinant types or recombinants, new combinations of parental traits, with yellow-wrinkled or green-round seeds.



- The physical basis of recombination between unlinked genes is the random orientation of homologous chromosomes at metaphase I of meiosis, which leads to the independent assortment of the two unlinked genes

Independent Assortment of alleles results in genetic recombination

- A 50% frequency of recombination is observed for any two genes located on different (nonhomologous) chromosomes.
- The physical basis of recombination between unlinked genes is the random orientation of homologous chromosomes at metaphase I of meiosis, which leads to the independent assortment of alleles.
- The F1 parent (YyRr) produces gametes with four different combinations of alleles: YR, Yr, yR, and yr.
- The orientation of the tetrad containing the seed-color gene has no bearing on the orientation of the tetrad with the seed-shape gene.



Recombination of Linked Genes is a result of Crossing Over

In a cross test experiment, the occurrence of parental types with a frequency greater than 50% indicates that the genes are linked.

- Linked genes (genes located on the same chromosome) tend to move together through meiosis and fertilization.
- Under normal Mendelian genetic rules we would not expect linked genes to recombine into assortments of alleles not found in the parents.

– Morgan discovered that genes can be linked, but the linkage was incomplete, as evident from recombinant phenotypes.

- Morgan proposed that some process must sometimes break the physical connection between genes on the same chromosome.
- That mechanism was the **crossing over** of homologous chromosomes.

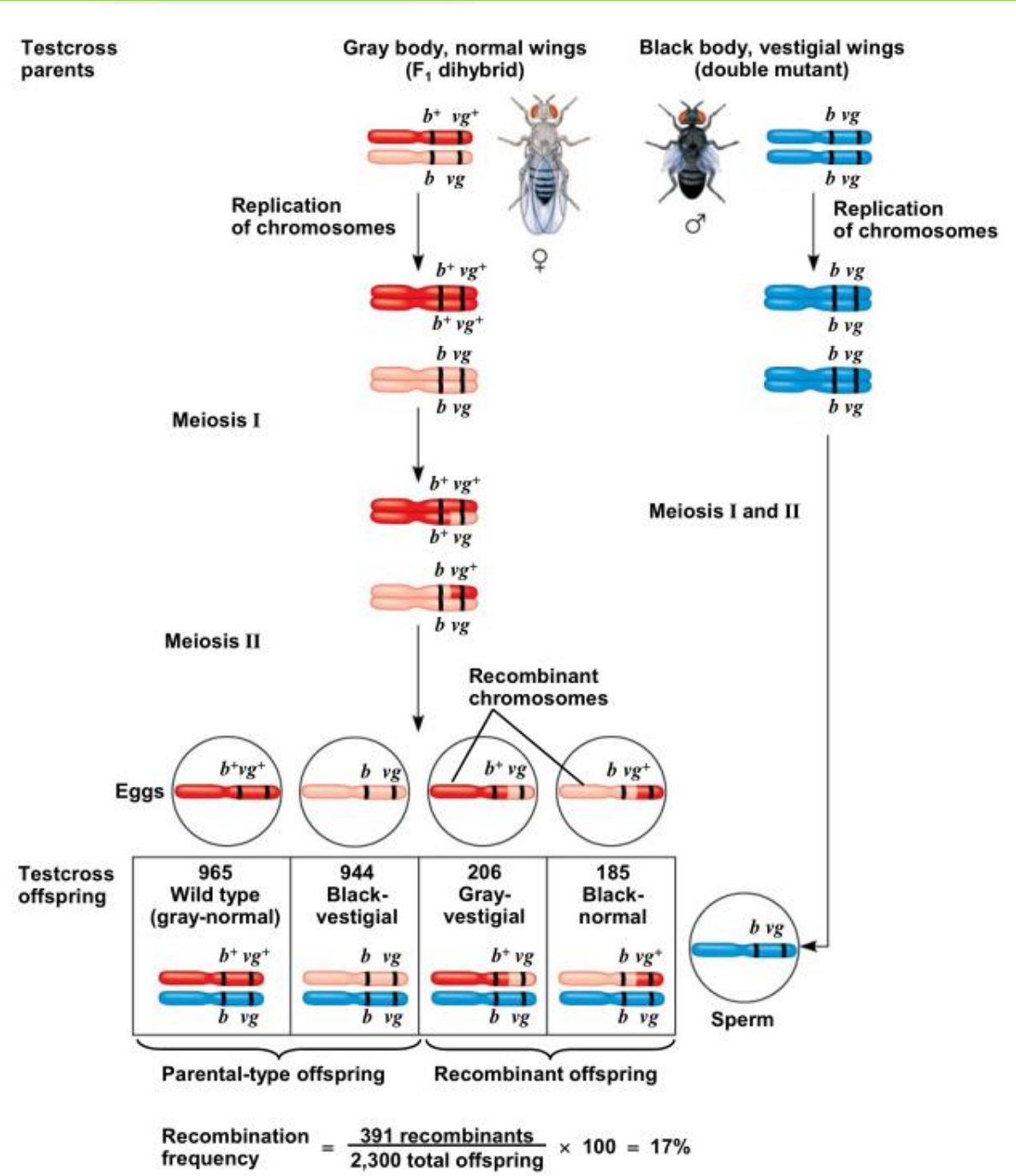
Crossing Over produced recombinant offspring

1. During Meiosis I crossing over between b and vg loci produces new allele combinations in some but not most egg producing cells. Therefore most eggs will harbor the parental type chromosomes.

2. During Meiosis II separation of the chromatids produces recombinant gametes (eggs) with the new allele combinations.

3. Fertilization of the eggs by the bbvgvg sperm will give rise to some recombinant offspring.

The recombination frequency is the percentage of recombinant flies in the total pool of offspring.



New Combinations of Alleles: Variation for Normal Selection

Meiosis and random fertilization generate genetic variation among offspring of sexually reproducing organisms due to independent assortment of chromosomes, crossing over in meiosis I, and the possibility of any sperm fertilizing any egg.

- Recombinant chromosomes bring alleles together in new combinations in gametes
- Random fertilization increases even further the number of variant combinations that can be produced
- This abundance of genetic variation is the raw material upon which natural selection works.

If the traits conferred by particular combinations of alleles are better suited for a given environment, organisms possessing those genotypes will be expected to thrive and leave more offspring, ensuring the continuation of their genetic complement. In the next generation, the alleles will be shuffled anew. Ultimately, the interplay between environment and phenotype (and thus genotype) will determine which genetic combinations persist over time.

Alterations of chromosome number or structure cause some genetic disorders

• Small-scale random mutations are the source of all new alleles and lead to new phenotypic traits.
Large-scale chromosomal changes can also affect an organ ism's phenotype.

• Physical and chemical disturbances as well as errors during meiosis can damage chromosomes in major ways or alter the number of chromosomes in a cell.

• Large-scale chromosomal alterations often lead to spontaneous abortions (miscarriages) or cause a variety of developmental disorders.

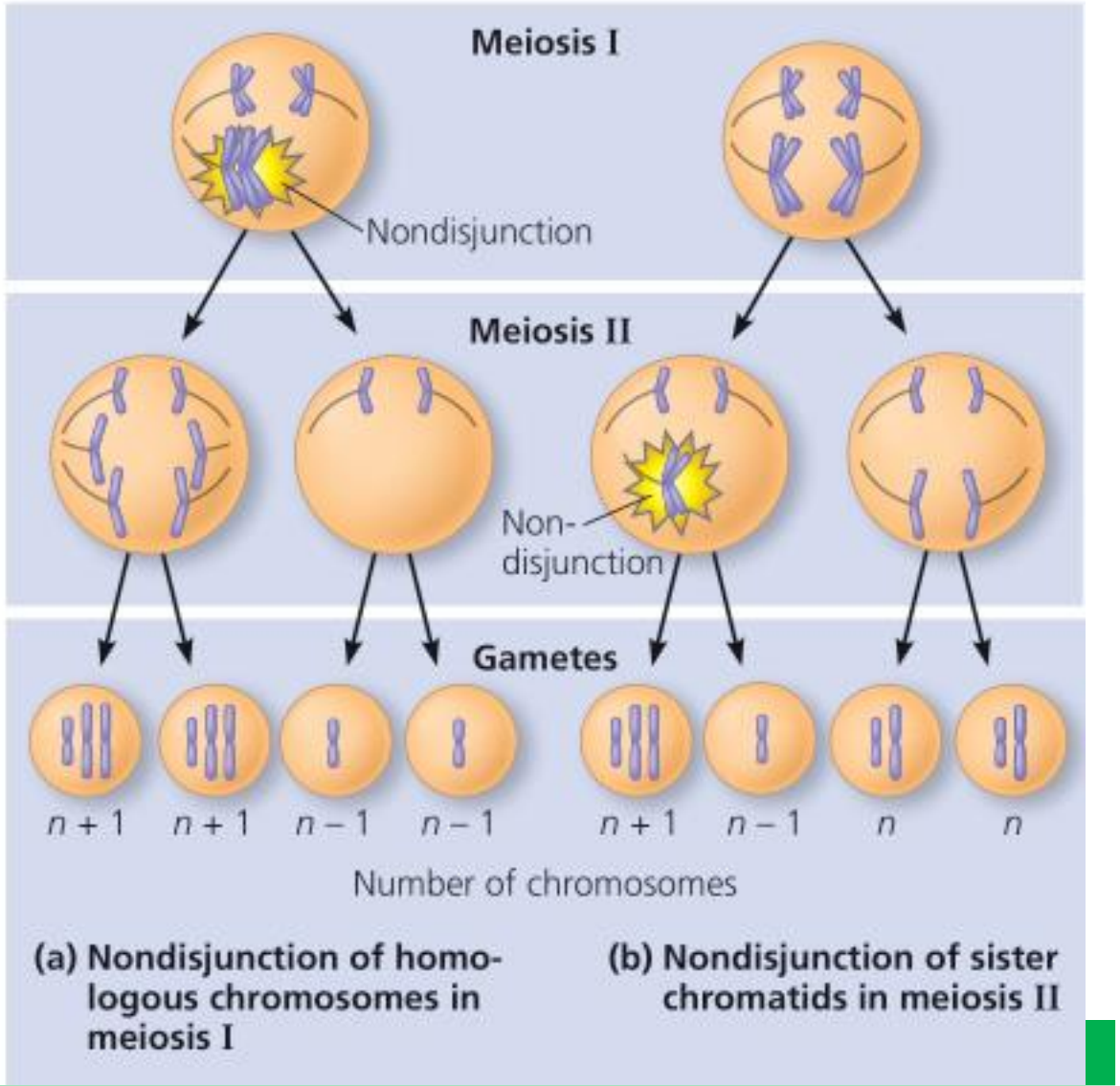
– Plants tolerate genetic defects to a greater extent than do animals.

Abnormal Chromosome Number

Nondisjunction occurs when problems with the meiotic spindle cause errors in separation of homologous chromosomes during meiosis I or separation of sister chromatids during meiosis II.

In nondisjunction, one gamete receives two of the same type of chromosome and another
gamete receives no copy. The other chromosomes are usually distributed normally.

▼ **Figure 15.13 Meiotic nondisjunction.** Gametes with an abnormal chromosome number can arise by nondisjunction in either meiosis I or meiosis II. For simplicity, the figure does not show the spores formed by meiosis in plants. Ultimately, spores form gametes that have the defects shown. (See Figure 13.6.)



Offspring produced by nondisjunction

If a gamete with abnormal number of chromosomes unites with a normal gamete in fertilization, the zygote will also have an abnormal number of a particular chromosome, a condition known as **aneuploidy**.

- **Trisomic cells** have three copies of a particular chromosome and have 2n + 1 total chromosomes. e.g. Down Syndrome
- **Monosomic cells** have only one copy of a particular chromosome and have 2n – 1 chromosomes.
- Nullisomicsomic cells have no copies of a particular chromosome and have 2n -2 total chromosomes.
- If the organism survives, aneuploidy typically leads to a distinct phenotype.

Offspring produced by nondisjunction

Polyploid organisms have more than two complete sets of chromosomes in all somatic cells.
– **Triploidy**-3 sets (3n) of chromosomes
One way a triploid cell may arise is by the fertilization of an abnormal diploid egg produced by non-disjunction of all its chromosomes.
– **Tetraploidy**-4 sets (4n) of chromosomesTetraploidy could result from the failure of a 2n zygote to divide after replicating its chromosomes. Subsequent normal mitotic divisions would
then produce a 4n embryo.

Polyploidy is relatively common among plants and much less common among animals, although it is known to occur in fishes. amphibians and even mammals. e.g. Burrowing rodent
•The spontaneous origin of polyploid individuals plays an important role in the evolution of plants.
•Many crop plants are polyploid. For example, bananas are triploid (3n) and wheat is hexaploid (6n).
•Polyploids are more nearly normal in phenotype than aneuploids.
– One extra or missing chromosome apparently upsets the genetic balance during development more than does an entire extra setof chromosomes.

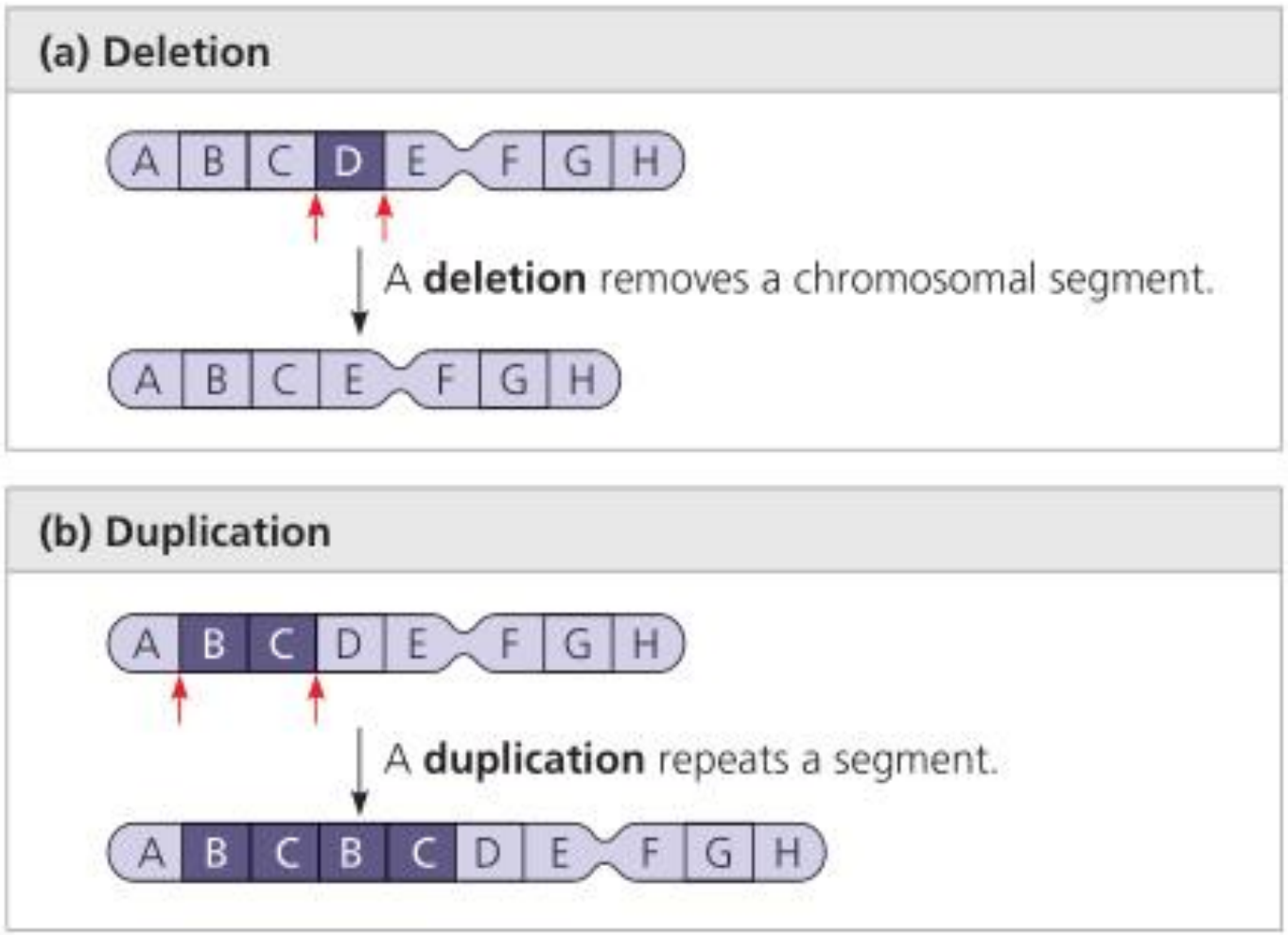
Alterations of Chromosome Structure

Errors in meiosis or damaging agents such as radiation can cause breakage of a chromosome, which can lead to four types of changes in chromosome structure

– Deletion removes a chromosomal segment
– Duplication repeats a segment
– Inversion reverses a segment within a chromosome
– Translocation moves a segment from one chromosome to another

Alterations of Chromosome Structure

▼ Figure 15.14 Alterations of chromosome structure. Red arrows indicate breakage points. Dark purple highlights the chromosomal parts affected by the rearrangements.



Human Disorders Due to Chromosomal Alterations

- Alterations of chromosome number and structure are associated with some serious disorders
  - Most result in miscarriage of the fetus
- Some types of aneuploidy appear to upset the genetic balance less than others, resulting in individuals surviving to birth and beyond
- These surviving individuals have a set of symptoms, or syndrome, characteristic of the type of aneuploidy

Down Syndrome (Trisomy 21)

- Down syndrome is usually the result of an extra chromosome 21, so that each body cell has a total of 47 chromosomes.

- Although chromosome 21 is the smallest human chromosome, trisomy 21 severely alters an individual's phenotype in specific ways.

- The frequency of Down syndrome increases with the age of the mother, a correlation that has not been explained.

Trisomy 21 may be linked to some age-dependent abnormality in a meiosis I checkpoint that normally delays anaphase until all the kinetochores are attached to the spindle.





Aneuploidy of Sex Chromosomes

• Nondisjunction of sex chromosomes produces a variety of aneuploid conditions
• **Klinefelter syndrome** is the result of an extra chromosome in a male, producing **XXY** individuals
–These individuals have male sex organs but abnormally small testes and are sterile.
–Although the extra X is inactivated, some breast enlargement and other female characteristics are common.

• **Monosomy X**, called Turner syndrome, produces **X0** females, who are sterile; it is the only known viable monosomy in humans because their sex organs do not mature.
–When given estrogen replacement therapy, girls with Turner syndrome develop secondary sex characteristics.

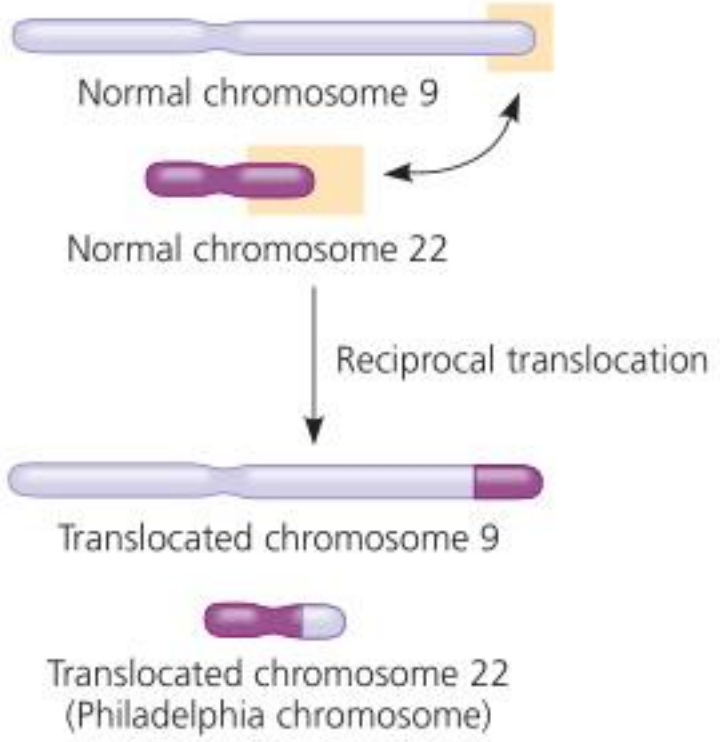
• Trisomy X (XXX) are generally healthy and have no unusual physical features other than being slightly taller than average. Females with trisomy X are at risk for learning disabilities.
• Males with extra Y chromosome (XYY) tend to besomewhat taller than average

Disorders Caused by Structurally Altered Chromosomes

• The syndrome **cri du chat** ("cry of the cat"), results from a specific deletion in chromosome 5
– A child born with this syndrome is mentally retarded and has a catlike cry; individuals
usually die in infancy or early childhood.

• Chromosomal translocations can also occur during mitosis; some have been implicated in certain cancers.
i.e. Chronic myelogenous leukemia (CML), are caused by translocations of chromosomes.

▼ **Figure 15.16 Translocation associated with chronic myelogenous leukemia (CML).** The cancerous cells in nearly all CML patients contain an abnormally short chromosome 22, the so-called Philadelphia chromosome, and an abnormally long chromosome 9. These altered chromosomes result from the reciprocal translocation shown here, which presumably occurred in a single white blood cell precursor undergoing mitosis and was then passed along to all descendant cells.



Some inheritance patterns are exceptions to the standard chromosome theory

- There are two normally occuring exceptions to Mendelian genetics.
- One exception involves genes located in the nucleus, and the other exception involves genes located outside the nucleus.

- In both cases, the sex of the parent contributing the allele is a factor in the pattern of inheritance.

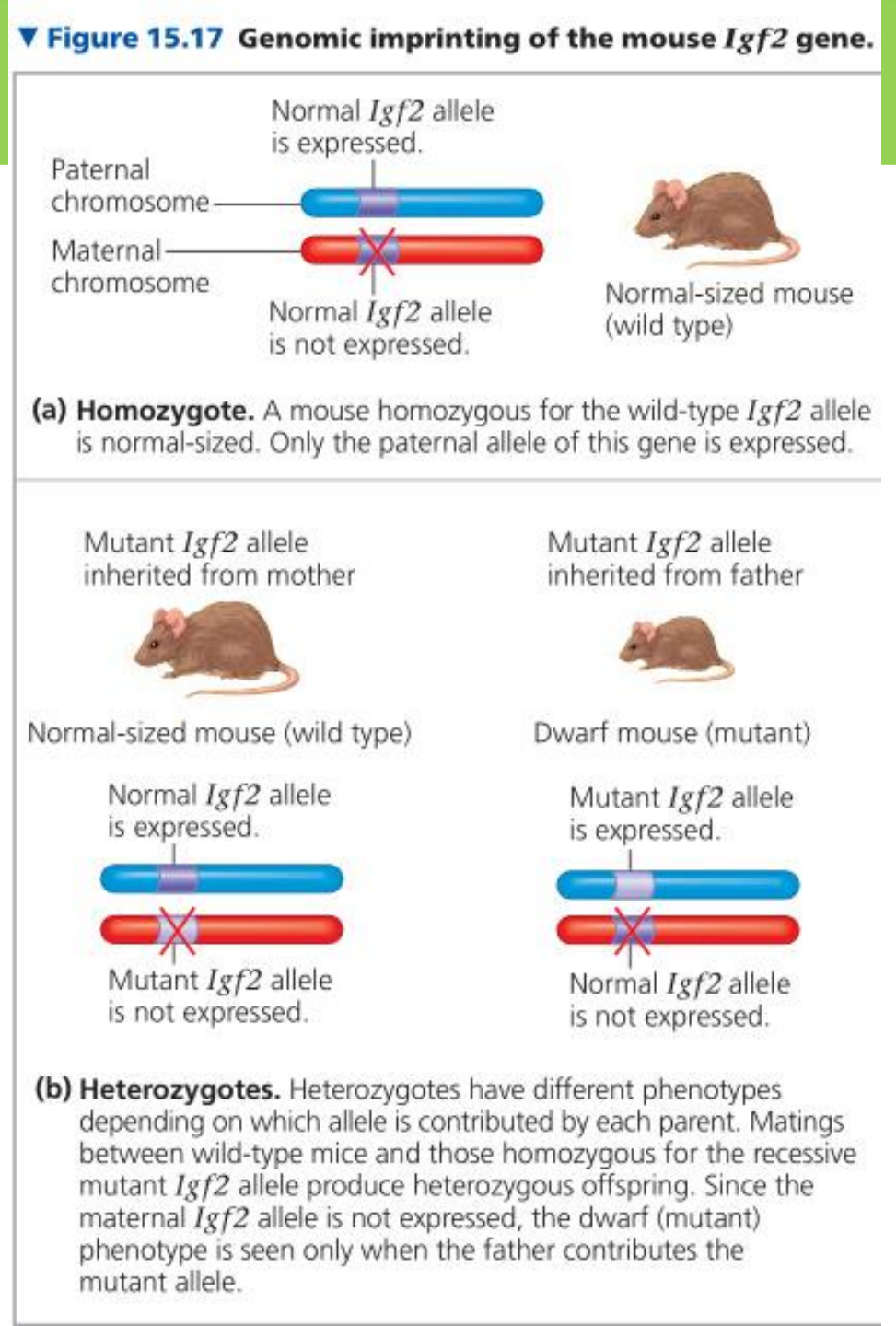
Genomic Imprinting

• Variation in phenotype depending on whether an allele is inherited from the male or female parent is called **genomic imprinting.** (Note that unlike sex-linked genes, most imprinted genes are on autosomes.)
Using newer DNA sequence-based methods, about 100 imprinted genes have been identified in humans.

Genomic imprinting occurs during the formation of gametes and results in the silencing of the imprinted genes.
– Because different genes are imprinted in sperm and eggs, some genes in a zygote are maternally imprinted and others are paternally imprinted.
– For a maternally imprinted gene, only the paternal allele is expressed.
– For a paternally imprinted gene, only the maternal allele is expressed.
•The maternal and paternal imprints are transmitted to all body cells during early development.
•Although only a few genes are imprinted most of these genes are critical for embryonic development.

Genomic Imprinting

- The gene for insulin-like growth factor 2 (Igf2) was one of the first imprinted genes to be identified.
  – Although the growth factor is required for normal prenatal growth, only the paternal allele is expressed.
- Evidence that the Igf2 allele is imprinted initially came from crosses between wild-type mice and dwarf mice homozygous for a recessive mutation in the Igf2 gene.
- The phenotypes of heterozygous offspring differ, depending on whether the mutant allele comes from the mother or the father.
- The Igf2 allele is imprinted in eggs, turning off expression of the imprinted allele.
- In sperm, the Igf2 allele is not imprinted and functions normally.



▼ **Figure 15.17** **Genomic imprinting of the mouse *Igf2* gene.**

(a) Homozygote. A mouse homozygous for the wild-type *Igf2* allele is normal-sized. Only the paternal allele of this gene is expressed.

(b) Heterozygotes. Heterozygotes have different phenotypes depending on which allele is contributed by each parent. Matings between wild-type mice and those homozygous for the recessive mutant *Igf2* allele produce heterozygous offspring. Since the maternal *Igf2* allele is not expressed, the dwarf (mutant) phenotype is seen only when the father contributes the mutant allele.

Genomic Imprinting

In many cases, the imprint seems to consist of methyl (¬CH3) groups that are added to cytosine nucleotides of one of the alleles.

Genomic imprinting is thought to affect only a small fraction of mammalian genes

- Most imprinted genes are critical for embryonic development

Inheritance of Organelle Genes

Not all of a eukaryotic cell's genes are located on nuclear chromosomes, or even in the nucleus; some genes are located in organelles in the cytoplasm. Because they are outside the nucleus, these genes are sometimes called **extranuclear genes** or **cytoplasmic genes**.

Mitochondria, chloroplasts, and other plant plastids carry small circular DNA molecules.

Because a zygote inherits all its mitochondria from egg, all mitochondrial genes in most animals and plants demonstrate maternal inheritance.

– The products of mitochondrial genes make up the protein complexes of the electron transport chain and ATP synthase.
– Some defects in mitochondrial genes prevent cells from making enough ATP and result in diseases that affect the muscular and nervous systems
– For example, mitochondrial myopathy and Leber's hereditary optic neuropathy

Some rare human disorders are produced by mutations to mitochondrial DNA

Tissues that require large energy supplies (the nervous system and muscles) may suffer energy deprivation from these defects.
– For example, a person with mitochondrial myopathy suffers weakness, intolerance of exercise, and muscle deterioration.
– Another mitochondrial disorder is Leber's hereditary optic neuropathy, which can produce sudden blindness in young adults.
• Other mitochondrial mutations may contribute to diabetes, heart disease, and other diseases of aging, such as Alzheimer's disease.
• Over a lifetime, new mutations gradually accumulate in mitochondrial DNA and contribute to the aging process.

- While nuclear DNA is inherited from both parents and recombines with every generation, mitochondrial DNA is inherited solely from your mother and can remain unchanged for tens of thousands of years. As such, mitochondrial DNA ("mitogenome") is a key tool for tracking genetic history.